ORE Studio User Manual

Marco Craveiro

Version 0.0.24 (local 0d172c852-dirty)

Contents

List of Figures

List of Tables

Part I

Getting Started

Everything you need to go from a fresh installation to a fully provisioned, operational OreStudio deployment: what the application is, how to connect to a backend, and how to bootstrap the system for first use.

1

Introduction

This chapter introduces ORE Studio and frames the manual that follows. It examines what the application is and the problem it solves, the open-source quantitative finance engines it builds upon, the audience it is written for, the functional areas through which it is organised, the boundaries of what it does and does not attempt, and the conventions used throughout this book. Its scope is orientation rather than operation: by the end the reader should understand where ORE Studio sits in the risk-technology landscape and how to navigate the remainder of the manual.

Overview

The chapter advances the argument that the reader should understand what ORE Studio is before learning how to use it, and it builds that understanding in layers. It begins by establishing identity in What is ORE Studio?, which explains the application and the open-source engines it rests upon; this is the premise the rest of the chapter depends on. From there it considers for whom the manual is written in Audience, before mapping the territory the application covers in Functional Areas. It then draws the boundary in What ORE Studio Is Not, distinguishing a learning environment from a production system, and shows how the book itself is laid out in How This Manual Is Organised. Finally it sets expectations with two cautions — the Early-Stage Software Notice on the maturity of the software and A Note on This Document on how the manual was produced. The Conclusion draws these steps together.

What is ORE Studio?

ORE Studio is a desktop application for exploring, configuring, and running quantitative finance calculations, backed by a set of headless services and a database rather than doing any of that work it-

self. The Qt client you interact with is a thin presentation layer: it renders what you see and forwards every action to the appropriate backend service, which does the actual work and persists the result. It provides a graphical environment built around the *Open Source Risk Engine* (ORE)[1, 2], a widely-used open-source library for pricing derivatives and measuring financial risk, which is itself built on QuantLib[3, 4]. ORE Studio handles the data — storing trades, market data, and model configurations, and presenting the results — while ORE and QuantLib provide the mathematics. You cannot understand ORE Studio without understanding these two engines beneath it, so it is worth being precise about what each one is and where ORE Studio sits relative to them.

[1] Open Source Risk Engine, project home: https://www.opensourcerisk.org/.

[2] Open Source Risk Engine, source: https://github.com/OpenSourceRisk/Engine.

[3] QuantLib, project home: https://www.quantlib.org/.

[4] QuantLib, source: https://github.com/lballabio/QuantLib.

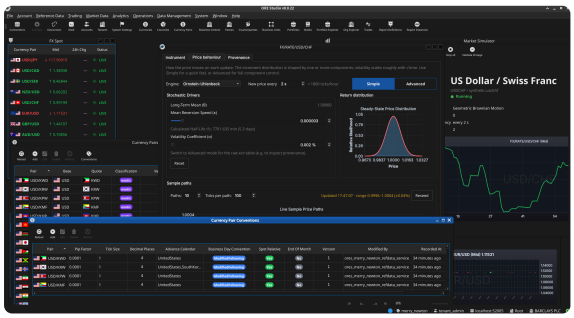


Figure 1.1: ORE Studio v0.0.22 — the main workspace showing the instrument and trade management views.

The Open Source Risk Engine

The *Open Source Risk Engine* is a C++ library for pricing and risk analytics maintained by Acadia, an LSEG (London Stock Exchange Group) business[5]. It builds on QuantLib and extends it with industry-grade valuation adjustments (XVA), sensitivities, regulatory scenarios, and a broad set of financial instruments, together with interfaces for trade and market data and system configuration via API and XML. ORE Studio is a graphical wrapper around ORE: it imports ORE's XML inputs — trade definitions, market data, model configuration, conventions, fixings, calendars — into a persistent database, drives ORE execution from the configuration you assemble in the interface, and renders ORE's outputs (net present value, XVA, sensitivities, scenarios) back to you. It does *not* reimplement or modify ORE's pricing and risk models; it inherits ORE's models, conventions, and limitations as they are. ORE Studio is independent of, and unaffiliated with, the ORE project.

[5] Acadia, an LSEG business, which maintains ORE: https://acadia.inc/.

QuantLib

QuantLib is a free, open-source library aimed at providing a comprehensive software framework for quantitative finance — modelling, trading, and risk management. It has been the de-facto standard quantitative-finance library in C++ for over twenty years, and ORE is built directly on top of it. For ORE Studio, QuantLib is *two hops upstream*: nothing in ORE Studio calls QuantLib directly, and QuantLib's

instruments, day-count conventions, and calendars appear in our data only because ORE surfaces them. As with ORE, ORE Studio neither extends nor modifies QuantLib, and inherits any of its conventions or limitations as-is.

Audience

This manual is written for several kinds of reader:

- **Analysts, traders, quants, and middle-office staff** who want to explore financial instruments and risk calculations through a graphical interface. No programming experience is required.
- **Students and researchers** learning quantitative finance. ORE Studio makes it possible to experiment with derivative pricing, yield curve construction, and risk measures without writing code.
- **LLM-based agents and coding assistants** that interact with ORE Studio on a user's behalf, help configure workspaces, interpret results, or draft analytical reports. Multimodal models — those capable of processing both text and images — are preferred, as the manual makes extensive use of screenshots to describe the user interface.

The focus throughout is on what you can do through the application itself: entering trades, configuring market data, running analytics, and understanding the results.

Functional Areas

ORE Studio is organised around a set of functional areas accessible from the main menu:

- *Reference data* — currencies, business calendars, counterparties, and market conventions that form the foundation for everything else.
- *Instruments and trades* — define, store, and manage financial instruments and their terms.
- *Market data* — yield curves, volatility surfaces, fixing histories, and other inputs to pricing and risk models.
- *Model configuration* — choose and parameterise pricing engines, simulation models, and sensitivity specifications.
- *Analytics* — run calculations and view results: net present value, sensitivities (Greeks), credit and funding valuation adjustments (XVA), Monte Carlo simulations, and stress tests.

All data is stored persistently, so trades, curves, and configurations can be saved, revisited, and reused across sessions.

What ORE Studio Is Not

ORE Studio is a learning and exploration environment, not a production trading or risk system. It is independent of and unaffiliated with ORE, QuantLib, or any financial institution. For the key valuation cases, the quantitative mathematics are provided by ORE and QuantLib; ORE Studio is the surface through which they are configured and their results explored. ORE Studio does reimplement a small subset of QuantLib for its own internal purposes, but that reimplementation is not used by the engine's valuation code. Users who need production-grade performance, real-time market data feeds, or regulatory reporting should look to enterprise risk platforms.

How This Manual Is Organised

The chapters follow the natural order of first use. After this introduction, *Connecting to ORE Studio* covers launching the application, establishing a connection to the backend, and orienting yourself in the main window; *Initial Setup* then covers provisioning the system, tenants, and parties. Later chapters cover each functional area in turn. You do not need to read the manual cover to cover; each chapter can be read independently once the system is up and running.

Early-Stage Software Notice

> **Early-stage software.** ORE Studio is currently in active development (version 0.x). Features, workflows, and this documentation are all evolving and may change between releases. Numbers produced by the system are for learning and exploration only — they must not be used for real trading, risk management, or any financial decision-making.

A Note on This Document

This manual was generated entirely by large language models (LLMs) working under human supervision. While every effort has been made to ensure accuracy, LLM-generated content can contain errors, omissions, or descriptions that do not match the actual software behaviour. If you find a discrepancy between this manual and the application, trust the application. Please report inaccuracies via the project issue tracker so they can be corrected.

Conclusion

The chapter set out to orient the reader before any operation begins, and it has done so in layers. It established what ORE Studio is — a desktop surface over the open-source ORE and QuantLib engines — and for whom this manual is written, then mapped the functional areas through which the application is organised and drew a

firm boundary around it as a learning and exploration environment rather than a production trading or risk system. With that frame in place it described how the book is arranged for first use and tempered expectations with two cautions: that the software is early-stage and evolving, and that the manual itself was generated by language models under human supervision. The takeaway is a sense of place: the reader now knows where ORE Studio sits in the risk-technology landscape and how to read the chapters that follow, which take up each functional area in turn.

2

Connecting to ORE Studio

Connecting to a backend is the first task of every session, and the subject of this chapter. It traces the path from launching the application in its disconnected state to an authenticated session against a running backend. Along the way it sets out the Connection Browser and the full lifecycle of connection definitions — adding, editing, organising, securing, and storing them — the connect-and-login sequence and the ways it can fail, the status bar's account of the live connection, and the supporting material around a session: command-line options, the application version, the system tray, and logging out.

Overview

This chapter advances the argument that reaching a working session means starting the application, teaching it about the backends it may talk to, authenticating against one of them, and learning to read the session's state — and it proceeds in that order. It begins with Getting started, which launches the application and explains the *disconnected* state in which it first appears, and The main window before login, which describes what is and is not available until a login succeeds. From there it turns to The Connection Browser, the central place where backends are defined, walking through adding, editing, and deleting a connection, organising connections into folders, protecting credentials with a master password, grouping by environments and tags, and where the connection store lives and how to back it up. With a connection in place it moves to Connecting and logging in — selecting the active connection, the login screen, filtering Quick Connect by label, the distinct ways a connection can fail, the first login and default account, and registering a new account. It then covers Connection status and reconnection and the vocabulary of The status bar, so the reader can tell the connection's state at a glance. Finally it gathers the operational periphery — Starting ORE Studio from the command line, Finding the application version, Running

in the background and the system tray, and Logging out — and the Conclusion draws these steps together.

Getting started

When you launch ORE Studio a splash screen appears briefly while the application initialises, showing the ORE Studio logo and the build version in its lower-right corner.

Figure 2.1: The ORE Studio splash screen shown during start-up. The build version appears in the lower-right corner.

Once initialisation completes you are greeted by the main window in its *disconnected* state. The application is running but has not yet established a connection to the backend service that performs calculations. Before you can work with trades, market data, or analytics, you need to connect to a running ORE Studio backend.

The main window before login

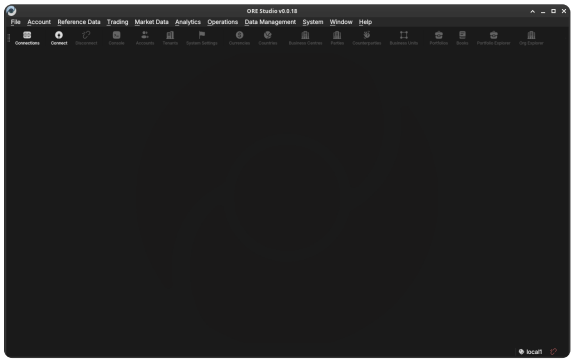


Figure 2.2: The ORE Studio main window immediately after launch, in the disconnected state. The toolbar shows the Connect and Connections buttons; the workspace area is empty until you log in.

In the disconnected state most of the application is inactive. The menu bar and toolbar are present but workspace panels, trade lists, and analytics views are all unavailable until a successful login. Two toolbar buttons are always accessible:

- **Connect** — opens the login dialog for the currently selected connection.
- **Connections** — opens the Connection Browser, where you configure the list of backends ORE Studio knows about.

The Connection Browser

The Connection Browser is the central place where you maintain the list of ORE Studio backends the application can connect to. You

might have a local development backend, a shared team backend, and a staging backend — each appears as a separate entry here. It lets you add, edit, delete, and organise those entries before you attempt a login.

Open it at any time from the toolbar **Connections** button or from the *File* menu.

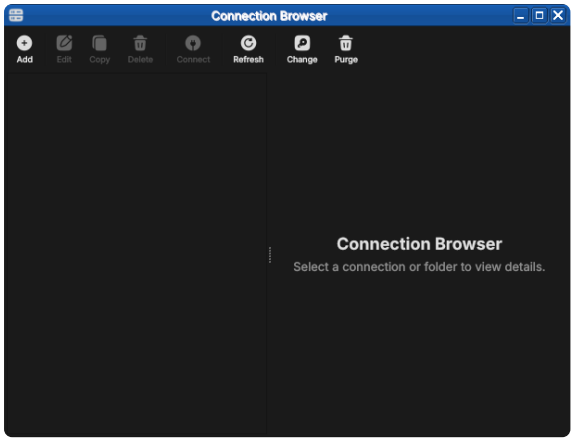


Figure 2.3: The Connection Browser open with no connections configured. The toolbar exposes Add, Edit, Copy, Delete, Connect, Refresh, Change and Purge; the unusable actions are greyed out until something is selected.

Adding a new connection

Click **Add** (or the "+" button) to open the *New Connection* form. Fill in the fields:

- **Name** — a free-text label you choose, used throughout the UI to identify this backend (e.g. "Local dev", "Team staging"). Must be unique among your saved connections.
- **Host** — the hostname or IP address of the machine running the ORE Studio backend (e.g. `localhost`, `192.168.1.10`, `ores-staging.example.com`).
- **Port** — the TCP port the backend is listening on. The default is `35900`; change it only if the backend was started on a different port.
- **Environment** — an optional grouping label (see Environments below). Helps you distinguish development, staging, and production backends at a glance.
- **Tags** — zero or more free-text labels (see Tags below). Useful for filtering and searching when you have many connections.

Click **Save** (or **OK**) to add the connection to the list. The new entry appears immediately in the Connection Browser list. No network contact is made at this point — ORE Studio simply stores the configuration.

Editing a connection

Select an existing connection in the list and click **Edit** (or double-click the row) to open the same form pre-populated with the saved

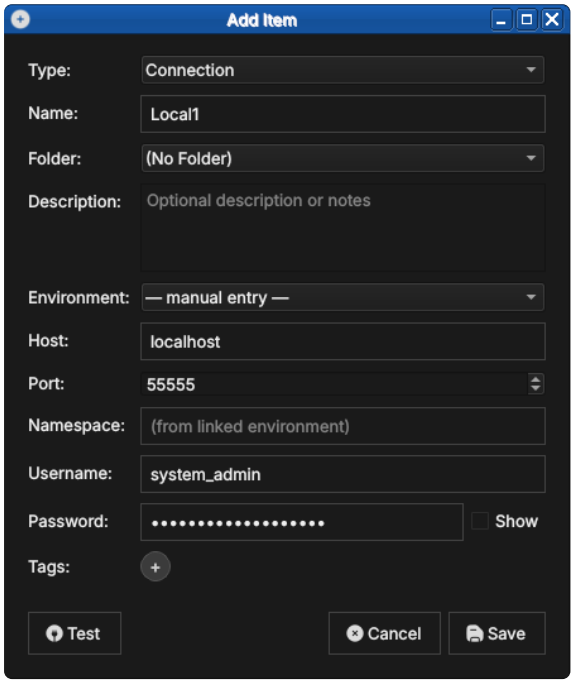


Figure 2.4: The New Connection form with example values filled in. *Type* is set to `Connection`; the *Environment* may be left as *manual entry* or linked to a defined environment.

values. Change any field and click **Save**. The connection list updates immediately.

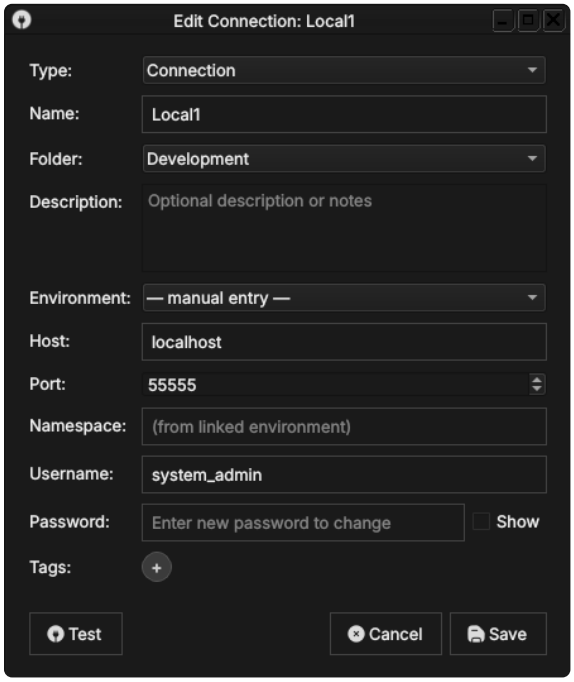


Figure 2.5: Editing a saved connection. The dialog title shows *Edit Connection*, the fields are pre-populated, and the password field reads *Enter new password to change* — the stored password is preserved unless you type a replacement.

Deleting a connection

Select one or more connections in the list and click **Delete** (or press the Delete key). A confirmation dialog asks you to confirm the removal.

Deleting a connection removes only the saved configuration; it has

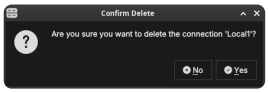


Figure 2.6: The delete confirmation dialog. ORE Studio names the connection being removed and waits for you to confirm with *Yes* or cancel with *No*.

no effect on the running backend or any data stored there.

Organising connections into folders

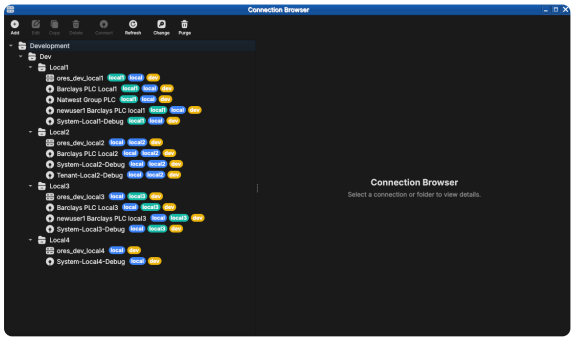


Figure 2.7: A populated Connection Browser. Connections are grouped under folders (here `Development → Dev → Local1...Local4`) and each carries coloured chips — the environment and tags — making the tier and purpose of every entry obvious at a glance.

When the list grows, *folders* keep it manageable. A folder is a named container you can nest, grouping related connections — by team, by client, or by environment tier. In the populated browser above, the connections sit under a `Development → Dev → Local1...Local4` folder tree.

To create one, click **Add** and set *Type* to `Folder`. Give it a name and an optional description, and choose a parent folder (or *No Folder* to place it at the top level). Connections are then assigned to a folder through their *Folder* field, or by dragging them in the tree.

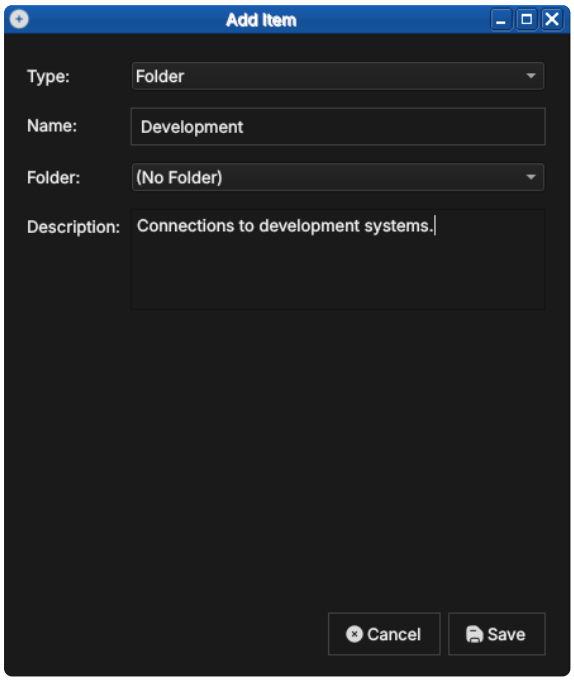


Figure 2.8: Creating a folder. With *Type* set to `Folder`, a folder needs only a name and an optional description; it can be nested inside another folder.

Protecting your credentials with a master password

The passwords you save with a connection are encrypted at rest using a *master password* that you choose. The first time you save a

credential, ORE Studio offers to create one.

Figure 2.9: Creating the master password. It encrypts every saved server password; there is no recovery if you forget it, so you may also leave it blank to store passwords unencrypted.

Type the same value into *New Password* and *Confirm Password*; the fields turn green when they match. *Show password* reveals what you typed.

Figure 2.10: The master password confirmed — both fields match and are outlined in green, ready to accept with *OK*.

When you next launch ORE Studio and open the Connection Browser, it prompts you to unlock your saved connections by entering the master password. Until you do, the encrypted passwords stay sealed.

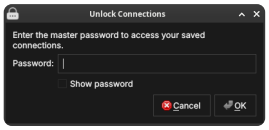


Figure 2.11: The unlock prompt shown on a later launch. Enter the master password to decrypt your saved connection credentials.

If you leave the master password blank, your connections still work but their passwords are stored unencrypted — acceptable on a personal machine, but not recommended on shared or portable systems.

Environments

An *environment* is a named grouping that you assign to a connection to indicate which tier of your infrastructure it belongs to. Typical values are `Development`, `Staging`, and `Production`, but the list is fully customisable — you can define any environment names that make sense for your setup.

Environments appear as a coloured badge or label next to the connection name in the list, making it immediately obvious which tier you are about to connect to. This is a safety feature: it is easy to accidentally connect to production when you meant staging; a clearly labelled environment badge prevents that.

See the populated Connection Browser figure under Organising connections into folders above — the same coloured chips shown there are the environment and tag labels described below.

To manage the available environment names, open the *Environments* section within the Connection Browser settings (or the dedicated menu item). From there you can:

- **Add** a new environment name and choose its display colour.

- **Rename** an existing environment (all connections using it update automatically).
- **Delete** an environment (connections that used it revert to *Unassigned*).

An environment is itself created from the Connection Browser: click **Add** and set *Type* to `Environment`. An environment carries its own host, port, HTTP port, namespace and tags — these become the connection details that any connection linked to it inherits.

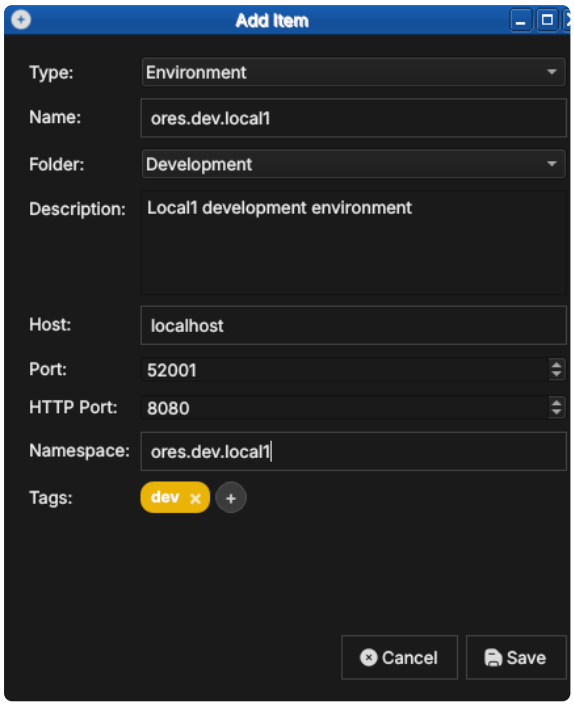


Figure 2.12: Creating an environment. With *Type* set to `Environment`, you give it a name, folder, host, ports, namespace, and tags (here a `dev` tag).

Once an environment exists, a connection can link to it by selecting it in the connection's *Environment* field instead of *manual entry*. The connection then inherits the environment's host, port, and namespace, so you maintain those details in one place.

Tags

Tags are free-text labels you attach to a connection to enable flexible filtering and search. Unlike environments (which represent a single tier), a connection can carry any number of tags. Examples: `personal`, `client-acme`, `read-only`, `high-memory`.

Tags are displayed inline with the connection name in the list. The Connection Browser provides a tag filter bar at the top of the list: type or select a tag to show only the connections that carry it.

To add or remove tags, open the Edit form for the entry and use the *Tags* field. Type a new tag name and press Enter (or comma) to add it; click the × on an existing tag chip to remove it. Tags are created on first use — there is no separate tag management screen.

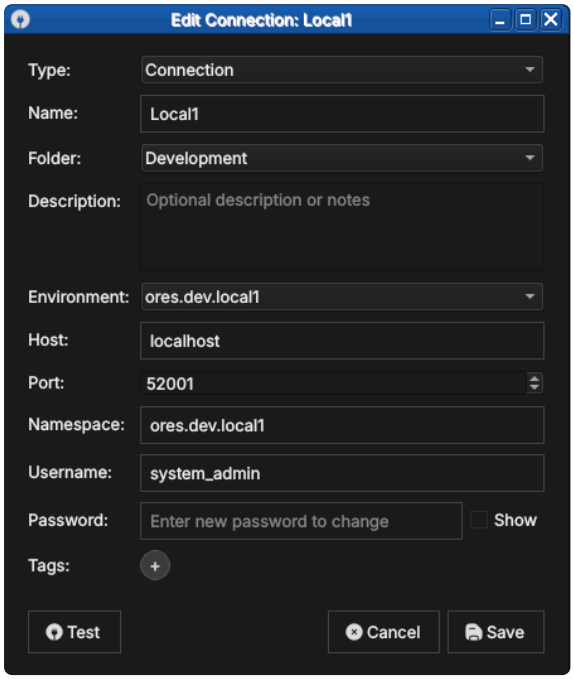


Figure 2.13: A connection linked to an environment. Selecting `ores.dev.local1` in the *Environment* field populates the host, port and namespace from that environment.

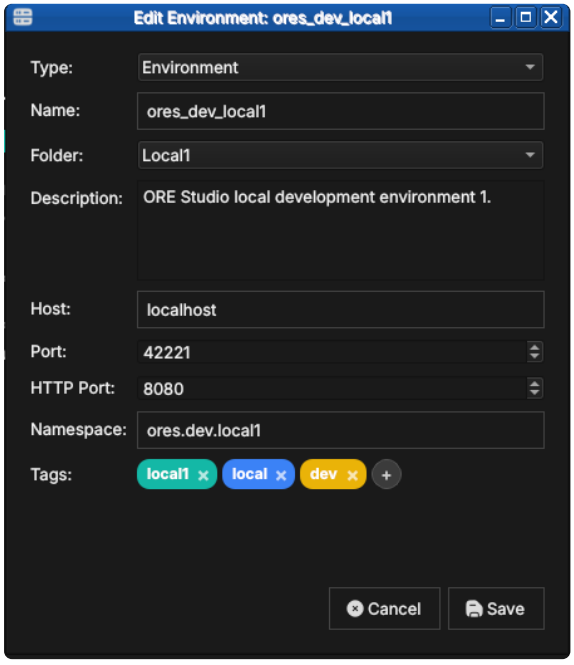


Figure 2.14: Tags shown as coloured chips in an entry's Edit form (here the `ores_dev_local1` environment, tagged `local1`, `local` and `dev`). Add tags in the *Tags* field, remove one with the × on its chip, or click + to add another.

Where your connections are stored

ORE Studio keeps your connections — together with their environments, folders, and tags — in a single local SQLite database file named `connections.db`. Your UI settings (preferences and window layout) are stored separately, in the operating system's standard settings store. Neither leaves your machine.

- **Linux**:
  - settings in `~/.config/OreStudio/`;
  - database at `~/.local/share/ores.qt/connections.db`.

- **macOS**:
  - settings in `~/Library/Preferences/`;
  - database at `~/Library/Application Support/ores.qt/connections.db`.
- **Windows**:
  - settings in registry: `HKCU\Software\OreStudio\OreStudio`;
  - database at `%APPDATA%\ores.qt\connections.db`.

If `connections.db` is missing when ORE Studio starts, it creates a new empty one — so the Connection Browser simply opens with no entries.

Backing up and restoring your connections

Because everything you configure here lives in that one `connections.db` file, backing it up is a file copy:

1. Quit ORE Studio, so the database is fully written and not locked.
2. Copy `connections.db` from the location above to a safe place — on Linux, for example:

```
cp ~/.local/share/ores.qt/connections.db ~/connections-
    backup.db
```

To restore, quit ORE Studio and copy the backup back over `connections.db`. To move your connections to another machine, copy the file to the same location there. The file is self-contained, so a single copy preserves all your connections, environments, folders, and tags.

Connecting and logging in

Once you have at least one connection configured, you can log in.

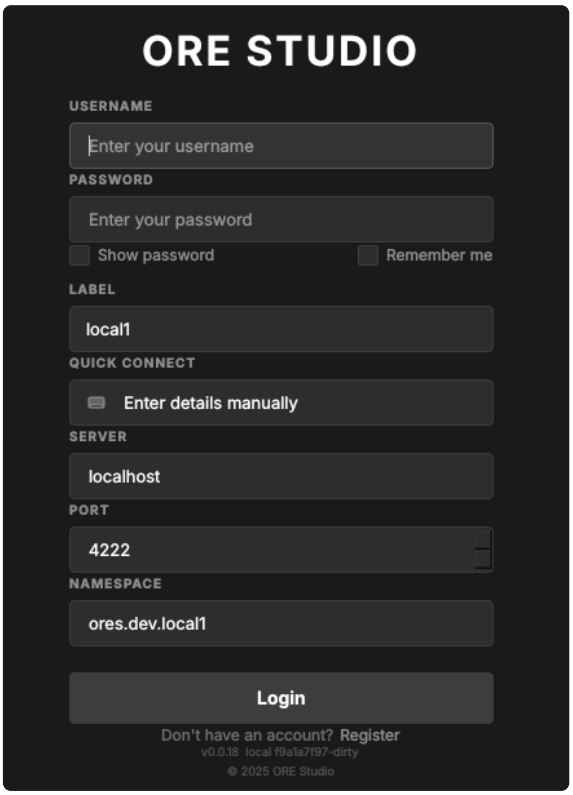
Selecting the active connection

In the Connection Browser, select the connection you want to use and click **Set as active** (or double-click it). The selected connection becomes the target for the **Connect** button in the main toolbar. The status bar at the bottom of the main window shows the name of the active connection.

You do not have to choose in advance, though: clicking **Connect** opens the login dialog directly, and its *Quick Connect* selector lets you pick the connection there (see Filtering Quick Connect with labels below).

The login screen

With an active connection selected, click **Connect**. ORE Studio attempts to reach the backend. If the backend is reachable, the *Login* dialog appears.



Enter your **username** and **password** and click **Login**. Tick *Remember me* to have ORE Studio recall the username next time. ORE Studio authenticates against the backend's user database. On success the dialog closes and the main window transitions to the *connected* state: the workspace panels become active, the menu bar is fully enabled, and the status bar shows your username and the connected backend name.

If your account spans more than one party, **Log in to default party** (ticked by default) skips the party-selection step and logs you straight in to the party configured as your account's default. Untick it if you want to choose the party explicitly on this login instead.

The fields below the credentials describe *which* backend you are logging in to:

- **Label** — the saved connection's name (e.g. `local1`).
- **Quick Connect** — pick a saved connection to fill *Server*, *Port* and *Namespace* in one step, or leave it on *Enter details manually* to type them yourself.
- **Server**, **Port**, **Namespace** — the backend address the login targets.

The *Label* and *Quick Connect* fields work together. The *Label* names the environment — it defaults from the connection (or from the

`--instance-name` you launched with); *Quick Connect* then offers the connections for that label, and picking one fills in *Server*, *Port* and *Namespace* so you do not have to type them.

If you do not yet have an account, the **Register** link in the footer opens account registration. If authentication fails, an error message is shown below the password field. Check that you are using the correct credentials for this backend; each backend maintains its own user database independently.

Figure 2.15: The *Label* and *Quick Connect* fields. The Label selects the environment; Quick Connect offers that environment's connections and fills in the server details.

Filtering Quick Connect with labels

With only a handful of connections, *Quick Connect* is easy to scan. But a real deployment soon accumulates many — several environments, each with its own system, tenant and per-party logins. Left unfiltered (the *Label* set to *All*), the selector lists every one:

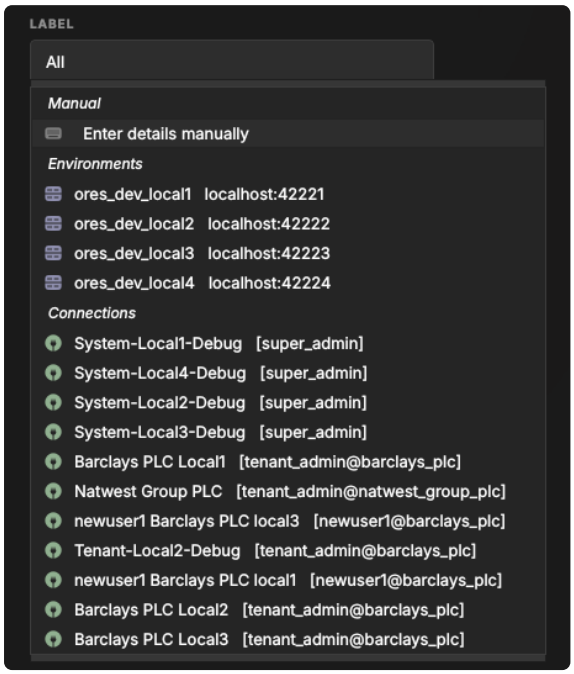


Figure 2.16: Quick Connect with the *Label* set to *All* — every saved connection across all environments and tenants. Hard to pick the right one at a glance.

Each connection carries a *label* — its environment tag, such as `dev`, `local1` or `local2`. The *Label* selector lists those labels; choosing one filters Quick Connect to just the connections that carry it:

With `local1` selected, Quick Connect shows only the `local1` entries — the environment's own database, its system and tenant logins, and its per-party connections — so you reach the right backend without scrolling past unrelated ones:

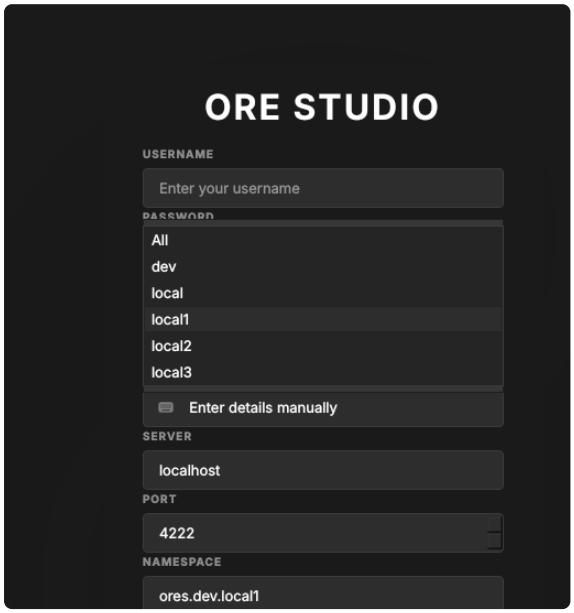


Figure 2.17: The *Label* selector lists the labels found on your connections. Pick one to narrow Quick Connect to that environment.

You normally run one client *per environment*: a client built for `local1` should connect to `local1`, not to staging or production. Rather than pick the label by hand each time, set it when you launch the client — the `--instance-name` command-line option (see Telling windows apart) opens the client with the *Label* already set, so Quick Connect is pre-filtered to that environment. The `compass client start` command does this for you, taking the label from `ORES_CHECKOUT_LABEL` in your `.env`.

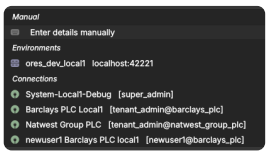


Figure 2.18: Quick Connect filtered to the `local1` label: only that environment's connections remain.

When the connection fails

Not every login failure is a wrong password. ORE Studio talks to the backend over a messaging server (NATS), and if it cannot reach that server — or the services behind it — it tells you which is wrong rather than just reporting "login failed".

If the messaging server itself is unreachable, ORE Studio reports that it cannot connect, along with the host and port it tried:

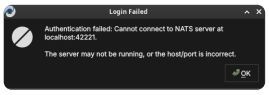


Figure 2.19: A login failure caused by an unreachable messaging server. The host and port ORE Studio tried are shown; the server is likely not running, or the host/port is wrong.

If the messaging server is running but the application services behind it have not started, ORE Studio says so explicitly — the fix is to start the backend services, not to change your connection:

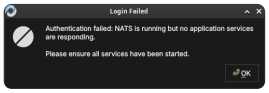


Figure 2.20: A login failure where the messaging server is up but no application services are responding. Ensure the backend services have been started.

In both cases your saved connection is fine; correct the backend (start the server or its services) and click **Connect** again.

First login and the default account

On a freshly initialised backend the only account that exists is the default administrator account. Your system administrator will have provided the initial credentials. After first login it is strongly recommended to change the password immediately via *Settings* → *Account*.

Once a tenant has been provisioned (covered in the next chapter,

Initial Setup), you log in with that tenant's administrator account — the username takes the form `user@tenant_code`. Picking the connection from *Quick Connect* fills in the server, port and namespace for you:

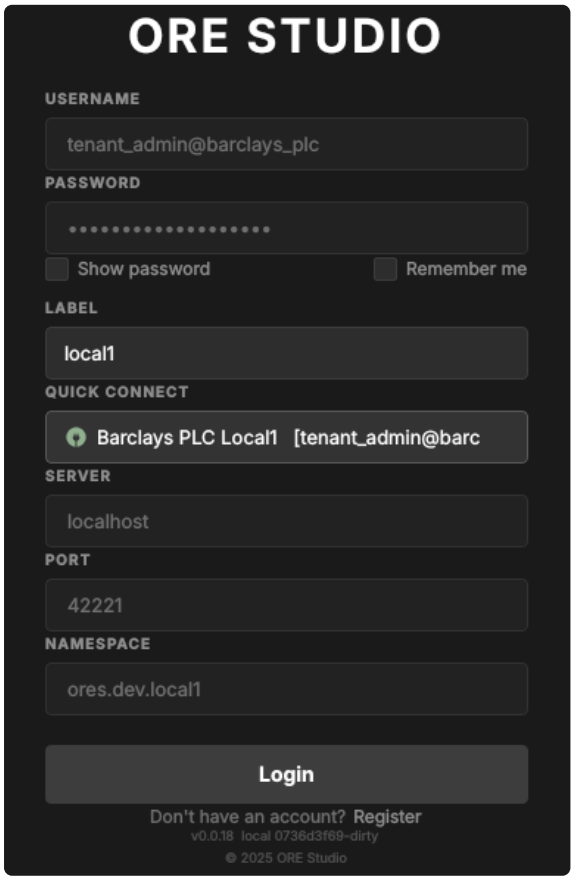


Figure 2.21: Logging in as a tenant administrator (`tenant_admin@barclays_plc`), with the connection chosen from *Quick Connect*.

Registering a new account

If your deployment allows it, the **Register** link in the login dialog footer opens the *Create Account* form. Fill in a username, email, and password (with confirmation), check the *Server* and *Port* point at your backend, and click **Create Account**. The *Log in* link returns to the login dialog.

Self-registration is not always available: the backend must be set up to permit new users to register themselves. Where it is disabled, accounts are created for you by an administrator instead (see First login and the default account).

> Account registration currently fails with `Account creation failed: NATS connect failed: SSL Error` — the registration path's NATS/TLS connection does not succeed. Until this is fixed, ask an administrator to create your account. The fix is tracked in Fix account registration NATS SSL error.

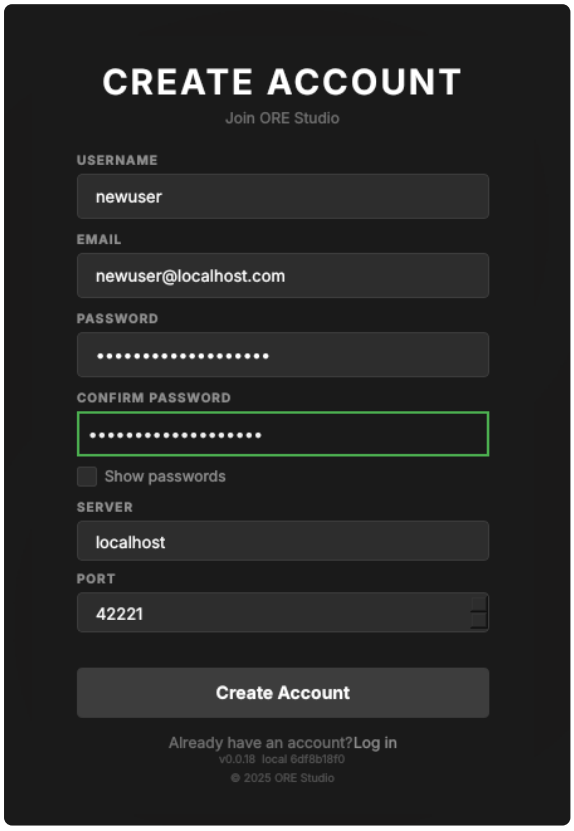


Figure 2.22: The Create Account form, opened from the *Register* link in the login dialog footer.

Connection status and reconnection

The status bar always shows whether you are connected. After a deliberate log-out — or once a stopped backend is running again — use the **Connect** button to re-establish the session: your saved connection is retained, so you only re-enter your password.

> ORE Studio does not yet reliably detect a backend that drops *while you are logged in*. If the messaging server (NATS) or the services behind it are stopped mid-session, the client may keep looking connected until your next action fails or hangs. Detecting the drop and showing a clear connection-lost state is planned work, tracked in Detect and report NATS disconnection.

The status bar

The status bar runs along the bottom of the main window and provides a continuous read on the application's connection state. It is always visible regardless of what is open in the workspace.

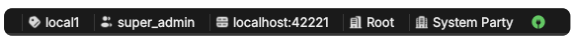


Figure 2.23: The status bar in the fully connected, logged-in state: from left to right, the environment marker (`local1`), the username (`super_admin`), the server (`localhost:42221`), the active party (Root / System Party), and the green connected indicator.

The status bar is divided into zones from left to right:

- **Connection name** — the name of the active connection as entered in the Connection Browser (e.g. "Local dev"). Clicking this zone opens the Connection Browser directly.

- **Environment badge** — the coloured environment label (e.g. `Production` in red) if one is assigned to the active connection. Absent when no environment is set. This is the most prominent safety indicator: always glance at the environment badge before performing any write operation.
- **Username** — the account name of the currently logged-in user. Absent in the disconnected state.
- **Server** — the backend address (host and port) the session is connected to, e.g. `localhost:42221`.
- **Active party** — the party context for the session (e.g. `Root / System Party`), set at login when your account spans multiple parties.
- **Connection indicator** — a coloured icon at the right showing the live link state (green when connected).

The status bar changes appearance to reflect the connection state:

- **Disconnected** (no active connection) — grey; shows "Not connected".
- **Connecting** (handshake in progress) — animated indicator; shows "Connecting to /name/...".
- **Connected and logged in** — normal; shows all zones as described above.

> A distinct *connection lost* state for a backend that drops mid-session is planned but not yet implemented — see Connection status and reconnection above.

In the disconnected state the strip is reduced to the environment marker and the connection label, with the broken-link icon on the right:

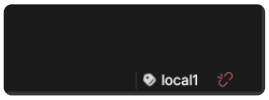


Figure 2.24: The status bar in the disconnected state, showing the environment marker, the connection label (`local1`), and the broken-link indicator.

The connected state is shown at the start of this section. Compare it with the disconnected strip above to recognise at a glance which state the application is in.

Starting ORE Studio from the command line

ORE Studio can be launched directly from a terminal, which is useful for scripting, for telling several windows apart, or for pointing the application at a non-default configuration directory. Run `ores --help` to see the full list of accepted options for your installed version; the most commonly used ones are documented below.

```
ores --help
```

Telling windows apart (instance colour and name)

When you run more than one ORE Studio window at once — for example against different backends, or from separate checkouts — you can give each window a distinct identity so you can tell them apart at a glance. This is *not* a light/dark theme; it is purely a per-window marker.

- `--instance-color HEX` draws a small coloured circle in the status bar in that colour (a 6-digit RGB hex, e.g. `F44336` for red). Give each window a different colour.
- `--instance-name NAME` (short form `-n`) labels the window with a name, also shown in the status bar, so you can see which window is which.

```
ores --instance-name "Local dev" --instance-color 2196F3
```

`--instance-name` does more than label the window: it also sets the login dialog's *Label* field on start-up, pre-filtering *Quick Connect* down to just that label's connections instead of showing every saved connection across every environment — see Filtering Quick Connect with labels above.

The colour and the name are independent: the colour is only a visual marker, and the name is what identifies the instance. If you launch via `compass client start`, its `--colour red|green|blue|<hex>` sets the marker colour and `--name` sets the instance name; when you omit `--name`, the name comes from `ORES_CHECKOUT_LABEL` in your `.env` — never from the colour.

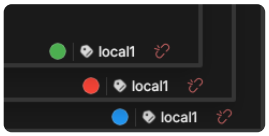


Figure 2.25: The status bars of three windows running the same `local1` instance, each given a different `--instance-color` (green, red, blue) so you can tell them apart at a glance.

Configuration directory

By default ORE Studio stores its UI settings — preferences and window layout — in the operating system's standard settings store. The exact per-OS locations are listed under Where your connections are stored; your saved connections live separately in `connections.db`. To use a different directory:

```
ores --config-dir /path/to/config
```

This is useful when running multiple isolated instances, or when keeping configuration under version control for team-shared settings.

Selecting a connection at startup

To bypass the Connection Browser and connect immediately to a named connection:

```
ores --connection "Local dev"
```

ORE Studio will start, set the named connection as active, and open the login dialog automatically. The connection name must match exactly (case-sensitive) a saved entry in the Connection Browser.

Logging and diagnostics

For troubleshooting, verbosity can be increased:

```
ores --log-level debug    # verbose output to stdout
ores --log-file /tmp/ores.log  # write log to a file instead
```

Log output includes connection lifecycle events, authentication attempts, and backend protocol messages. The `debug` level is intended for issue reporting and development; it produces high-volume output and should not be left enabled during normal use.

Finding the application version

Knowing exactly which build you are running matters when reporting an issue or checking client/backend compatibility. ORE Studio shows its version in several places.

The title bar of the main window always shows the version next to the application name:

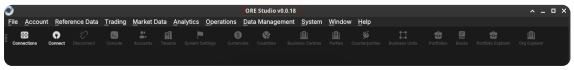


Figure 2.26: The version in the main window title bar, alongside the full menu bar and toolbar of the connected application.

It also appears in the lower-right corner of the splash screen at start-up:

Figure 2.27: The build version in the lower-right corner of the splash screen.

. . . and in the footer of the login dialog, below the *Register* link:

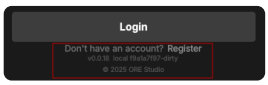


Figure 2.28: The version string in the login dialog footer, beneath the *Register* link.

From the command line, `ores --version` prints the client version string (e.g. `ORE Studio 0.0.19`) and exits — handy in scripts or when filing a bug report.

```
ores --version
```

Running in the background and the system tray

While ORE Studio is running it places an icon in your desktop's system tray (notification area). The icon keeps the application reachable when its window is minimised or hidden, and shows the instance name so you can tell multiple windows apart.

Figure 2.29: The ORE Studio icon in the system tray, labelled with the instance name so you can identify the window it belongs to.

Logging out

To end your session, choose *File* → *Log Out* or click the **Disconnect** toolbar button. ORE Studio closes the connection to the backend and returns to the disconnected main window. Your saved connection configurations are preserved; you can log back in at any time.

Logging out does not stop the backend service — it only terminates your client session. Other users connected to the same backend are unaffected.

To close the application entirely, quit the window (or choose *File* → *Exit*). ORE Studio asks you to confirm so you do not lose an active session by accident:

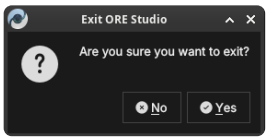


Figure 2.30: The exit confirmation. Click *Yes* to close ORE Studio, or *No* to keep working.

Conclusion

The chapter set out to show that reaching a working session follows from starting the application, defining the backends it may talk to, authenticating against one, and reading the session's state in turn, and it has traced exactly that path. It began at the disconnected main window, where nothing but the Connect and Connections actions is available, and used the Connection Browser to build up the set of known backends — adding and editing entries, organising them into folders, sealing their credentials behind a master password, and grouping them by environment and tag, all held in a single local store that travels with a file copy. Building on that, the connect-and-login sequence took a configured connection through to an authenticated session, with Quick Connect and labels narrowing the choice and the distinct failure messages distinguishing a wrong password from an unreachable server. The status bar then gave the session a continuous, glanceable account of its own state, and the command-line options, version markers, system tray, and log-out path filled in the operational periphery. The reader can now reach a live backend; the next chapter covers what must happen before a brand-new backend is usable at all: provisioning.

3

Initial Setup

Before first use, a fresh ORE Studio installation must be provisioned — a one-time sequence this chapter describes in full. It sketches the model the steps rest on — tenants as units of isolation, parties as the organisational hierarchy within them — then walks the fixed three-wizard sequence in order: the System Provisioner creates the platform administrator and the first tenant; the Tenant Provisioner seeds a tenant with reference data and its house party hierarchy; and the Party Provisioner equips each party with counterparties and report definitions. The chapter closes with party selection at login, the point where provisioning ends and daily use begins.

Overview

The chapter advances the argument that provisioning a fresh installation is a fixed sequence that follows from the data model it serves, and it proceeds in that order. It begins by sketching that model in The model in brief, establishing the tenant as the unit of isolation and the party as the organisational hierarchy within it; this is the premise the rest of the chapter builds on. From there it lays out the The provisioning sequence as a whole — the three steps and the login identity each one requires — before working through each wizard in turn. It walks the System Provisioner, which creates the platform administrator and the first tenant; then the Tenant Provisioner, which seeds a tenant with reference data and its house party hierarchy; and then the Party Provisioner, which equips a single party with counterparties and report definitions. Finally it shows where provisioning hands over to daily use in Choosing a party at login, and the Conclusion draws these steps together.

The model in brief

ORE Studio organises data around two concepts that the provisioning wizards exist to create. A *tenant* is a fully isolated organisational space — its users, reference data, and results are invisible to every other tenant; typically one per organisation. Within a tenant, *parties* form the organisational hierarchy: the *house* — your own legal entities, desks, and booking centres — alongside the external *counterparties* it trades with. Users log in to a specific party, and their position in the hierarchy determines what they can see. The full model — tenant types, the system tenant and system party, hierarchy visibility, and the house versus counterparty distinction — is covered in the Tenants chapter.

The provisioning sequence

A freshly installed ORE Studio backend goes through a fixed sequence before it is ready for normal use. Each step requires a specific login identity:

1. **System provisioning** — no login is needed; the backend is in *provisioning mode* and the System Provisioner wizard launches automatically the first time any client connects. This step creates the platform administrator account and provisions the first tenant and its tenant administrator. Until it completes, no normal user logins are accepted.

2. **Tenant provisioning** — log out of the system administrator session (or simply connect as a new user) and **log in as the tenant administrator** created in step 1. OreStudio detects that this account has never logged in before and launches the Tenant Provisioner automatically. This step seeds the tenant with reference data and creates the house party hierarchy.

3. **Party provisioning — log in to a specific house party** using the tenant administrator account, or any operational account that holds party setup permissions for that party. OreStudio detects that the party has not yet been provisioned and launches the Party Setup wizard. Repeat this step for each party in the hierarchy that needs its own counterparties, books, and reports.

Each step has a corresponding wizard in the ORE Studio UI. The following sections walk through each one.

The System Provisioner wizard

The first time you connect to a backend that has never been initialised, ORE Studio detects *provisioning mode* and launches the **New System Provisioner** automatically. This one-time wizard creates the

platform administrator account, chooses a single- or multi-tenant layout, and provisions the first tenant together with its own administrator.

Welcome

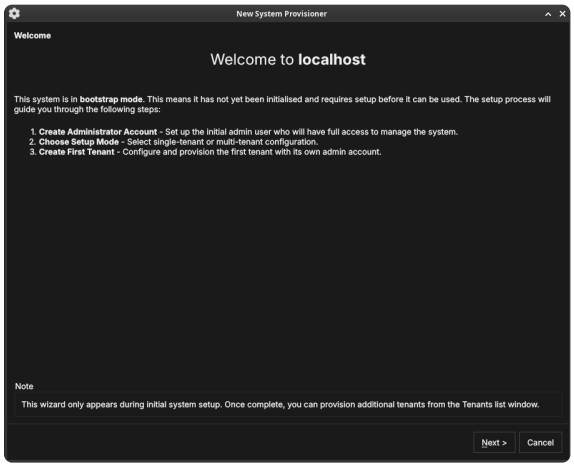


Figure 3.1: The System Provisioner welcome page, shown when the backend is in provisioning mode. It lists the three setup steps.

The welcome page confirms the system is in provisioning mode and outlines the steps: create the administrator account, choose the setup mode, and create the first tenant. Click **Next**.

Create the administrator account

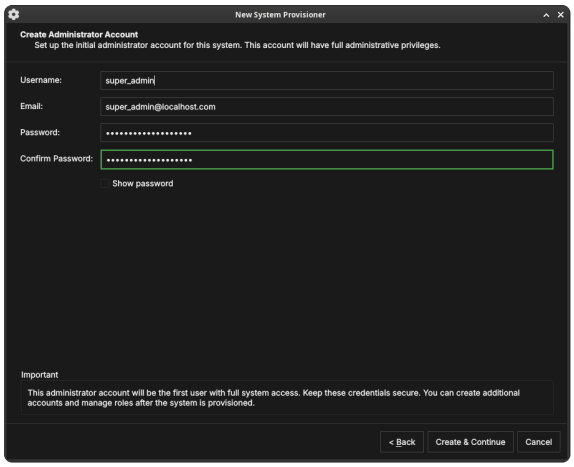


Figure 3.2: Creating the platform administrator account — the first account, with full administrative privileges over the whole deployment.

Enter a **username**, **email**, and **password** (with confirmation) for the platform administrator. Keep these credentials safe — this account has unrestricted access to the entire deployment. Click **Create & Continue**.

Choose the setup mode

Choose how the deployment is organised:

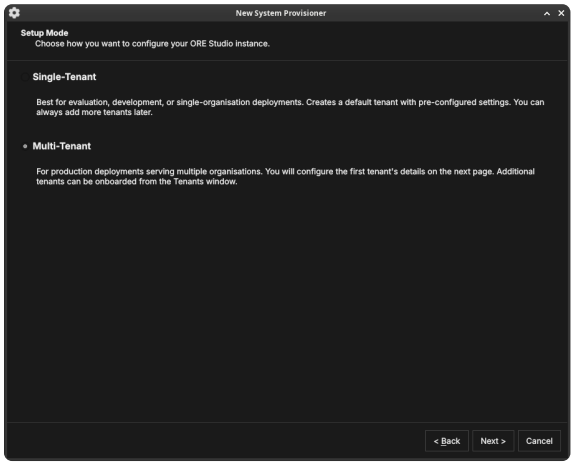


Figure 3.3: The Setup Mode page. *Single-Tenant* creates a default tenant with pre-configured settings; *Multi-Tenant* lets you configure the first tenant in detail and onboard more later.

- **Single-Tenant** — best for evaluation, development, or a single organisation. Creates a default tenant with sensible defaults; you can add more tenants later.
- **Multi-Tenant** — for production deployments serving several organisations. You configure the first tenant's details on the next page and onboard additional tenants from the Tenants window.

Click **Next**.

Tenant details

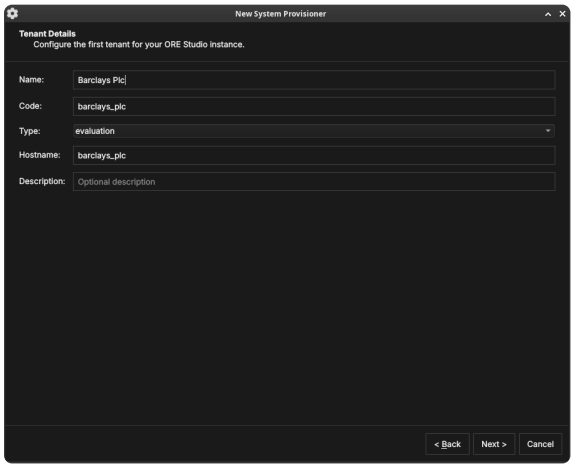


Figure 3.4: The first tenant's details: a display name, a short code, a tenant type, and a hostname.

Configure the first tenant:

- **Name** — the organisation's display name (e.g. "Barclays Plc").
- **Code** — a short machine code used internally (e.g. `barclays_plc`).
- **Type** — the tenant type (see Tenant types): *System*, *Production*, *Evaluation*, or *Automation*.
- **Hostname** — the host identifier for the tenant.

Click **Next**.

Tenant administrator account

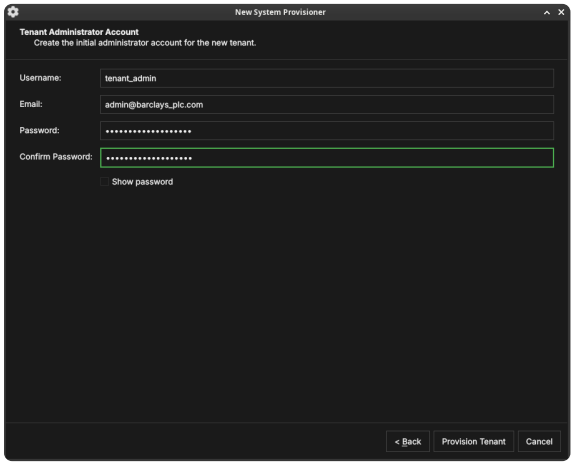


Figure 3.5: The tenant administrator account. This account administers the new tenant and is independent of the platform administrator.

Create the initial administrator for the new tenant — a **username**, **email**, and **password**. This account is given the `TenantAdmin` role: full control within the tenant, but no cross-tenant access. Click **Provision Tenant**.

Provisioning

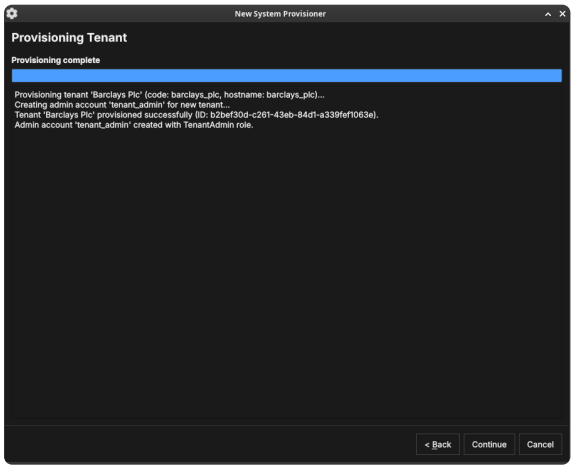


Figure 3.6: The provisioning step. ORE Studio creates the tenant and its administrator, logging progress as it goes.

ORE Studio provisions the tenant and creates its administrator account, logging each step. When it finishes, click **Continue**.

Setup complete

The final page summarises what was created and tells you the next step: log in as the tenant administrator to run the Tenant Provisioner. To create further tenants later, use *System* → *Identity* → *Tenants* and click **Onboard**. Click **Finish** — the backend leaves provisioning mode and the login dialog appears.

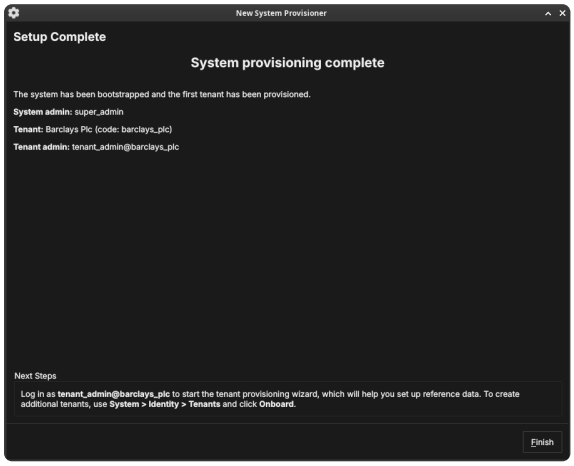


Figure 3.7: The System Provisioner summary: the platform admin, the first tenant, and the tenant admin that were created, with next-step guidance.

The Tenant Provisioner wizard

The first time you log in as a tenant administrator, ORE Studio launches the **New Tenant Provisioner**. It seeds the tenant with reference data and, optionally, an initial party hierarchy. You can skip it with **Cancel** and set everything up by hand later from the Data Librarian and Parties windows.

Welcome

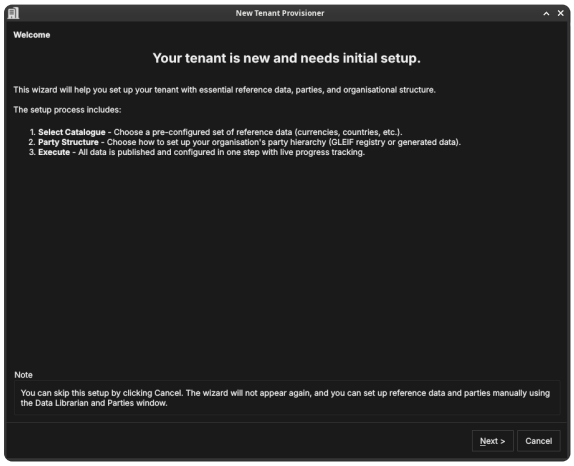


Figure 3.8: The Tenant Provisioner welcome page, listing its three steps: select a reference-data catalogue, choose a party structure, and execute.

The welcome page outlines the steps: select a catalogue, choose how to populate the party hierarchy, and execute. Click **Next**.

Select a catalogue

A *catalogue* is a ready-made bundle of reference data. The *Base System* catalogue provides industry-standard ISO and FpML data — country codes, currency codes, and financial-market standards — suitable for production use. Choose a catalogue and click **Next**.

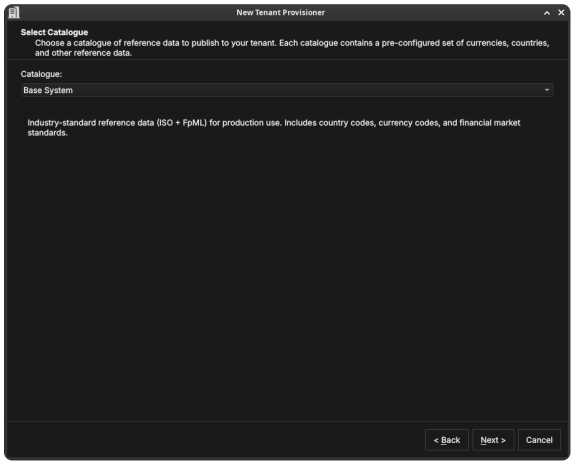


Figure 3.9: Selecting a reference-data catalogue — a pre-configured set of currencies, countries, and market standards.

Choose a data source

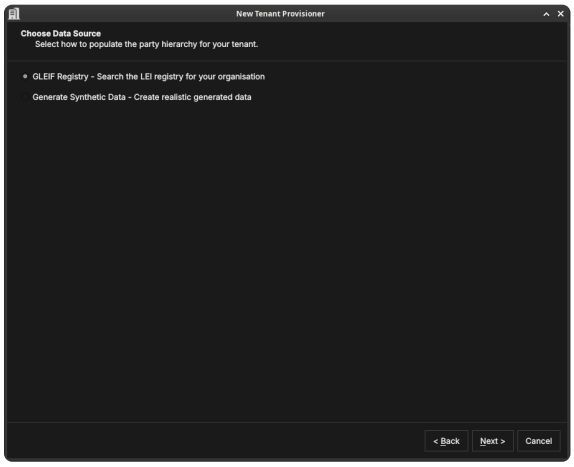


Figure 3.10: Choosing how the tenant's internal party hierarchy — your organisation's own legal entities and business units — will be seeded.

This step seeds the *internal* party hierarchy: the legal entities and business units that represent your own organisation — what practitioners call *the house*. As described in Parties, these are the operational parties that own trades, books, and analytics results.

The data source choice does **not** affect counterparties (the external entities your organisation trades with — those are imported separately in the Party Setup wizard). It only determines how the house structure is created.

In a group context — for example a bank holding company with separate London and New York subsidiaries — the house hierarchy reflects the corporate structure. The root party is the top-level legal entity; its children are the subsidiary entities; their children are trading desks, books, or booking centres. Each node in this tree represents a real organisational unit, and users log in to the node that corresponds to their own entity.

Two seeding methods are available:[1]

- **GLEIF Registry** — searches the global LEI registry for your real organisation. The entity you select becomes the root party, and its corporate descendants in the GLEIF hierarchy — subsidiaries,

[1] GLEIF, the Legal Entity Identifier system and BIC codes are described in more detail in the Parties chapter's "What is a Party?" section.

branches, and related entities — are created automatically as child parties. This is the recommended approach for production deployments: the GLEIF data provides real legal entity names, LEI codes, and verified corporate relationships.

- **Generate Synthetic Data** — creates a realistic but fictional party hierarchy using generated names. Use this for evaluation tenants, demonstrations, or testing where you want a plausible structure without importing real organisational data.

Click **Next**.

Party setup (optional)

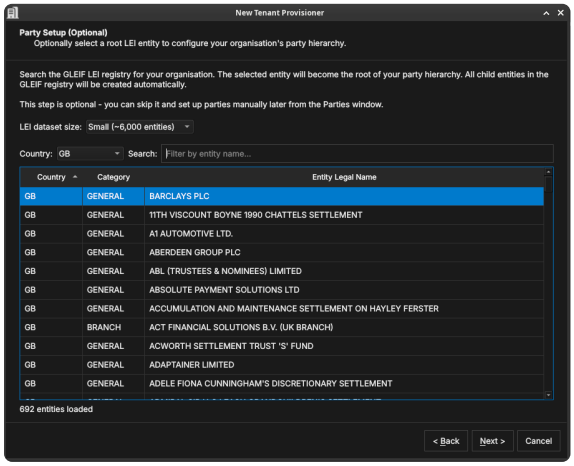


Figure 3.11: The optional Party Setup step. Search the GLEIF LEI registry for your root entity (here Barclays PLC); its corporate descendants are created as child parties automatically.

It helps to be clear about what this step does and does not do.

What it creates: the party *nodes* — the named entries in the hierarchy that represent your legal entities and business units. Each node gets a name, a LEI code, and a position in the tree. After this step OreStudio knows that "Barclays PLC" exists as a party, that it is the parent of "Barclays Bank UK PLC", and so on. The hierarchy structure is in place.

What it does not create: the operational *content* within each party — the books, portfolios, business units, counterparties, and report definitions that each party uses day-to-day. That content is added separately, after provisioning, by the Party Setup wizard (described in The Party Provisioner wizard below). Think of this step as building the org chart; the Party Setup wizard then stocks each office.

To use the GLEIF registry, pick an **LEI dataset size** (which controls how much of the registry is loaded for searching), filter by **Country** if needed, and search by entity name. The entity you select becomes the *root house party* — the top of your internal hierarchy. Its corporate children and grandchildren in the GLEIF registry (subsidiaries, branches, booking centres) are created automatically as child operational parties. You do not need to build the structure by hand; the GLEIF corporate hierarchy data does it for you.

This step is optional — skip it to create parties manually later from the Parties window. Click **Next**.

Publishing

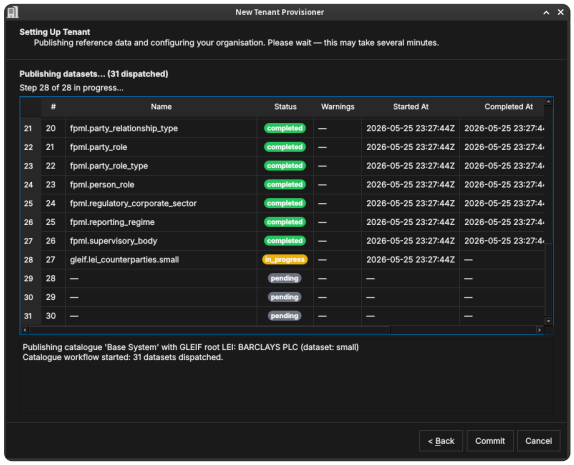


Figure 3.12: The publishing step, showing 31 datasets being dispatched in dependency order. Dataset 28 (gleif.lei_counterparties.small) is in progress; datasets 29–31 are pending.

OreStudio runs a multi-step publishing pipeline to populate the tenant with all the data selected in the previous steps. The pipeline dispatches datasets in the correct dependency order — classification tables before the entities that reference them, catalogue data before GLEIF data — and shows live progress in the table.

Each row in the table is one dataset. The columns are:

- **#** — the dataset's position in the dependency graph (zero-based). Datasets with lower numbers are prerequisites for those with higher numbers.
- **Name** — the dataset identifier, in `source.dataset` form. The prefix identifies the data source: `iso.*` datasets contain ISO standard data (currencies, countries); `fpml.*` datasets contain FpML reference classifications (party types, roles, regulatory sectors); `gleif.*` datasets contain GLEIF LEI entity data (counterparties, house parties).
- **Status** — `pending` (not yet started), `in_progress` (actively loading), or `completed` (finished successfully). A red status indicates a failure.
- **Warnings** — any non-fatal issues encountered during that dataset's load, displayed as a count. Click the row to see details.
- **Started At** / **Completed At** — UTC timestamps for the dataset's execution window.

The log panel at the bottom provides a running narrative: which catalogue is being published, how many datasets were dispatched, and the final outcome for each phase (reference data, organisation, activation).

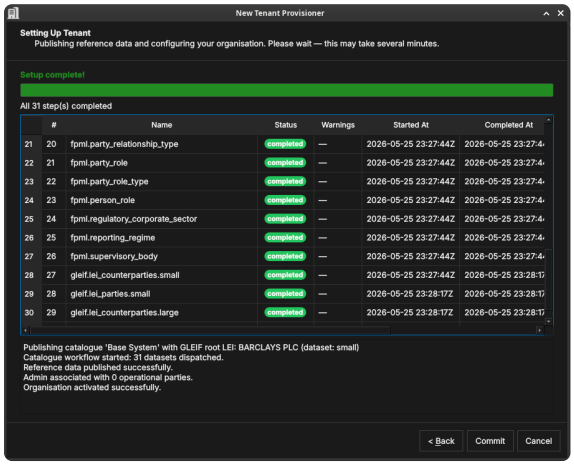


Figure 3.13: All 31 datasets completed. The log confirms: reference data published, organisation associated with parties, and the tenant activated.

When all rows show `completed` and the log ends with "Organisation activated successfully", click **Commit** to finalise the setup, or **Back** to change your selections.

Setup complete

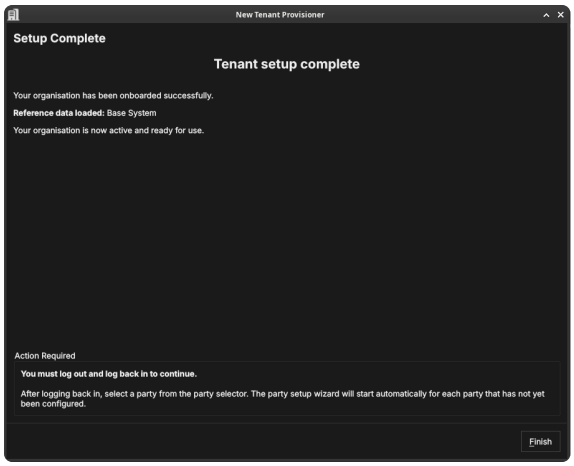


Figure 3.14: The Tenant Provisioner summary, confirming the reference data and parties that were set up.

The final page confirms what was created. Click **Finish**. The tenant now has its reference data and party hierarchy in place.

The Party Provisioner wizard

Where the Tenant Provisioner establishes the party *hierarchy*, the **Party Setup** wizard configures the operational structure *within* a party: it imports counterparties, then publishes every party-scoped data bundle the party needs — risk management (business units, portfolios, books, and a standard set of risk-report definitions), synthetic market data configuration, and curated FX driver rates. It runs for a party that needs initial setup, and can be skipped with **Cancel** (counterparties and bundles can be configured later from the application menus).

Welcome

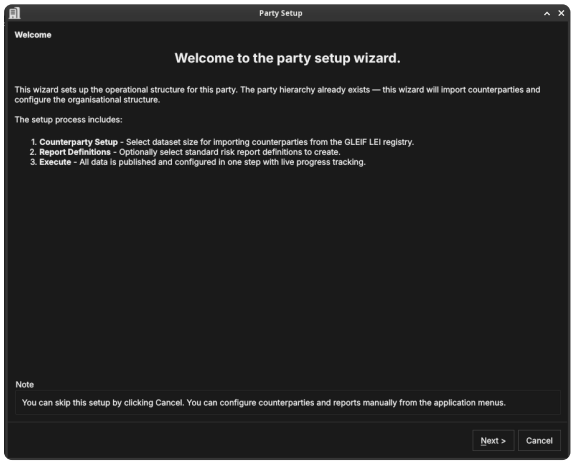


Figure 3.15: [SCREENSHOT OUTDATED — retake needed] The Party Setup welcome page, listing its steps: counterparty import and execute.

By this point the house hierarchy — the party nodes representing your organisation's legal entities — already exists. This wizard populates one specific party within that hierarchy: it attaches the external counterparties that this party trades with, then publishes its operational structure, risk reports, and market data configuration in one step. Click **Next**.

Counterparty import

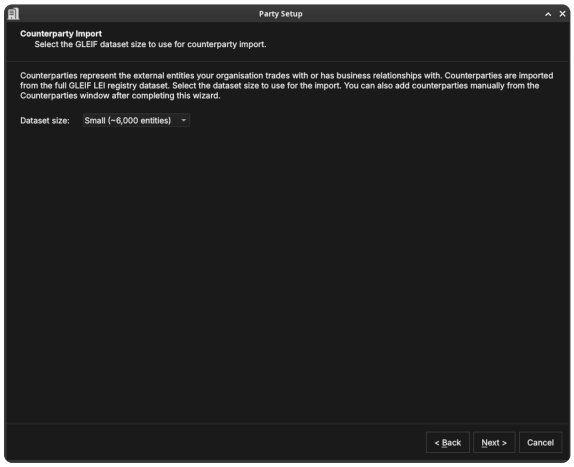


Figure 3.16: Choosing the GLEIF dataset size for importing counterparties. The dataset size controls how many real-world legal entities are loaded from GLEIF as counterparties.

Counterparties are the external legal entities your organisation trades with — banks, broker-dealers, corporates, funds. Where parties (the house) represent your own organisational structure, counterparties represent the other side of your trades. The distinction is important: house parties own books and positions; counterparties appear on trade tickets as the facing entity.

OreStudio imports counterparties from the GLEIF LEI registry. This is a deliberate design choice: GLEIF contains over two million real legal entities with verified names, LEI codes, countries of incorporation, and corporate relationships. Importing from GLEIF means

your OreStudio instance starts with a realistic, authoritative counterparty population rather than a hand-crafted list. In an evaluation or demonstration tenant this is especially valuable — trades booked against real counterparty names (Deutsche Bank, JP Morgan, BlackRock) look and behave exactly as they would in production, making scenario testing meaningful.

The **dataset size** controls how many GLEIF entities are loaded: smaller sizes load a representative subset quickly; larger sizes import a more complete global population. You can add individual counterparties later from the Counterparties window regardless of which size you choose. Click **Next**.

Execute

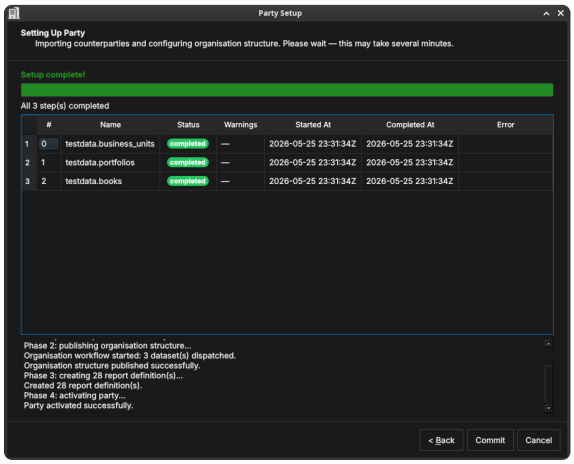


Figure 3.17: [SCREENSHOT OUTDATED — retake needed] The execute step showing the risk_management bundle's datasets completed: testdata.business_units, testdata.portfolios, testdata.books, and ore.report_definitions. The log below shows the five phases completed in sequence.

OreStudio runs the party setup in five phases, shown in the log panel:

1. **Phase 1: importing counterparties** — loads the selected GLEIF counterparty dataset into the party's counterparty table. The progress table shows one row per dataset (e.g. `gleif.lei_counterparties.small`), with the same Status / Started At / Completed At columns as the Tenant Provisioner publishing step.

2. **Phase 2: publishing risk management** — publishes the `risk_management` bundle in full: business units (`testdata.business_units`), portfolios (`testdata.portfolios`), books (`testdata.books`) — the lowest-level containers that own individual trades and positions — and the standard set of risk-report definitions (`ore.report_definitions`), since reports reference the book/portfolio tree.

3. **Phase 3: publishing synthetic market data configuration** — publishes the `synthetic_realistic_2026` bundle in full: the "2026 Realistic" theme's FX spot and IR curve generation configs, and any further asset class added to the bundle in future. Two other themes exist – `synthetic_ore_samples_2016` ("2016 ORE Samples", matching the conventions ORE's own sample data comes from)

and `synthetic_uniform_demo` (a simple, non-vintage demo/exercise archetype) – publish those separately (e.g. via `ores.shell`) if you want them available too. Only one theme's feeds should ever run at once: the Market Simulator's Start action at the collection root prompts you to pick a theme rather than starting them all.

4. **Phase 4: publishing FX driver rates** — publishes the curated `marketdata.reference_vintage_2016_02_05` bundle so the party has real market series/observations to browse.

5. **Phase 5: activating party** — marks the party as operational so users can log in to it and begin booking trades.

When the log ends with "Party activated successfully", click **Commit**.

Setup complete

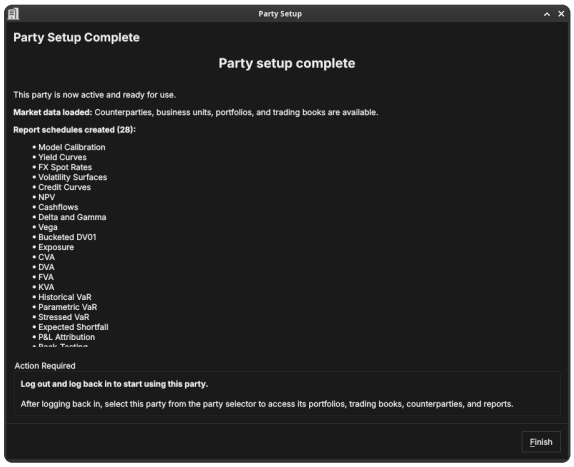


Figure 3.18: [SCREENSHOT OUTDATED — retake needed] The Party Setup summary, confirming the party is active and ready for use.

The final page confirms the party is now fully operational: it has counterparties, an organisational structure, scheduled risk reports, and market data configuration.

Choosing a party at login

Some accounts are associated with more than one party — for example a group administrator who can act for several entities. When you log in with such an account, ORE Studio asks which party context to use for the session.

Choose **System Party** for administrative work, or **Operational Party** to work within a specific business entity. Filter the list by booking **centre**, or type in the search box to narrow it.

Select a party and click **Select**. Your data visibility for the session is determined by the party you choose and its position in the hierarchy.

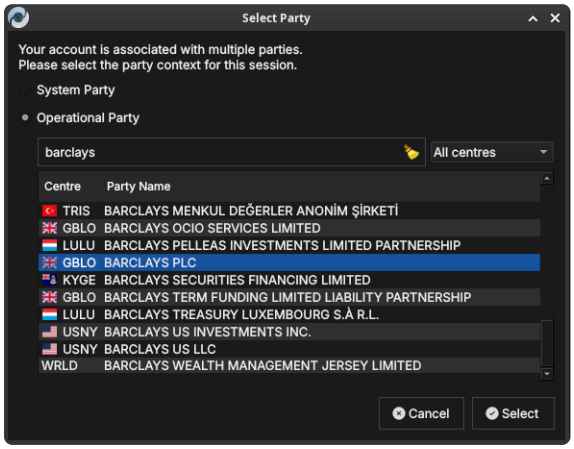


Figure 3.19: The Select Party dialog. Choose the System Party or an Operational Party; filter by booking centre or search by name.

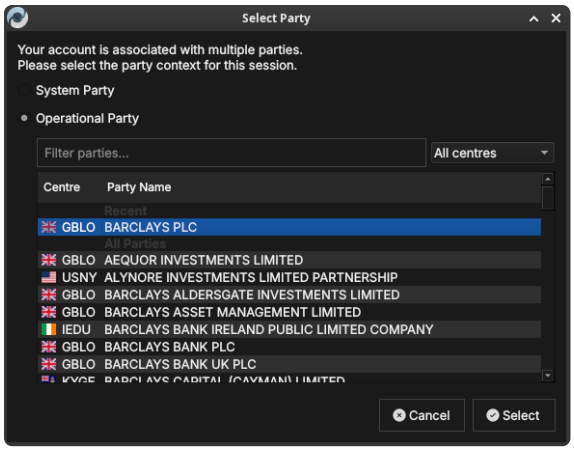


Figure 3.20: The Select Party dialog with a recently used party listed under *Recent* at the top for quick access.

Conclusion

The chapter set out to show that provisioning a fresh installation is a fixed sequence that follows from the data model it serves, and it has traced exactly that path. The tenant and party model came first — the tenant as the unit of isolation, the party as the hierarchy within it, with the full treatment reserved for the Tenants chapter — because the provisioning steps exist to create precisely those structures. Building on that premise, the three wizards did the work in turn: the System Provisioner created the platform administrator and the first tenant, the Tenant Provisioner seeded the tenant with reference data and its house party hierarchy, and the Party Provisioner equipped each party with counterparties, an organisational structure, and report definitions. Each step required its own login identity, and each ran once. Provisioning ends where daily use begins — choosing a party at login — and the accounts the wizards created, like every account after them, are managed through the windows described in the next chapter.

4

Provisioning from the Shell

Everything the wizards do, the shell can do too. This chapter covers the same ground as the provisioning wizards from the command line, where a complete system can be provisioned with three commands — or with one script, run end to end without touching the desktop application. It sets out the `provision` commands and how each maps onto its wizard, the supporting commands they are built from, the script library that ties them together, and a complete worked example, showing that scripted provisioning is repeatable: the same script against a fresh installation always produces the same system.

Overview

The chapter advances the argument that provisioning a complete system from the command line means understanding the client that runs the commands, then each provisioning command in the order it must be run, then the smaller commands they are built from, and finally how to tie them together into a repeatable script — and it proceeds in that order. It begins by establishing the client and the rules it enforces in The shell and provisioning, which explains how the shell speaks to the backend and why provisioning follows a fixed sequence; this is the premise the rest of the chapter builds on. From there it walks the sequence one stage at a time — Provisioning the system, then Provisioning a tenant, then Provisioning a party — each command standing in for the wizard it replaces. It then opens up the machinery in The supporting commands, the smaller commands the `provision` phases call and that you can run by hand to inspect the system or recover a failed run, before showing in Scripts how the shell runs those commands unattended from a file. Finally A complete example runs the whole sequence from a single shipped script, and the Conclusion draws these steps together.

The shell and provisioning

`ores.shell` is ORE Studio's interactive command-line client. Like the desktop application it speaks to the backend services over the message bus, so anything it does respects the same validation, audit trail, and permissions. Start it, connect, and type `help` to see the available commands; every command described here also explains its own arguments through the shell's built-in help.

Provisioning from the shell follows exactly the sequence described in the previous chapter — system, then tenant, then party — including the logouts and logins between the stages, because each login is what refreshes your session's view of the system's state. The difference is that each wizard collapses into a single command whose options carry the same defaults the wizard pre-fills, so the common case needs very few of them.

Provisioning the system

`provision system` performs the System Provisioner's work: it checks the installation is in bootstrap mode, creates the platform administrator, logs in as that account, and provisions the first tenant together with its tenant administrator.

```
provision system super_admin Secure-Password-123
    admin@localhost.com --tenant-admin-password Secure-Password
    -123 --tenant-hostname default
logout
exit
```

The tenant administrator's password is the only option without a default — everything else mirrors the wizard's single-tenant mode: tenant code `default`, name *Default Tenant*, type `evaluation`, and the tenant administrator named `tenant_admin` with the e-mail address `admin@<code>.com`. To provision a custom tenant instead — what the wizard calls multi-tenant mode — override what you need with `--tenant-code`, `--tenant-name`, `--tenant-type`, `--tenant-hostname`, `--tenant-description`, `--tenant-admin` and `--tenant-admin-email`.

The command validates everything before touching the backend, refuses to run when you are already logged in or the system is not in bootstrap mode, and on success prints the login for the next stage:

```
 Connected to nats://localhost:42222
 WARNING: System is in BOOTSTRAP MODE
ores-shell> [1/3] Creating initial admin account 'super_admin
    '...
  Account created (ID: b88d04ec-ac25-4137-981a-ddefb1d92592).
[2/3] Logging in as 'super_admin'...
 Login successful!
[3/3] Provisioning tenant 'default'...
 System provisioned. Tenant 'Default Tenant' (ID: 93b5d425-...)
    , admin 'tenant_admin'.
```

```
Next: logout, then: login tenant_admin@default <password>   the
     tenant is in bootstrap mode; run provision tenant.
ores-shell>  Logged out successfully.
ores-shell> Bye!
```

Note the login principal is the username at the tenant's *hostname* (`tenant_admin@default` above), not its display name.

Provisioning a tenant

`provision tenant` performs the Tenant Provisioner's work for the tenant you are logged in to. Log in as the tenant administrator the previous stage created — the tenant is in bootstrap mode, which is exactly what the command requires — and run:

```
login tenant_admin@default Secure-Password-123
provision tenant --source synthetic --seed 42
logout
exit
```

```
ores-shell>  Login successful!
ores-shell> Using bundle 'base' (first available).
[1/4] Publishing bundle 'base'...
  Dispatched 31 dataset(s); workflow instance: 7c059b7f-...
 All 31 step(s) completed.
[2/4] Generating synthetic organisation...
 Synthetic organisation generated (seed 42):
  parties:             5
  counterparties:      10
  ...
[3/4] Associating 'tenant_admin' with the operational parties
    ...
  5 parties associated.
[4/4] Finalizing tenant provisioning...
 Tenant provisioned: bundle 'base', 5 parties associated.
ores-shell>  Logged out successfully.
ores-shell> Bye!
```

With no options this publishes the first available reference data catalogue and uses the GLEIF registry as the data source, just as the wizard's defaults do. The options mirror the wizard's pages:

- `--bundle <code>` selects a specific catalogue (`bundles list` shows what is available).
- `--source gleif` (the default) optionally takes `--root-lei <lei>` to build the house hierarchy from a real organisation; find the LEI with `lei countries` and `lei entities <country> --filter <text>`.
- `--source synthetic` generates a realistic artificial organisation instead. All the generation controls from the wizard's synthetic page are available as options with the same defaults (`--party-count`,

`--counterparty-count`, `--portfolio-leaf-count` and so on); `--seed <n>` makes the result reproducible — the same seed always generates the same organisation, names included.

The command publishes the catalogue and waits while the backend works through its datasets, printing each step as it completes; generates the synthetic organisation when selected; associates your administrator account with every operational party; and finally marks the tenant active. As with the wizard, log out and back in afterwards so your session picks up the now-active tenant.

Provisioning a party

`provision party` performs the Party Provisioner's work for one party. Unlike the wizard — which runs for the party you selected at login — the command always names its target explicitly, either by its full name or by its identifier; `parties list --category Operational` shows the candidates:

```
login tenant_admin@default Secure-Password-123
provision party "Lloyds Wealth Management Ltd"
logout
exit
```

```
ores-shell>  Login successful!
ores-shell> [1/5] Importing counterparties (dataset small)...
 All 1 step(s) completed.
[2/5] Publishing organisation structure and risk reporting...
 All 4 step(s) completed.
[3/5] Publishing synthetic market data configuration...
 All 2 step(s) completed.
[4/5] Publishing FX driver rates...
 All 1 step(s) completed.
[5/5] Activating party 'Lloyds Wealth Management Ltd'...
 Party 'Lloyds Wealth Management Ltd' provisioned and active.
ores-shell>  Logged out successfully.
ores-shell> Bye!
```

The only option is `--dataset-size small|large`, selecting the counterparty import size (small is the default). The command imports the counterparties, then publishes every party-scoped bundle in turn – risk management (business units, portfolios, books, and report definitions; reports reference the book/portfolio tree, so both publish together), synthetic market data configuration (FX, IR curve, and any future asset class – adding one is a data change in `dq_dataset_bundle_member_populate.sql`, never a shell code change), and the curated FX driver-rate dataset – and finally activates the party.

The supporting commands

The `provision` commands are built from smaller commands you can use on their own — to inspect the system, to recover when a step fails, or to assemble a custom flow. Each `provision` phase prints which of these it is performing, so a failed run tells you where to pick up by hand.

Command	Purpose
`bundles list`	The reference data catalogues available for publication.
`bundles publish <code> [--wait]`	Publish a catalogue; `--wait` blocks until the backend finishes.
`workflow steps <id>` / `workflow wait <id>`	Inspect or wait on a long-running backend operation.
`lei countries` / `lei entities <country>`	Browse the GLEIF registry for a `--root-lei` value.
`synthetic generate`	Generate a synthetic organisation (all controls as options).
`parties list`	List parties, with `--category` and `--status` filters.
`account-parties add <account> <party>`	Grant an account access to a party.
`tenants complete-provisioning`	Mark the logged-in tenant's provisioning complete.
`reports templates`	The standard report definitions seeded in the risk_management bundle.

The `provision` commands call these in sequence; you can call them individually to inspect the system or to recover a failed run from where it stopped.

Scripts

The shell runs scripts with the `load` command: one command per line, `#` starts a comment, and blank lines are ignored. A script stops at the first command that fails — so a run that reaches the end has genuinely succeeded — and reports the failing line and command. When you want a script to press on regardless, for example while exploring, pass `--continue-on-error`:

```
load my_script.ores
load my_script.ores --continue-on-error
```

ORE Studio ships a small library of provisioning scripts in `projects/ores.shell/scripts/`. Each `.ores` script in the library is generated from a documentation file alongside it that explains, step by step, what the script does and

what it expects of the system — read that file before running a script for the first time. Treat the shipped scripts as templates: copy one and adjust the copy rather than editing the original, which the build regenerates.

A complete example

The library's `acme_system_provision.ores` provisions a complete system from a fresh installation: platform administrator, an Acme tenant with a reproducible synthetic organisation, and the house's root party with its risk management, synthetic market data, and FX driver-rate bundles. (Its sibling `barclays_system_provision.ores` does the same from the GLEIF golden copy instead of synthetic data.) It expects a fresh installation in bootstrap mode, all services running, and a connected, logged-out session.

```
load projects/ores.shell/scripts/library/acme_system_provision.
    ores
exit
```

An error-free run ends with a fully provisioned system (the excerpt below elides most of the run — the full output reports every dataset, generation count and report definition as it lands):

```
ores-shell> Loading script: projects/ores.shell/scripts/library
    /acme_system_provision.ores
> provision system super_admin Secure-Password-123
    admin@localhost.com ...
 System provisioned. Tenant 'Default Tenant' ...
...
> provision party "Lloyds Wealth Management Ltd"
[5/5] Activating party 'Lloyds Wealth Management Ltd'...
 Party 'Lloyds Wealth Management Ltd' provisioned and active.
> logout
 Logged out successfully.
Script complete: 9 commands executed.
```

Two details are worth pausing on. The seed pins the generated organisation, which is why stage 3 can name *Lloyds Wealth Management Ltd* — change the seed and you must discover the new root party with `parties list --category Operational` and update the script. And the logouts between stages are not ceremony: they are how the session learns that the tenant has become active and the parties exist, exactly as the wizards instruct.

Log in — from the shell or the desktop application — as `tenant_admin@default`, select a party, and daily use begins, just as at the end of the previous chapter.

Conclusion

The chapter set out to show that a complete system can be provisioned from the command line by understanding the client, the provisioning commands in sequence, the supporting commands beneath them, and the scripts that tie them together, and it has traced exactly that path. The shell speaks to the same backend services as the desktop application, so it enforces the same validation, audit trail, and permissions, and provisioning follows the same fixed sequence — system, then tenant, then party — with the logouts and logins between stages that refresh the session's view of the system. Each `provision` command collapses a wizard into a single line whose options carry the same defaults, and each is built from smaller supporting commands that can be run by hand to inspect the system or recover a failed run. The `load` command then runs those commands unattended from a file, and the shipped script library showed the whole sequence executing end to end from a fresh installation. Because the same script against the same starting point always produces the same system, scripted provisioning is the natural tool for development environments, testing, and automation — everything the wizards do, reproducibly and without the desktop.

Part II

Administration

Administration covers the ongoing care of an OreStudio installation once the provisioning wizards have done their one-time work: creating and managing the accounts people and processes log in with, the roles and permissions that govern what they may do, and the tenants that keep each organisation's data isolated.

5

Accounts and Roles

Every person and process that talks to OreStudio does so through an *account*. This chapter examines day-to-day account management: the Accounts window and how it differs between administrator identities, the account detail dialog and each of its tabs, the flow for creating a new account, and the roles and permissions that govern what an account may do — OreStudio's industry-standard Role-Based Access Control model, which ties identity to capability.

Overview

The chapter advances the argument that managing accounts well means understanding the population of accounts you administer, the individual record and what it carries, how a new one is brought into being, and finally the model that decides what any of them may do — and it proceeds in that order. It begins by surveying the body of records in The Accounts window, where the list reflects who you are logged in as and updates live as others make changes; this establishes the administrator's vantage point the rest of the chapter works from. From there it narrows to a single record in The account detail dialog and its six tabs — general identity, security, login status, roles, parties, and provenance — the dialog that serves equally for inspection and editing. It then turns to bringing a new account into being in Creating an account, the same dialog in create mode, with the password, role, and party steps and the warnings that guard them. Finally it sets out the model underneath it all in Roles and permissions — the Role-Based Access Control scheme by which accounts hold roles and roles bundle permissions — distinguishing the service roles that run OreStudio's own machinery from the domain roles modelled on a financial institution's desks. The Conclusion draws these steps together.

The Accounts window

Open the Accounts window from the *Administration* submenu of the *System* menu. The Administration menu is only present for accounts that hold administrative permissions.

The provisioning wizards described in the Initial Setup chapter create the first accounts for you — the platform administrator during system provisioning and the tenant administrator during tenant creation. Those wizards are a one-time bootstrap: every account after that is created and managed in the Accounts window described here. Accounts come in four types, shown as coloured badges throughout the UI: *User* for people, *Service* for OreStudio's own backend services, *Algorithm* for automated processes, and *LLM* for large language model agents. What an account is allowed to do is determined by the *roles* assigned to it; which data it can see is determined by the *tenant* it belongs to and the *parties* it is assigned (see Initial Setup for the tenant and party model).

The window lists every account visible to your login identity, one row per account. Alongside the username and email, badge columns show the account type, the login status (*Online*, *Recent*, *Old*, or *Never* for accounts that have never logged in), and whether the account is locked. The remaining columns carry the record version and provenance — who last modified the record and when.

The toolbar follows the standard entity-window layout (*Reload*, *Add*, *Edit*, *Delete*, *History*) and adds four account-specific actions: *Lock* and *Unlock* to suspend and restore login access, *Reset Pwd* to set a new password, and *Sessions* to inspect the account's login sessions.

What you see depends on who you are

Account visibility follows tenant isolation. Logged in as the platform administrator on the system tenant, the window shows the full machinery of the platform — the `super_admin` account plus the service accounts that OreStudio's backend components use to talk to each other:

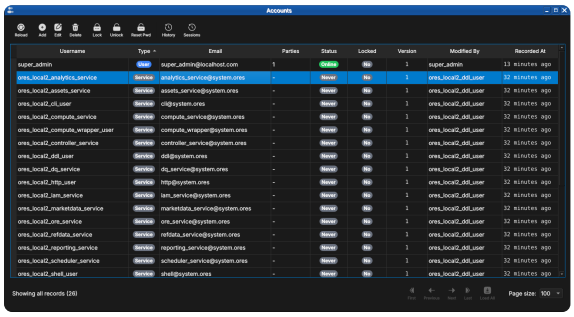


Figure 5.1: The Accounts window as seen by the platform administrator. Most rows are service accounts — one per backend component — shown with the teal *Service* type badge. Note the gray *Never* login-status badges: service accounts authenticate with certificates rather than interactive logins.

Logged in as a tenant administrator, the same window shows only the accounts belonging to that tenant — a freshly provisioned tenant contains exactly one, the tenant administrator itself:

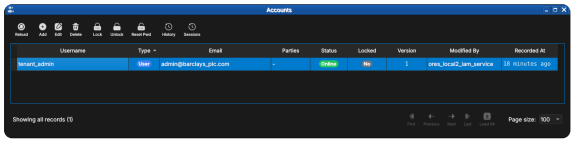


Figure 5.2: The same window as seen by a tenant administrator immediately after tenant provisioning. Only the tenant's own accounts are visible — here just `tenant_admin`, currently *Online*.

Live updates

Like all OreStudio entity windows, the Accounts list updates in response to changes made elsewhere — another administrator creating an account in a different session, for example. When unseen changes are pending, the *Reload* button carries a yellow stale indicator; recently changed rows are highlighted until you have seen them.

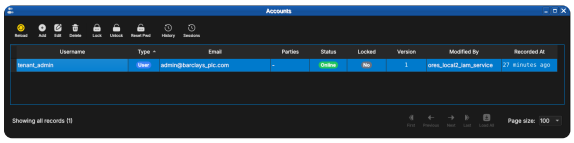


Figure 5.3: The Reload button showing the yellow stale indicator: the account list has changed since it was last loaded.

After a reload, rows that changed since you last looked stay highlighted until acknowledged, so a batch of new arrivals stands out from the records you have already reviewed:

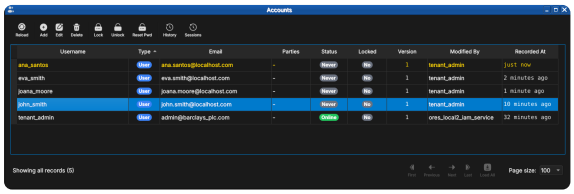


Figure 5.4: Newly created accounts highlighted in the list. The `ana_santos` row was recorded "just now" and remains highlighted until acknowledged.

The account detail dialog

Double-click an account (or select it and press *Edit*) to open the detail dialog. The dialog has six tabs; the same dialog serves creation, editing, and read-only inspection of historical versions.

- *General* — username, email, and the account type. The type is chosen at creation time and is read-only afterwards.
- *Security* — set or change the account password (see the new-account flow below).
- *Login Status* — read-only login telemetry: whether the account is online, locked, its failed login count, last login time, and last known IP addresses.
- *Roles* — the roles assigned to this account (next section).
- *Parties* — the parties this account may log in to.
- *Provenance* — who changed this record version, when, and why.

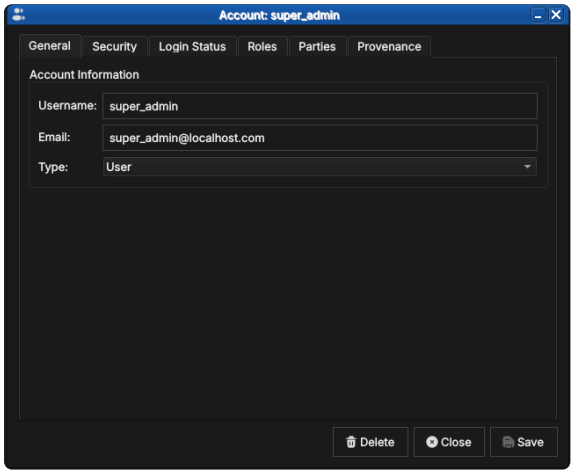


Figure 5.5: The General tab for `super_admin`, showing the username, email address, and account type.

Roles and parties tabs

The Roles tab lists the roles currently assigned to the account, and is where capabilities are granted and revoked: pick a role in the combo box below the list and add it, or select an assigned role and remove it. Changes take effect on the account's next login.

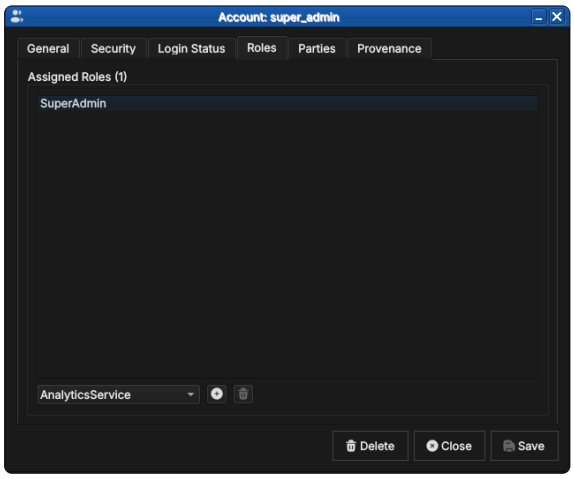


Figure 5.6: The Roles tab for `super_admin`, holding the single `SuperAdmin` role. The combo box below the list adds further roles; the buttons alongside add and remove the selection.

The Parties tab works the same way but governs visibility rather than capability: the parties listed here are the party contexts the account may select at login, as described in Initial Setup.

An account with no party assignment cannot log in to any party context — the dialog warns you if you try to save one (see the new-account flow below).

For an account assigned to more than one party, the **Default Party** combo below the list picks which of the assigned parties is offered automatically at login — the party the **Log in to default party** checkbox on the login dialog selects without prompting (see Connecting to ORE Studio). Leaving it unset falls back to the manual Select Party dialog every time.

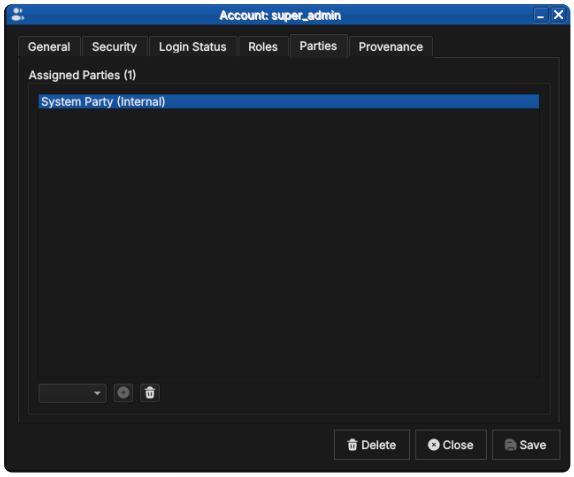


Figure 5.7: The Parties tab for `super_admin`, assigned to the internal System Party. Party assignment determines which party contexts the account may select at login.

Provenance

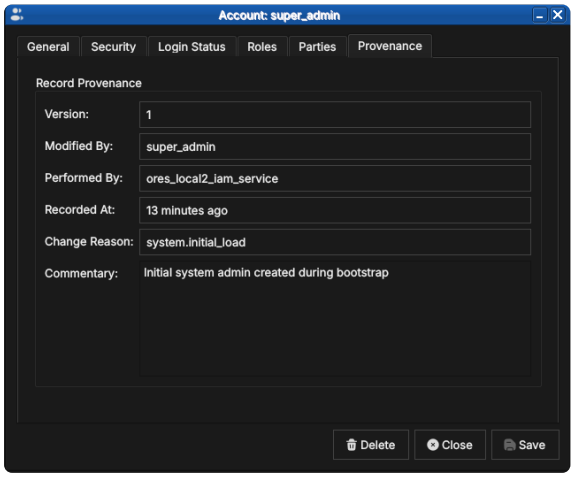


Figure 5.8: The Provenance tab for `super_admin` at version 1: modified by `super_admin`, performed by the IAM service, with the `system.initial_load` change reason and the bootstrap commentary.

Every account record is versioned and carries full provenance, following the same conventions as all OreStudio entities — see the Provenance section of the Reference Data chapter. Note the distinction visible in the screenshot: *Modified By* is the account on whose behalf the change was made, while *Performed By* records the backend service that executed it.

Creating an account

Press *Add* in the Accounts window toolbar to open the same detail dialog in creation mode, titled *New Account*.

Passwords are set on the *Security* tab. The confirmation field outlines green once both entries match; tick *Show passwords* to verify what you typed before saving.

Passwords are entered masked; ticking *Show passwords* reveals both fields, useful when setting an initial password you are about to hand to the new user:

Next, assign at least one role on the Roles tab — without one the

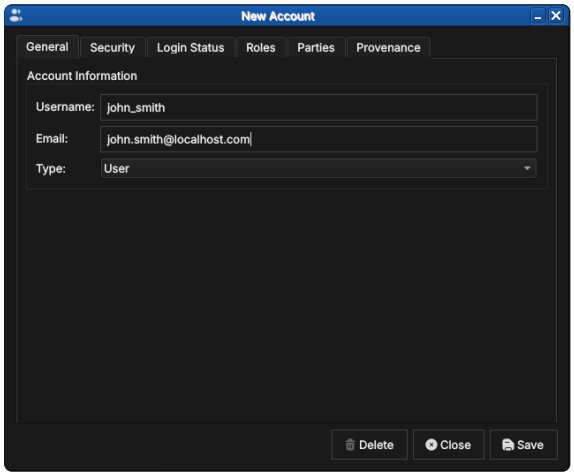


Figure 5.9: Creating `john_smith`: username, email, and account type on the General tab.

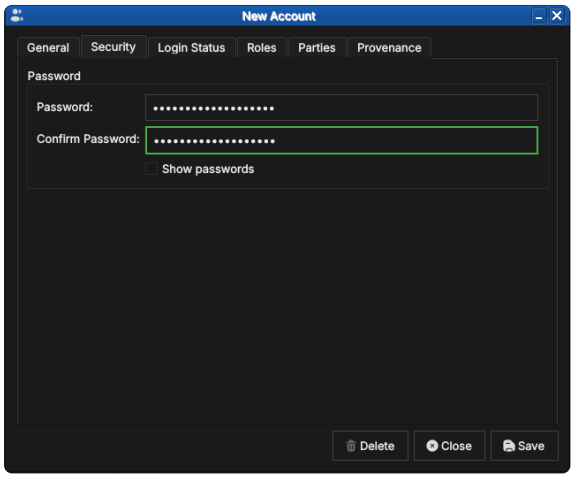


Figure 5.10: Setting the initial password. The green outline on the confirmation field indicates the two entries match.

account will hold no permissions (see the note below):

For accounts that should log in to a party context, assign at least one party on the Parties tab; the combo box offers the tenant's party hierarchy:

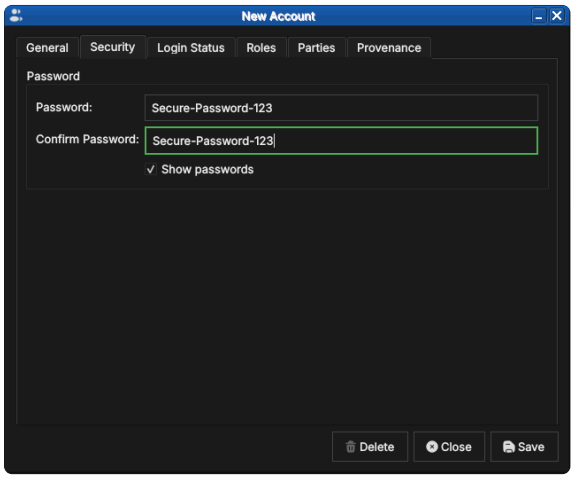


Figure 5.11: The same tab with *Show passwords* ticked, revealing the password text for visual verification.

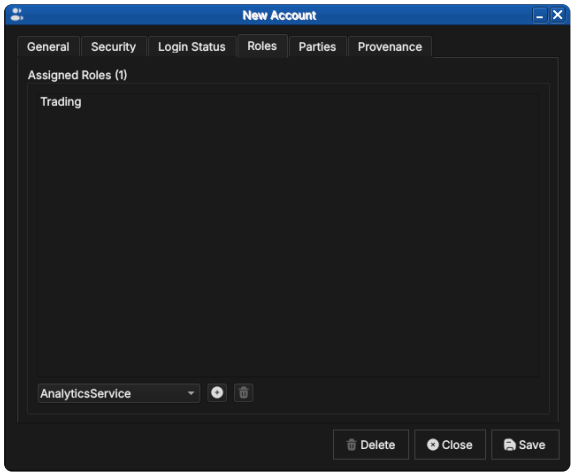


Figure 5.12: Assigning the `Trading` role to the new account. The combo box offers every role defined in the tenant.

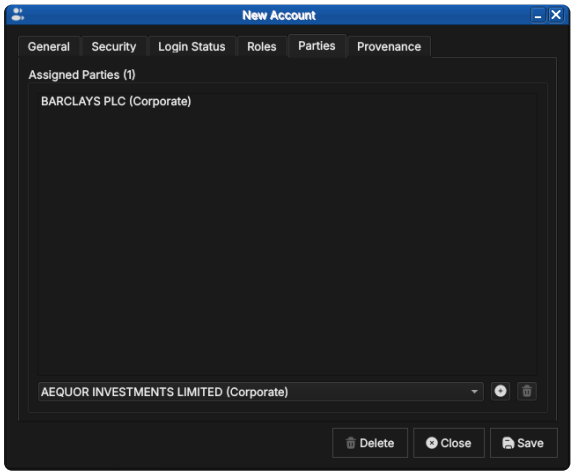


Figure 5.13: Assigning the new account to `BARCLAYS PLC`. The combo box lists the tenant's party hierarchy.

If you save without assigning a party, OreStudio asks for confirmation — such accounts exist but cannot enter any party context at login:

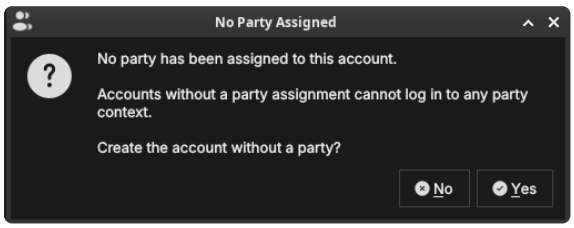


Figure 5.14: The confirmation shown when saving an account with no party assignment.

> At present there is no equivalent warning for *roles*: OreStudio lets you save a new account with no role assignment and stays silent. Such an account can log in but holds no permissions, so almost every operation will be denied. Until a warning is added, double-check the Roles tab before saving a new account.

Roles and permissions

OreStudio implements industry-standard *Role-Based Access Control* (RBAC). RBAC separates three concepts that are easy to conflate:

- An *account* is an **identity** — it answers "who is this?". It carries credentials (a password for users, certificates for services) and is the thing that logs in, appears in provenance records, and gets locked or unlocked. An account grants nothing by itself.
- A *permission* is the **atomic unit of capability** — it answers "what single thing may be done?". Permissions are fine-grained strings following a `domain::resource:action` pattern: `refdata::currencies:read` allows reading currency records, `iam::tenants:create` allows creating tenants. A trailing wildcard covers a whole domain — `analytics::*` grants every analytics permission — and the bare wildcard `*` grants everything.
- A *role* is a **named bundle of permissions** — it answers "what job does this identity do?". Roles are the only bridge between the two: accounts never hold permissions directly, they hold roles, and every permission an account exercises arrives through one of its roles.

This indirection is what keeps administration tractable: when a new permission is needed by everyone in trading, it is added to the `Trading` role once rather than to every trader's account, and an account's capabilities can be read at a glance from its role list.

Open the Roles window from the *Administration* submenu of the *System* menu to see the bundles:

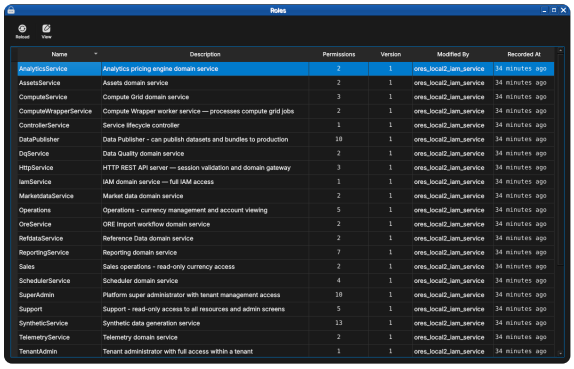


Figure 5.15: The Roles window listing the built-in roles — one per backend domain service plus the administrative and operational roles — with their permission counts and provenance.

Service roles

Every backend component runs under a dedicated service account (the teal-badged accounts seen earlier in the system tenant), and each service account holds exactly one matching service role — `IamService`, `RefdataService`, `AnalyticsService`, `ComputeService`, `WorkflowService` and so on, one per component. The bundle gives the component full access to its own domain and read access to whatever else it legitimately needs: the `AnalyticsService` role seen below holds `analytics::*`

plus `iam::tenants:read`, because the pricing engine owns the analytics domain but only ever reads tenant records. This is the principle of least privilege applied to OreStudio's own machinery — a compromised or misbehaving component cannot reach beyond its role.

Domain roles

For human users, OreStudio seeds operational roles modelled on the desks and functions of a financial institution:

Role	Intended for
`SuperAdmin`	Platform administrators
`TenantAdmin`	Tenant administrators
`Trading`	Traders
`Sales`	Sales desks
`Operations`	Middle/back office
`Support`	Support staff
`Viewer`	Auditors, casual consumers
`DataPublisher`	Data stewards

Table 5.1: The seeded domain roles and their intended audiences.

Each role's capability profile:

- `SuperAdmin` — everything, plus the tenant lifecycle: create, suspend, terminate, reset.
- `TenantAdmin` — everything within their own tenant.
- `Trading` — read reference data; create, modify, archive and delete workspaces.
- `Sales` — read-only reference data.
- `Operations` — manage reference data (read, write, delete); view accounts.
- `Support` — read-only across resources and admin screens: accounts, roles, login info.
- `Viewer` — basic read-only access to domain data and workspaces.
- `DataPublisher` — publish curated datasets and bundles to production tables.

The profiles embody a simple gradient: `Viewer` and `Sales` observe, `Trading` works within its own workspaces, `Operations` maintains the shared data everyone depends on, and `Support` sees administrative state without being able to change it. The two administrator roles differ in scope rather than degree — `TenantAdmin` is omnipotent inside one tenant, while `SuperAdmin` additionally manages the tenants themselves from the system tenant.

The seeded roles are a starting point, and their capability profiles will grow as more of the system is brought under permission control. Roles are inspected through a read-only detail dialog:

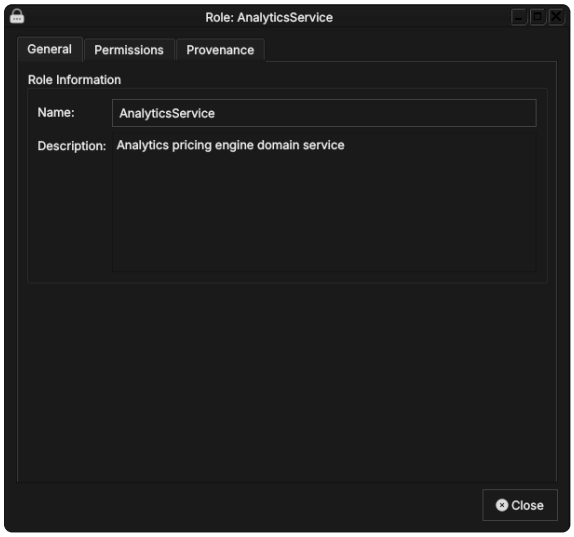


Figure 5.16: The `AnalyticsService` role: name and description.

The General tab carries only the role's name and a description of its purpose — a role has no other state of its own. The substance lives on the Permissions tab, which lists the permission strings the role bundles. Reading them is the quickest way to understand exactly what a role grants — here, `analytics::*` (every analytics permission) plus `iam::tenants:read` (read-only access to tenant records), the least-privilege profile of the pricing engine discussed above.

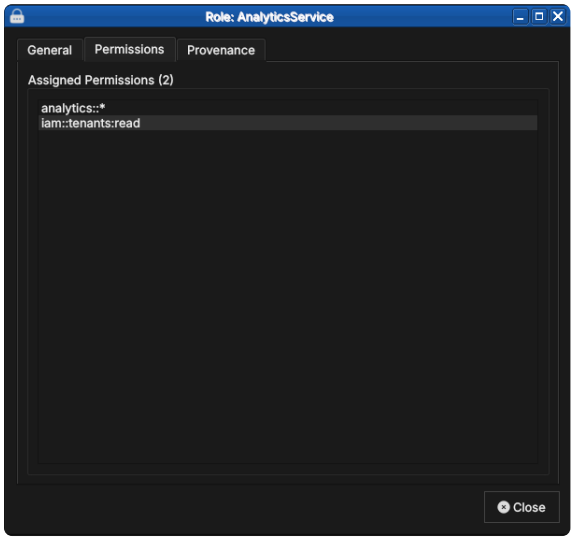


Figure 5.17: The role's two permissions, following the `domain::resource:action` pattern.

Finally, the Provenance tab shows that role records are versioned with full provenance like every other entity — changes to what a role grants are part of the audit trail.

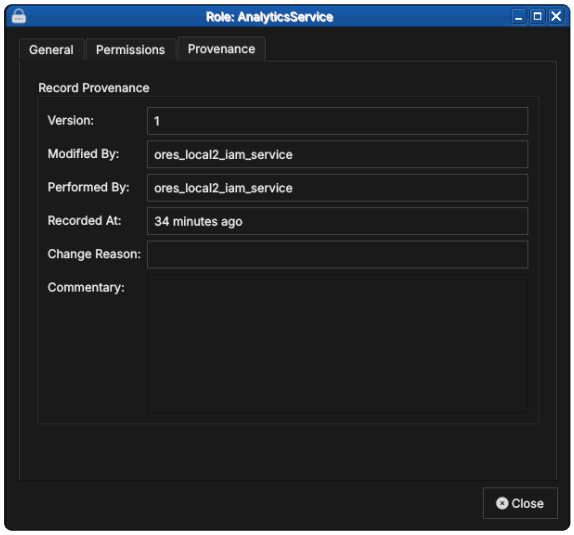


Figure 5.18: Role records carry the same versioned provenance as every other entity.

Conclusion

The chapter set out to show that managing accounts well follows from understanding the population you administer, the individual record, the flow that creates one, and the model that governs them all in turn, and it has traced exactly that path. The Accounts window presented every account your login identity may see — the platform administrator sees the system tenant's service machinery, a tenant administrator only their own tenant — and the detail dialog then managed a single account's credentials, roles, parties, and provenance across its six tabs. Creating an account proved to be the same dialog in create mode, with the party warning (and the silent role gap noted above) to watch for. Underneath it all, the RBAC model tied identity to capability: accounts are identities, permissions are atomic capabilities, and roles are the named bundles that bridge them — service roles for OreStudio's own components, domain roles for the desks and functions of a financial institution. The next chapter completes the administration picture with the tenants those accounts live in.

See also

- Initial Setup — the tenant and party model, the provisioning wizards that create the first accounts, and party selection at login.
- Tenants — managing the tenants that accounts belong to.
- Reference Data — the provenance and change-reason conventions shared by all entity windows.
- Role-Based Access Control — the general RBAC model behind OreStudio's roles and permissions.

6

Tenants

This chapter examines the *tenant* as the unit of isolation in OreStudio and the canonical reference for the organisational model built around it: the multi-tenant, multi-party hierarchy of the house and its counterparties, the type taxonomy and lifecycle that govern a tenant's operational posture, and the Tenants window and detail dialog through which tenants are managed. Tenant administration is a platform concern — the window is available to the platform administrator on the system tenant.

Overview

The chapter advances the argument that administering a tenant well means first understanding the organisational model it anchors, then the taxonomy that fixes its operational posture, and finally the interface that governs its lifecycle — and it proceeds in that order. It begins by establishing that model in The multi-tenant, multi-party model, which sets out tenant isolation and the party hierarchy of the house and its counterparties; this is the premise the rest of the chapter builds on. From there it turns to Tenant types, the four-way taxonomy that distinguishes the system, production, evaluation, and automation tenants by the controls each carries. It then surveys the body of records in The Tenants window, explaining the list view and its lifecycle status badges, before narrowing to a single record in The tenant detail dialog and its General and Provenance tabs, where an administrator moves a tenant through its lifecycle under the same audit regime as every other entity. This chapter picks up once tenants and parties exist; the Initial Setup chapter walks through the wizards that create them. The Conclusion draws these steps together.

The multi-tenant, multi-party model

A tenant's isolation extends to everything it contains: its own users, its own reference data (currencies, counterparties, books), and its own analytics results. Data belonging to one tenant is completely invisible to another tenant; there is no cross-tenant data leakage by design.

A typical deployment has one tenant per organisation. If you are running ORE Studio for your own firm, you will create one tenant representing your organisation. A managed-service provider running ORE Studio on behalf of multiple clients might create one tenant per client.

The **system tenant** is special — it is created automatically during system provisioning and cannot be deleted. It is the home of the platform administrator accounts. There is exactly one system tenant per deployment. All other tenants are created by platform administrators working from the system tenant.

Tenants come in several types with different operational controls; the next section covers the taxonomy.

Parties

Within a tenant, a *party* is a business unit — a legal entity, a branch, a trading desk, or any other organisational subdivision. Parties form a hierarchy: a parent party can see all data belonging to its children and grandchildren; a child party sees only its own data and that of its own descendants.

Every tenant has exactly one **system party**, created automatically when the tenant is provisioned. The system party is the administrative home for tenant administrator accounts. It is not a business entity and should not be used for trading activity.

Business parties — the entities that actually own trades, books, and analytics results — are **operational parties**. You create these after the tenant is provisioned, using the Party Provisioner.

In financial industry usage, the set of operational parties representing your own organisation is called *the house*. The house hierarchy models your corporate structure: the root party is your top-level legal entity; its children are subsidiaries, branches, or regional offices; their children are individual trading desks or booking centres. Trades, positions, and risk reports all belong to a specific party within the house.

Distinct from the house are *counterparties* — the external legal entities your house trades with (banks, broker-dealers, corporates, funds). Counterparties are reference data: they are attached to trade tickets to identify the facing entity, but they do not own books or belong to the party hierarchy. The house versus counterparty distinction appears throughout OreStudio's UI and data model; keeping it clear is important when working with trades and risk reports.

A user logs in to a specific party within the house. Their data

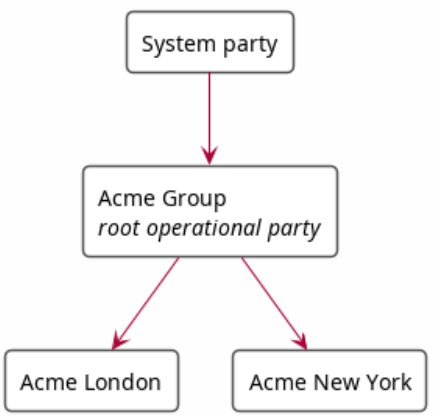


Figure 6.1: A simple party hierarchy showing the house: the system party at the top, the root operational party (Acme Group) below it, and two child operational parties (Acme London, Acme New York).

visibility is determined by their position in the hierarchy: a user logged in at "Acme Group" sees data from both "Acme London" and "Acme New York"; a user logged in at "Acme London" sees only their own data.

Tenant types

ORE Studio supports four tenant types, each with different operational controls:

- **System** — platform administration; one per deployment; platform-managed.
- **Production** — real customer organisations with live data; strict controls.
- **Evaluation** — demos, UAT, training, and exploratory testing; relaxed controls.
- **Automation** — programmatic test infrastructure; not for human use; no controls.

The **system tenant** is created automatically during system provisioning and cannot be deleted. It is the home of the platform administrator accounts; all other tenants are created by platform administrators working from it.

Production tenants are for live operations. They enforce strict controls including four-eyes authorisation for sensitive operations and KYC-gated counterparty onboarding.

Evaluation tenants provide a realistic but relaxed environment for demonstrations, user acceptance testing, and training. Bulk data import from external sources (such as the GLEIF/LEI registry) is available in evaluation tenants but not in production tenants.

Automation tenants exist for programmatic test infrastructure. They carry no operational controls and are not intended for human use.

The Tenants window

Open the Tenants window from the *Administration* submenu of the *System* menu.

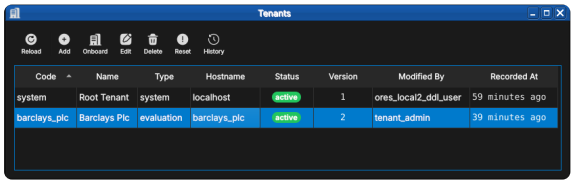


Figure 6.2: The Tenants window on a freshly provisioned installation: the `system` root tenant and one business tenant, `barclays_plc`, both *active*. Each row shows the tenant code, display name, type, hostname, lifecycle status, and provenance.

Each tenant carries:

- *Code* — the stable identifier (`system`, `barclays_plc`).
- *Name* — the human-readable display name.
- *Type* — one of the four tenant types described above; `system` for the root tenant, here `evaluation` for the business tenant.
- *Hostname* — the hostname the tenant's users connect through, which is how OreStudio routes a login to its tenant.
- *Status* — the lifecycle state, shown as a badge: *bootstrapping* while a newly created tenant awaits its provisioning wizard run, *active* once it is in service, with *suspended* and *terminated* closing out the lifecycle.

The toolbar extends the standard entity-window actions with *Onboard* — which creates a tenant and walks it through provisioning — and *Reset*, which returns a tenant to its post-provisioning state.

> Tenant operations are platform-level and far-reaching: deleting or resetting a tenant affects every account and every record within it. The change-reason dialog applies here as everywhere, so destructive operations leave an audit trail — but there is no undo.

The tenant detail dialog

Double-click a tenant (or select it and press *Edit*) to open the detail dialog.

The General tab carries the fields described above; the status combo is how an administrator moves a tenant through its lifecycle. The second tab holds the record's provenance:

Tenant records are versioned with full provenance like every other entity — see the Provenance section of the Reference Data chapter. The *History* toolbar action shows a tenant's full version history.

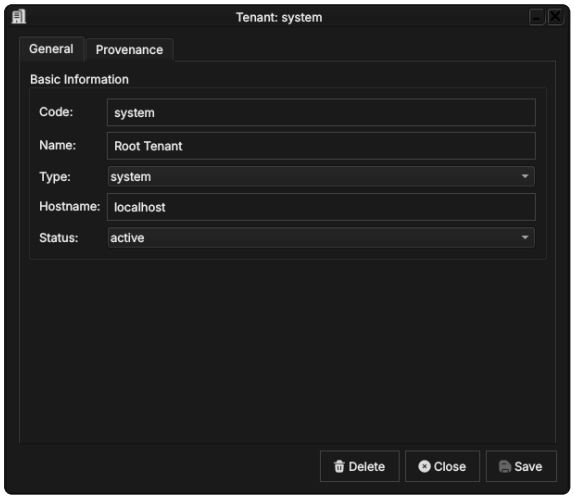


Figure 6.3: The `system` root tenant: code, name, type, hostname, and lifecycle status. The status combo is how an administrator suspends or reactivates a tenant.

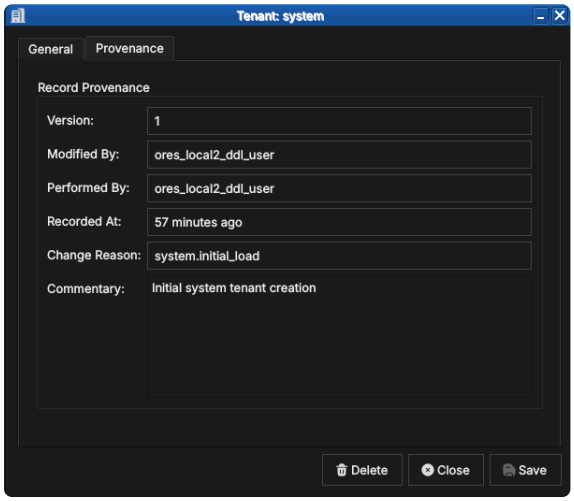


Figure 6.4: The familiar provenance tab: version, modifier, performing service, change reason and commentary — here the `system.initial_load` of the root tenant.

Conclusion

The chapter set out to show that administering a tenant well follows from understanding its organisational model, its type taxonomy, and its interface in turn, and it has traced exactly that path. The multi-tenant, multi-party model supplied the foundation — the tenant as the unit of isolation and the party hierarchy within it, the house and its counterparties, with visibility following the hierarchy. Building on that, the four tenant types fixed the operational posture of each tenant, ranging from the strictly controlled production tenant to the uncontrolled automation tenant. The Tenants window then took the record from the list view, where lifecycle status reads at a glance from the badges, down to the detail dialog, where an administrator edits a tenant's identity and moves it through its lifecycle under the same provenance regime as every other entity. With connection, provisioning, and administration now covered, the manual turns from the system itself to the data it manages: reference data.

See also

- Initial Setup — the tenant model and the provisioning wizards.
- Accounts and Roles — the accounts that live inside each tenant.

Part III

Reference Data

Reference data is the slowly-changing master data that underpins all financial activity in OreStudio — the currencies, counterparties, books, and market conventions that trades and analytics refer to. This part documents each reference data entity: what it represents, how to manage it through the Qt interface, and how to work with it from the shell and CLI.

7

Reference Data

Reference data, and the quality framework that governs it, are the subject of this chapter. It establishes what reference data is and how it differs from market data and trades, then sets out the framework that keeps it trustworthy: the six industry-standard DQ dimensions, the bitemporal storage model, versioning and immutable history, provenance tracking, and the structured change reason system. These are the shared foundations on which every subsequent entity chapter rests.

Overview

The chapter advances the argument that managing reference data well depends first on knowing what it is, and then on the quality framework that governs how it changes — and it proceeds in that order. It begins by establishing the category itself in What is Reference Data?, setting reference data apart from market data and trades and surveying the entity types OreStudio manages; this is the premise the rest of the chapter builds on. From there it turns to Data Quality, the heart of the chapter, which opens with the six industry-standard DQ dimensions and then examines the mechanisms that realise them: bitemporality and its two independent time dimensions, versioning and immutable history, provenance tracking who changed what and why, and the structured change reasons that make every modification an auditable event. The Conclusion draws these foundations together and points to the entity chapters that build on them.

What is Reference Data?

Reference data is the static or slowly-changing master data that all financial activity depends on. It defines the vocabulary of the system — the currencies a trade is denominated in, the counterparty on the other side of a deal, the book it settles into, the country a legal

entity is incorporated in. Without it, nothing else can be described precisely.

OreStudio's reference data covers a wide range of entity types:

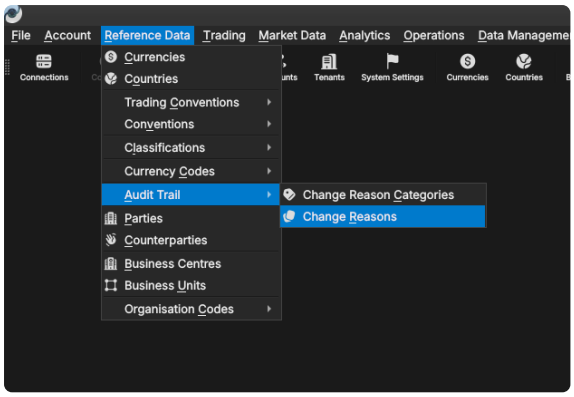


Figure 7.1: The Reference Data menu showing the full set of entity types managed by OreStudio. The Audit Trail sub-menu provides access to Change Reasons and Change Reason Categories.

- **Currencies** — ISO 4217 codes, display formatting, and rounding rules.
- **Countries** — ISO 3166 alpha-2 codes and names.
- **Trading conventions and calendars** — holiday calendars, settlement day rules, date rolling conventions, and tenor definitions that drive date arithmetic in trade processing and valuation.
- **Counterparties** — the external legal entities your organisation trades with.
- **Books** — the internal trading books that own positions.
- **Business units and portfolios** — the organisational structure within a party.
- **Party types and statuses** — the classification vocabulary for parties.
- **Classifications** — supplementary taxonomies such as currency market tiers and monetary natures.

Reference data is distinct from the other two main categories of data in the system:

- **Market data** — prices, FX rates, yield curves, and volatility surfaces that update continuously throughout the trading day. Market data is time-stamped and immutable once recorded; a new observation does not overwrite an old one.
- **Trades** — specific financial transactions between parties with defined cashflows, valuation models, and lifecycle states. Trades reference the reference data (they need a currency, a counterparty, a book) but they are not reference data themselves.

The key property of reference data is that it changes slowly and deliberately. A currency's rounding convention is not something that shifts intraday — it is corrected through a controlled, audited process. OreStudio enforces this with a comprehensive data quality framework described in this chapter.

GLEIF and the Legal Entity Identifier (the LEI standard, and how OreStudio uses it to seed parties and counterparties from real organisational data) is covered in the Parties chapter rather than here — legal-entity identification is a property of the parties and counterparties themselves, not of reference data generally.

Data Quality

Financial calculations are only as reliable as the data they consume. A misspelled currency name is a cosmetic nuisance; an incorrect rounding rule, a wrong counterparty identifier, or a stale market tier classification can silently propagate into risk figures, margin calls, collateral calculations, and regulatory reports. The consequences range from minor reconciliation breaks to significant financial loss or regulatory censure.

OreStudio's reference data layer is designed around the definition of data quality from ISO 8000 and the DAMA Data Management Body of Knowledge (DAMA-DMBOK): *data quality is the measure of how well a dataset satisfies the requirements of its intended business use*. In financial markets this means data must be not only accurate but also complete, consistent, timely, valid, and unique — the six industry-standard DQ dimensions described below.

The Six Dimensions

Accuracy is the degree to which data values agree with their authoritative golden source. OreStudio seeds currencies from ISO 4217, countries from ISO 3166, and counterparty identifiers from the GLEIF LEI registry. Where a discrepancy arises between a record in OreStudio and its source, the source takes precedence; the system provides tooling to re-import from authoritative datasets.

Completeness means all mandatory attributes are present. The service layer rejects saves that omit required fields and returns a validation error; the Qt UI marks missing mandatory fields visually before a save is attempted.

Consistency is the absence of contradictions between related entities. A currency's rounding type must be drawn from the governed rounding-types table; its market tier from the currency-market-tiers table. Foreign-key constraints in the database enforce this at the persistence layer.

Timeliness means data is available when calculations need it. The tenant provisioner imports standard catalogues at setup time so foundational reference data is ready before any trading activity

begins. NATS events propagate updates to all connected clients in real time.

Validity is adherence to business rules and formats. ISO code lengths, numeric code ranges, currency format strings, and rounding precision bounds are all validated at save time.

Uniqueness ensures no duplicate records exist for the same entity. Primary key constraints at the database level prevent duplicates; the service layer surfaces a clear error if a duplicate is attempted.

Bitemporality

The most important architectural property of OreStudio's reference data layer is its use of *bitemporality* — the recording of two independent time dimensions for every stored fact.

The first dimension is **valid time**, recorded in `valid_from` and `valid_to` columns on every row. Together they form an open interval `[valid_from, valid_to)` bounding the period during which this version of the record is authoritative. The live (current) row carries `valid_to = 9999-12-31 23:59:59` — a sentinel meaning "no known expiry". When an update is saved, the database trigger closes the existing row by setting `valid_to` to the current time, and inserts a new row with `valid_from = now` and `valid_to = sentinel`. The application never writes `valid_from` or `valid_to` directly — the trigger owns both columns.

The second dimension is **transaction time**, surfaced to the application as the `recorded_at` field. It records when this version was written to the database — independently of what period the version is valid for. Transaction time answers the question "what did the system believe at a given moment?"

All timestamps in OreStudio are stored and processed as UTC throughout. The Qt UI converts timestamps to your local timezone for display only; every timestamp you see in the application is a local-time rendering of a UTC value stored in the database.

The combination of the two time dimensions allows OreStudio to answer questions that neither alone could answer:

- *What is the current record for USD?* — the live row has `valid_to = sentinel`.
- *What did the USD record look like on a specific past date t?* — find the row where `valid_from` < t < valid_to=.
- *When did we first record the current name for a currency?* — inspect the `recorded_at` of the version where the name field changed.
- *Revert to an earlier version* — inserts a new row copied from the historical version, leaving all intermediate versions intact.

Bitemporality is particularly important for **calculation reproducibility**. Being able to reconstruct the exact state of all reference data as it existed on a past valuation date is a prerequisite for explaining

historical calculation results, for regulatory back-testing, and for resolving disputes about previously reported figures.

Versioning and Immutable History

Every save to a reference data entity creates a new, numbered version. The version counter starts at 1 on the first save and increments monotonically. History is immutable: no version is ever deleted or modified. Even a "revert" operation creates a new version — it does not roll the counter back or remove intermediate history.

This immutability guarantee is deliberate. Compliance frameworks such as MiFID II and EMIR require that firms demonstrate the state of their data at any past point in time. OreStudio's history tables provide this proof unconditionally.

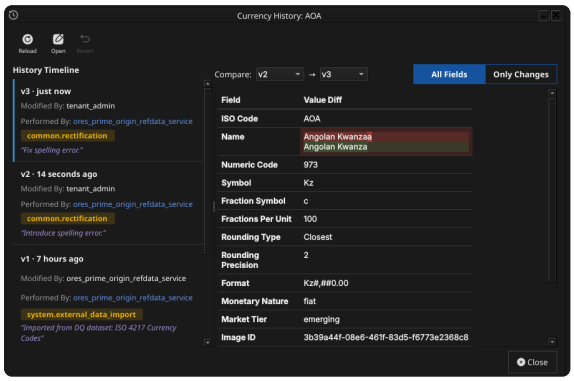


Figure 7.2: The Currency History dialog for the Angolan Kwanza, showing three versions. The All Fields/Only Changes toggle shows a field-by-field diff for the selected version against its predecessor.

The history dialog, accessible from every detail dialog's title bar or the list window's right-click menu, shows the full version list. Selecting a version populates a detail panel. The **Changes** tab shows a field-by-field diff; the **Full Details** tab shows the complete record at that version. The **Revert** button reinstates a historical version as a new current version.

Provenance

Every version of every reference data record carries six standard provenance fields. The **Provenance** tab in every detail dialog shows them read-only; it is disabled in create mode since no provenance exists before the first save.

The table below is the at-a-glance map from what you see on the tab to where each value comes from. Note the split: the two facts that must be tamper-proof — the version number and the moment of writing — are set by a database trigger and can never be supplied by the application; the four that carry business meaning are set by the application from your session and your answers to the change reason prompt.

`version` Monotonically increasing integer starting at 1, incremented by the database trigger on every save. Never written by the application.

UI label	Field	Set by
Version	`version`	DB trigger
Modified By	`modified_by`	Application
Performed By	`performed_by`	Application
Recorded At	`recorded_at`	DB trigger
Change Reason	`change_reason_code`	Application
Commentary	`change_commentary`	Application

Table 7.1: The six provenance fields, as labelled on the Provenance tab, and where each value originates.

`modified_by` The username of the account that submitted the save request.

`performed_by` The username on whose behalf the change was performed. Usually the same as `modified_by`; they differ in automated workflows, where a service account submits a change on behalf of a human operator — `modified_by` then identifies the service and `performed_by` the human, preserving accountability at both layers.

`recorded_at` The UTC wall-clock timestamp at the moment the row was written. Like all timestamps in OreStudio it is stored as UTC and displayed in your local timezone.

`change_reason_code` A structured code drawn from the change reasons table identifying the business justification for the change. See Change Reasons below.

`change_commentary` A free-text note accompanying the change. Mandatory for some reason codes, optional for others. Displayed in the history dialog and stored permanently with the version.

Change Reasons

Before every save OreStudio prompts for a *change reason* — a structured code identifying the business justification. Change reasons are organised into categories accessible from *Reference Data → Audit Trail*.

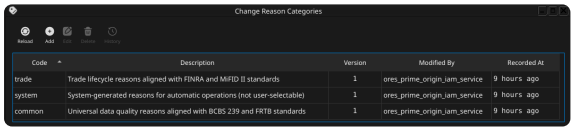


Figure 7.3: The Change Reason Categories window showing the three built-in category groups and their regulatory alignment.

Change reasons serve two purposes: they act as a forcing function against silent undocumented changes, and they make the audit trail machine-readable and regulatorily aligned. The category structure maps directly to BCBS 239, FRTB, MiFID II, and FINRA reporting obligations.

Each category below opens with a matrix of its codes: which operations each code applies to (create, amend, delete) and whether the code demands commentary — codes marked *required* will not save without a note, the rest take "—" as optional. The full code as stored in the audit trail is the category-qualified form, `category.code` — for

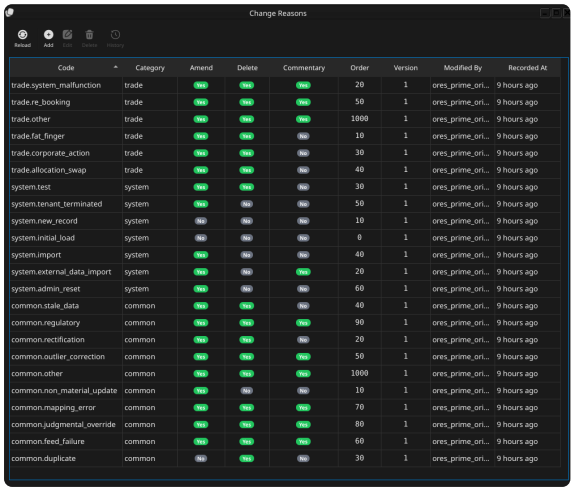


Figure 7.4: The Change Reasons list window (Reference Data → Audit Trail → Change Reasons), showing change reason codes across the built-in category groups.

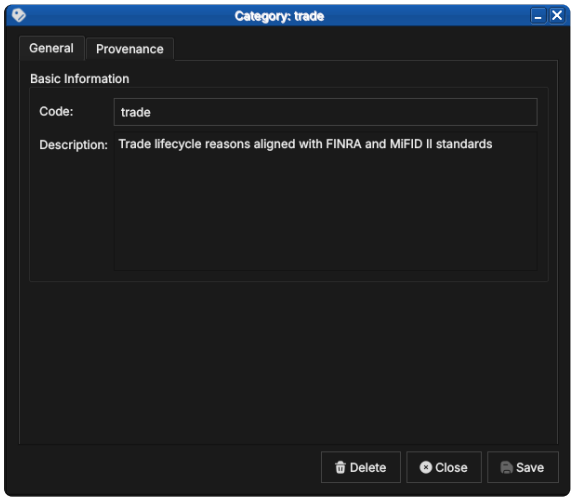


Figure 7.5: The detail dialog for the `trade` category, showing its regulatory description.

example `trade.fat_finger`; the matrices and descriptions omit the prefix within each category section.

1. The System Category

   System reasons are assigned automatically by the application and services. They are not shown in the change reason prompt and cannot be selected manually.

Code	Create	Amend	Delete	Commentary
`initial_load`	x			—
`new_record`	x			—
`external_data_import`		x		required
`import`	x	x		—
`test`	x	x	x	—
`tenant_terminated`		x		—
`admin_reset`		x		—

Table 7.2: The `system` category codes, the operations each applies to, and whether commentary is required.

`initial_load` Initial system provisioning or database migration. Used once during deployment.

`new_record` Normal operational record creation by the application or a service.

external_data_import Import from an external data source (ISO feed, GLEIF, vendor file). Commentary must record the data lineage source.

import Data loaded via the CLI import command.

test Test data created by automated test suites.

tenant_terminated Applied when a tenant is marked as terminated.

admin_reset Data reset by a system administrator during re-provisioning.

2. The Common Category

Common reasons are the universal data quality reasons, aligned with BCBS 239 and FRTB standards. These are the reasons most frequently used by operators during day-to-day data maintenance.

Code	Create	Amend	Delete	Commentary
non_material_update		x		—
rectification		x	x	—
duplicate			x	—
stale_data		x	x	—
outlier_correction		x	x	required
feed_failure		x	x	required
mapping_error		x	x	required
judgmental_override		x	x	required
regulatory		x	x	required
other		x	x	required

Table 7.3: The common category codes, the operations each applies to, and whether commentary is required.

non_material_update A cosmetic or administrative change with no economic impact — a "touch" to refresh a timestamp or correct capitalisation.

rectification Correction of a user or booking error. The most commonly used reason for fixing a data-entry mistake.

duplicate Removal of a duplicate record where an equivalent entry already exists.

stale_data Data not updated within the required liquidity horizon, requiring a refresh or removal.

outlier_correction Manual override following a plausibility check failure — a value outside acceptable bounds overridden by an operator. Commentary must identify the plausibility rule that triggered the review.

feed_failure Correction caused by an upstream vendor or API data issue. Commentary must identify the failed feed.

mapping_error Incorrect identifier translation — for example a wrong ISIN-to-FIGI mapping. Commentary must describe the mapping error.

judgmental_override Expert judgment applied when market prices or reference values are unavailable and an operator must supply a value manually.

regulatory Mandatory compliance adjustment required by a regulatory obligation or instruction.

other Exceptional changes that do not fit any other category. The commentary must fully explain the reason — this code exists as a last resort and should be used sparingly.

3. The Trade Category

Trade reasons are aligned with FINRA and MiFID II trade lifecycle reporting requirements.

Code	Create	Amend	Delete	Commentary
`fat_finger`		x	x	—
`system_malfunction`		x	x	required
`corporate_action`		x	x	—
`allocation_swap`		x	x	—
`re_booking`		x	x	required
`other`		x	x	required

Table 7.4: The `trade` category codes, the operations each applies to, and whether commentary is required.

fat_finger Erroneous execution — a trade entered with the wrong quantity, price, or instrument due to a keying error.

system_malfunction Change caused by a technical glitch or algorithmic error in an execution system. Commentary must identify the system and the nature of the malfunction.

corporate_action Adjustment following a corporate action such as a stock split, dividend reinvestment, or merger.

allocation_swap Reallocation between a house account and a client sub-account, or between sub-accounts.

re_booking Correction of a wrong legal entity booking — a trade entered under the wrong counterparty or book. Commentary must identify the correct entity.

other Exceptional trade lifecycle changes that do not fit any other code.

4. Extending Change Reasons

Change reason categories and individual codes are stored as versioned reference data and can be added through the same Qt UI and CLI tools used for other entities. In principle, a tenant administrator can create additional categories and codes for firm-specific workflows.

In practice this should be done cautiously:

- The standard codes are aligned with regulatory frameworks (BCBS 239, FRTB, MiFID II, FINRA). Non-standard codes risk creating audit trail entries that do not map cleanly to regulatory reports.
- `common.other` and `trade.other` with mandatory commentary cover most exceptional cases without adding custom codes.

- Custom codes cannot be distinguished from standard codes by the system, so the tenant must maintain its own record of which codes are standard and which are custom.
- Removing or renaming a code after it has been used in the audit trail leaves historical records referencing a code whose meaning is no longer documented.

Conclusion

The chapter set out to show that managing reference data well follows from understanding what it is and the framework that governs how it changes, and it has traced exactly that path. Reference data is the slowly-changing vocabulary of the system, distinct from streaming market data and transactional trades, and OreStudio governs it with the six industry-standard data quality dimensions. Building on that definition, the bitemporal storage model gives every record an immutable version history; provenance records who changed what, when, through which service, and why; and the structured change reason system turns every modification into an auditable event. These mechanisms are not specific to any one entity — they appear identically in every entity window. The chapters that follow, starting with currencies, document each entity on top of this shared foundation.

See also

- ISO 8000 — international standard for data quality.
- DAMA-DMBOK — data management body of knowledge, including the DQ-6 framework.
- Temporal database — background on transaction time, valid time, and the bitemporal model.
- *Time and Timestamps: Architecture and Conventions* — internal architecture document (`doc/knowledge/architecture/time-architecture.org`).
- BCBS 239 — Basel Committee principles for effective risk data aggregation and reporting.
- GLEIF — the Global Legal Entity Identifier Foundation; publishes the LEI registry and the LEI-to-BIC mapping dataset.

8

Currencies

This chapter examines the *currency* as a reference-data entity in ORE Studio. Currencies are identified by the international ISO 4217 standard; the chapter sets out that standard and the gaps it leaves, the extensions ORE Studio layers on top — display formatting, rounding rules, market tiers, and monetary natures — and the complete lifecycle of managing currency records through the Qt interface, the interactive shell, and the command-line tool.

Overview

The chapter advances the argument that managing a currency well means understanding what a currency actually *is*, its identity, the extensions that make it usable, the interface that governs its lifecycle, and the programmatic equivalents — and it proceeds in that order. It begins with the conceptual foundation in Money, currency, and cash, distinguishing three terms routinely conflated in casual usage; this framing is what makes the rest of the chapter's choices legible, not just a list of fields. It then turns to a currency's own birth-to-retirement Currency lifecycle, background the chapter draws on without OreStudio exposing it as a field. From there it establishes identity in ISO 4217 and its limitations, which sets out the codes that name a currency and the gaps that motivate ORE Studio's extensions. It then surveys the body of records in The Currencies window before narrowing to a single record in Currency Details and its General, Formatting, Rounding, and Provenance tabs. It then turns to changing a record under audit — Editing and Currency history — culminating in reverting to an earlier version, the point at which the bitemporal audit trail does its work. It then documents the auxiliary classification tables — rounding types, market tiers, and monetary natures — whose values a currency references, and finally shows the same operations automated from the shell and the CLI. The Conclusion draws these steps together.

Money, currency, and cash

Money, *currency*, and *cash* are used almost interchangeably in everyday speech, yet they name three distinct, if related, concepts, each nested within the one before it. Disentangling them clarifies what ORE Studio's `Currency` record is actually modelling, and, just as importantly, what it deliberately is not.

Money is the broadest of the three terms: any item or verifiable record generally accepted as payment for goods and services, or as repayment of debts, within a particular society. It is best understood not as a physical thing but as a social convention, one that functions precisely because a sufficiently large group of people believes it functions, and which ceases to function the moment that belief collapses. What qualifies as money is best captured by what it *does* rather than what it *is*: it serves as a medium of exchange, offering an efficient alternative to barter; as a unit of account, letting value be expressed numerically and compared across otherwise incommensurable goods; as a store of value, preserving purchasing power across time; and as a standard of deferred payment, so that an amount agreed today retains roughly the same value when it falls due in the future. Anything fulfilling these functions counts as money, which is why money has, historically, taken several distinct forms. Commodity money derives its value from the scarcity and physical usefulness of the substance embodying it, gold and silver being the paradigm cases. Fiat money, by contrast, carries no intrinsic value of its own; it is money purely because a government or a mutual agreement between parties declares it so. Bank money is more abstract still, consisting entirely of records: it is created the moment a bank extends a loan by crediting the borrower's deposit account, and destroyed the moment that loan is repaid. Despite the popular association of "money" with notes and coins, bank money constitutes the overwhelming majority of the money supply in any modern economy, physical cash typically accounting for a small single-digit percentage of the total. Cryptocurrencies form a fourth, more recent and non-standard category: digital assets used as a medium of exchange, with ownership recorded cryptographically and, typically, no central issuing authority standing behind them.

Currency narrows money down to a system of money in common use within a defined area of circulation, usually a country or a group of countries sharing a monetary union. Two meanings coexist in ordinary usage, the physical representation of banknotes and coins on one hand, and the abstract system of money itself on the other, the latter being the more useful framing for a system such as ORE Studio, given how small a fraction of any modern money supply physical cash actually represents. Being standardised by ISO 4217 is not a precondition for counting as a currency in this conceptual sense; standardisation is better understood as a seal of approval than as a definition. The currencies of smaller nations not covered by the standard, and cryptocurrencies, which rely on informal, community-

driven symbols such as `BTC` or `XBT` for Bitcoin rather than an assigned ISO code, are no less currencies for lacking one. A currency also has a lifecycle of its own, distinct from the lifecycle of any individual note or coin — see Currency lifecycle below.

Cash narrows the concept furthest of the three. It is money in the physical form of banknotes and coins, or, in the bookkeeping sense more relevant to a system like ORE Studio, the broader class of current assets comprising currency or currency-equivalents that can be accessed immediately or near-immediately. A bank deposit falls under this broader definition: it is a financial asset rather than physical currency, yet liquid enough to count as cash for most practical purposes. The relationship between the three terms can be summarised as a chain of increasingly narrow specialisation: not all money is a currency in the systemic sense, since commodity money and cryptocurrencies stretch the definition in different directions; and not all currency holdings are cash, since a currency can equally be held in a comparatively illiquid financial-asset form.

ORE Studio's `Currency` reference-data record, the subject of the remainder of this chapter, models currency in precisely this narrower, systemic sense: a named, identified system of money, equipped with the extensions described in the next section that let ORE Studio put it to practical use in calculations. The chapter does not attempt to model money in the abstract, nor the accounting concept of cash; those notions are drawn upon, where relevant, by the trading and booking chapters that build upon the currency reference data documented here.

Currency lifecycle

A currency is born when created and put into circulation, lives while in general use, and dies when eventually withdrawn — whether replaced by a successor currency, absorbed into a monetary union, or simply discontinued — though records of transactions conducted during its life must still be retained long after that withdrawal. This is background the rest of the chapter builds on, not a field OreStudio exposes directly: unlike a book, a currency record carries no `status` column of its own; a withdrawn currency simply stops being referenced in new records; its history remains intact under the same bitemporal audit trail as any other change.

The same birth-life-death shape recurs at three nested levels: the currency itself, its individual denominations (a given note or coin value can be introduced or discontinued independently of the currency surviving), and individual physical instances. Recognising the pattern as fractal is what keeps "currency lifecycle" from being mistaken for a single, currency-wide, all-or-nothing event.

Redenominations, currency replacements, and monetary unions forming or dissolving occur more often than a system designer might expect, and rarely afford the luxury of careful advance planning;

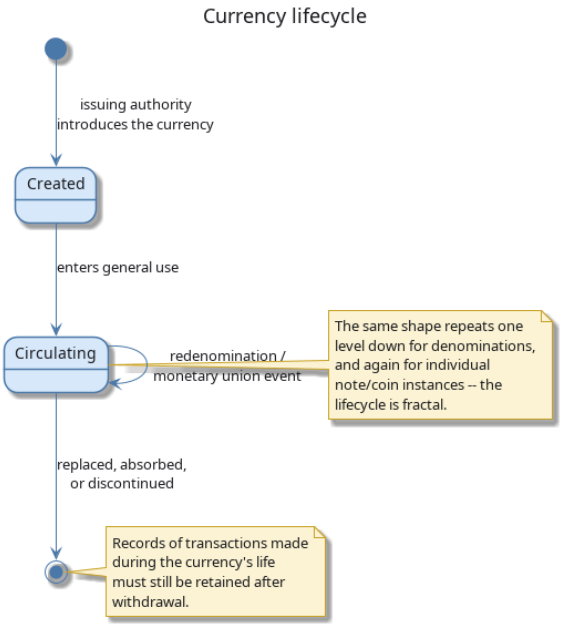


Figure 8.1: Currency lifecycle: creation, circulation (looping through redenomination and monetary-union events), and eventual withdrawal.

a system not built to anticipate such events — positions, reference data, and historical records all still keyed to a code that has just been withdrawn or redefined — can find itself significantly disrupted when one eventually occurs.

ISO 4217 and its limitations

The international standard for currency codes is ISO 4217. It defines a three-letter alphabetic code (`USD`, `EUR`, `GBP`), a three-digit numeric code (`840`, `978`, `826`), a currency name, and the number of decimal places — for example, USD has 2 minor units, meaning amounts are expressed to the nearest cent.

ISO 4217 is widely adopted and ORE Studio uses it as its primary identifier. However, the standard has gaps that matter in practice:

- It does not classify currencies by **market tier** — the distinction between major G10 currencies and emerging-market currencies is economically important but absent from the standard.
- It does not cover **digital assets**. Cryptocurrencies such as Bitcoin or Ethereum are increasingly relevant in financial systems but have no ISO 4217 code.
- It says nothing about **display formatting** — whether a symbol precedes or follows the amount, what the thousands and decimal separators are, or how the fractional unit is labelled.
- It does not specify **rounding conventions** — whether a calculation result should be rounded up, down, or to the nearest unit.

ORE Studio extends the ISO 4217 model with additional fields to cover these gaps: monetary nature, market tier, currency symbol, fraction symbol, a printf-style format string, and configurable

rounding rules. The standard fields remain primary; the extensions are supplementary.

With the underlying model in place, both the standard fields ISO 4217 provides and the extensions layered on top of them, the rest of the chapter turns from what a currency record contains to where you go to work with one. That begins, as it does for any list-backed entity in ORE Studio, with the window listing every currency the tenant holds.

The Currencies window

Open the Currencies window from the *Reference Data* menu. It lists all currencies defined in the tenant, with one row per currency.

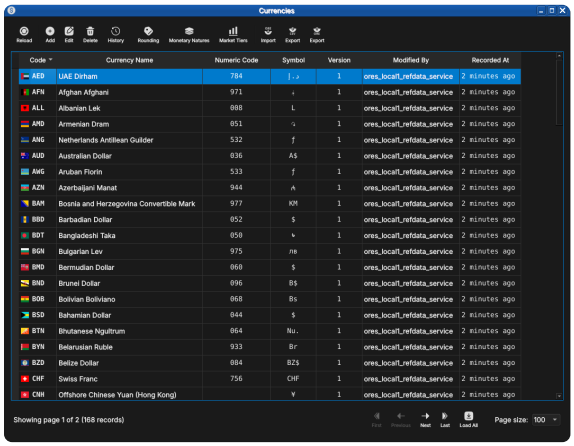


Figure 8.2: The Currencies window, showing the full list of currencies in the tenant. Each row shows the ISO code, name, numeric code, symbol, version number, the identity of the last modifier, and when the record was last recorded. The status bar shows the current page and total record count.

The toolbar buttons reload the list and adjust the display. The list is paginated; use the page controls at the bottom right to navigate. Double-clicking a row opens the **Currency Details** dialog.

Currency Details

The Currency Details dialog has four tabs: **General**, **Formatting**, **Rounding**, and **Provenance**. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the currency as a whole.

General

The General tab carries the core identity of the currency:

- **ISO Code** — the three-letter ISO 4217 alphabetic code. This is the primary key and cannot be changed after creation.
- **Name** — the full English name of the currency (e.g. *Angolan Kwanza*).
- **Numeric Code** — the ISO 4217 numeric code (e.g. 973 for AOA).

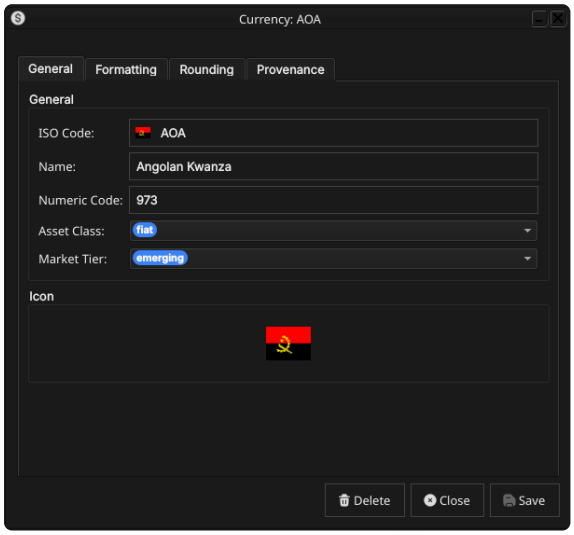


Figure 8.3: The General tab for the Angolan Kwanza (AOA). It shows the ISO code, full name, numeric code, monetary nature, market tier, and the country flag icon associated with the currency.

- **Asset Class** — the monetary nature of the currency: `fiat` for a government-issued currency, `crypto.major` for a major digital asset. See Monetary Natures below for the full list.
- **Market Tier** — the currency's liquidity classification: `g10` for the major currencies, `emerging` for others. See Market Tiers below.
- **Icon** — the country flag for the currency's issuing nation, populated automatically from the system's image dataset on import.

Formatting

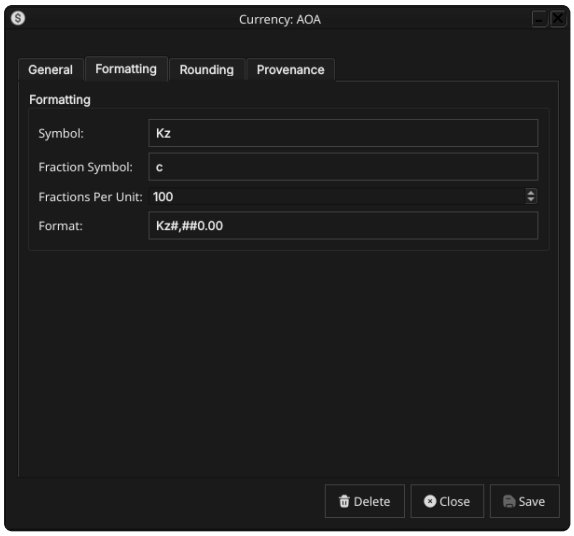


Figure 8.4: The Formatting tab for AOA, showing the currency symbol (Kz), the fractional unit symbol (c for centavo), fractions per unit (100), and the display format string.

The Formatting tab controls how currency amounts are displayed:

- **Symbol** — the currency symbol, e.g. `Kz` for the Kwanza or `$` for the US Dollar.
- **Fraction Symbol** — the symbol for the fractional unit, e.g. `c` for centavo.

- **Fractions Per Unit** — the number of fractional units in one whole unit. Most fiat currencies use `100`; the Kuwaiti Dinar uses `1000`; currencies with no fractional unit (e.g. Japanese Yen) use `0`; cryptocurrencies typically use `100000000`.
- **Format** — a printf-style format string controlling amount rendering, e.g. `Kz#,##0.00` produces `Kz1,234.56`.

Rounding

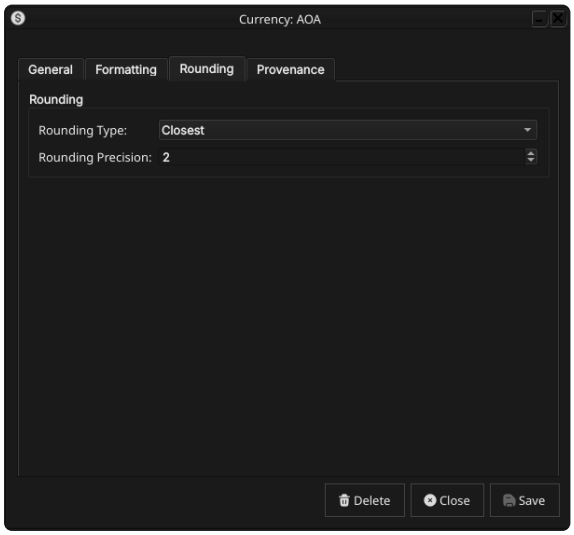


Figure 8.5: The Rounding tab for AOA, showing the rounding type (Closest) and rounding precision (2).

The Rounding tab controls how calculated amounts are rounded before storage and display:

- **Rounding Type** — the rounding algorithm applied to amounts in this currency. See Rounding Types below for the full list of options.
- **Rounding Precision** — the number of decimal places to round to. Typically `2` for most fiat currencies and `8` for cryptocurrencies.

Provenance

The Provenance tab is read-only and shows the audit metadata for the current version. See the Reference Data — Provenance section for a full description of the provenance fields common to all reference data entities.

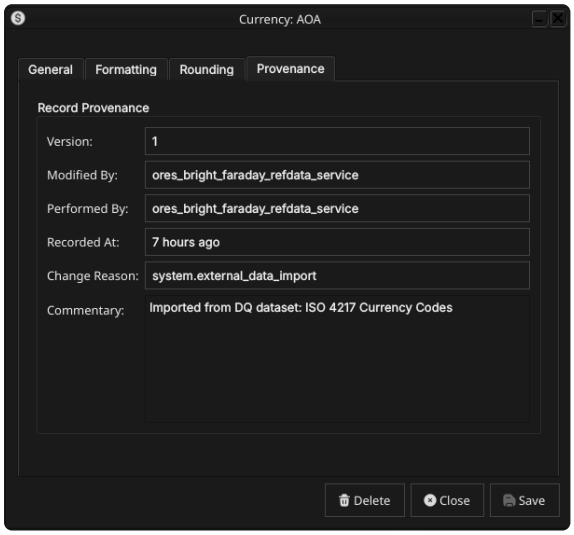


Figure 8.6: The Provenance tab showing record metadata: version number, the service that last modified the record, who performed the operation, when it was recorded, the change reason code, and a free-text commentary.

Editing a currency

To edit a currency, open it in the Currency Details dialog, make your changes, and click **Save**. Before the record is written, ORE Studio prompts for a change reason.

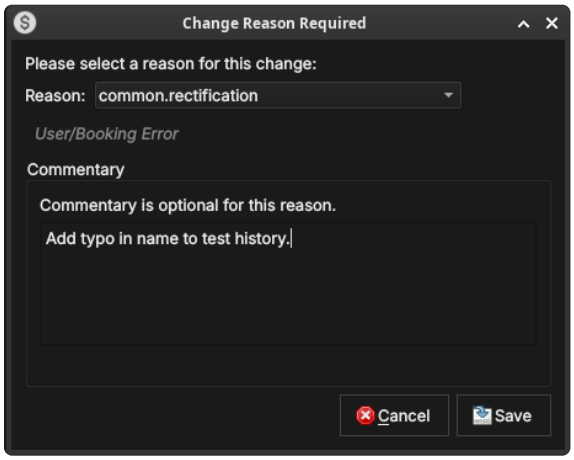


Figure 8.7: The Change Reason Required dialog, which appears before every save. A reason code must be selected; commentary is optional for some reason codes and required for others.

Select a **Reason** from the drop-down and add optional **Commentary**. Click **Save** to confirm. Every save creates a new version; the previous version is never overwritten.

Currency history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

Pick the two versions to compare from the **Compare** drop-downs. The **All Fields** / **Only Changes** toggle switches between showing every field and showing only the fields that differ between the two selected versions. The **Revert** button reinstates any historical version as a new version, preserving the full audit chain.

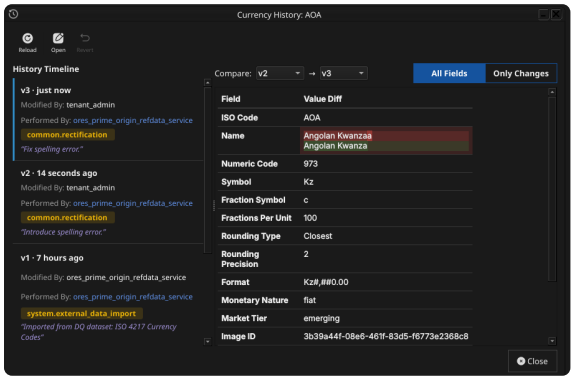


Figure 8.8: The History dialog for the Angolan Kwanza, showing three versions: the original ISO 4217 import (version 1), a test correction (version 2), and a final rectification (version 3, currently selected). The Only Changes toggle narrows the field list to what actually differs between version 2 and version 3.

Auxiliary Data

The three tables described in this section act as classification vocabularies for currencies. In a standard deployment their values are fixed at provisioning time and rarely change; ORE Studio still versions and audits them the same way as any other reference data. They are managed by the system tenant and are accessible from the *Reference Data* → *Currencies* sub-menu.

Rounding Types

Rounding types specify the algorithm used to round currency amounts in financial calculations. The values match the `roundingType` enumeration in ORE's XML schema, ensuring consistency with any ORE configuration files loaded into the system.

Open the Rounding Types window from *Reference Data* → *Currency Codes* or from the toolbar within the Currency Detail dialog.

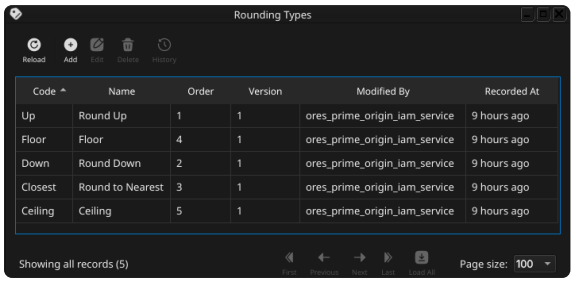


Figure 8.9: The Rounding Types list window showing the five standard rounding methods with their display order, version, and provenance.

Double-clicking a row opens the detail dialog, which shows the code, name, and a description that includes numeric examples at two decimal places.

There are five standard values:

1. Up

   Rounds away from zero regardless of the fractional part. The amount is always moved to the next representable value in the direction away from zero.

   Examples at 2 decimal places: `2.341` → `2.35`, `2.349` → `2.35`. For negative amounts: `-2.341` → `-2.35`.

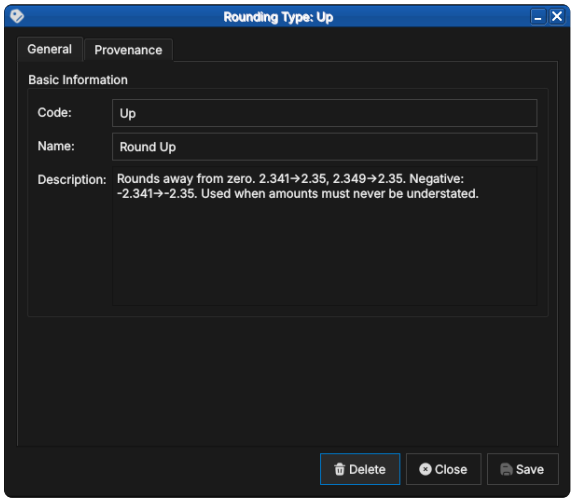


Figure 8.10: The detail dialog for the `Up` rounding type, showing the code, human-readable name, and a description with concrete examples.

Use `Up` when amounts must never be understated — for example, when calculating a fee that must cover the full cost.

2. Down

   Truncates toward zero. The fractional part is discarded.

   Examples: `2.349 → 2.34`, `-2.341 → -2.34`.

   Use `Down` in conservative contexts where overstating an amount would be the more serious error — for example, when reporting a liability that should not be exaggerated.

3. Closest

   Rounds to the nearest representable value, with halves rounded away from zero. This is the most common rounding mode for financial amounts and is the default for most fiat currencies.

   Examples: `2.344 → 2.34`, `2.345 → 2.35`, `-2.345 → -2.35`.

4. Floor

   Always rounds toward negative infinity — i.e., to the next lower value regardless of sign. Unlike `Down`, which rounds toward zero, `Floor` produces different results for negative amounts.

   Examples: `2.349 → 2.34`, `-2.341 → -2.35`.

   Use `Floor` when rounding must never produce a value higher than the unrounded input, even for negative amounts.

5. Ceiling

   Always rounds toward positive infinity — i.e., to the next higher value regardless of sign.

   Examples: `2.341 → 2.35`, `-2.349 → -2.34`.

   Use `Ceiling` when rounding must never produce a value lower than the unrounded input.

Market Tiers

Market tiers classify currencies by liquidity profile and market accessibility. The classification feeds into margin calculations, settlement conventions, and risk limits in downstream analytics.

Open the Currency Market Tiers window from *Reference Data* → *Classifications*.

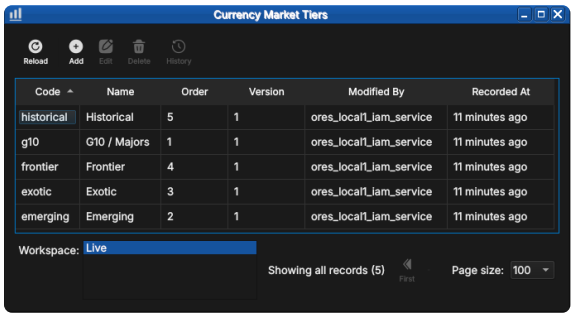


Figure 8.11: The Currency Market Tiers list showing all five standard tiers — historical, g10, frontier, exotic, and emerging — with their display order and provenance.

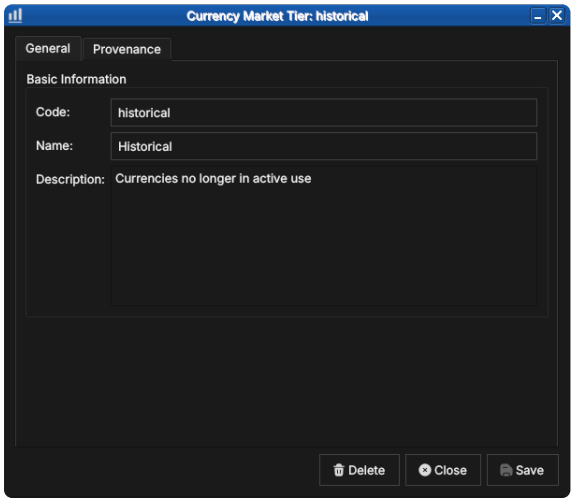


Figure 8.12: The detail dialog for the `historical` tier, showing the code, name, and description.

There are five standard tiers:

1. G10 / Majors

   The ten most liquid and widely traded global currencies: USD, EUR, GBP, JPY, CHF, AUD, NZD, CAD, SEK, and NOK. G10 currencies have deep interbank markets, tight bid/offer spreads, and continuous 24-hour liquidity. They are the primary currencies for derivatives trading and the dominant funding and collateral currencies in global finance.

2. Emerging

   Currencies from developing economies with growing but still maturing financial markets. They typically have moderate liquidity, wider spreads than G10, and may be subject to capital controls or central bank intervention. Examples include BRL (Brazilian Real), MXN (Mexican Peso), ZAR (South African Rand), INR (Indian Rupee), and CNY (Chinese Renminbi).

3. Exotic

   Thinly traded currencies with wide bid/offer spreads and limited liquidity outside their domestic market. Forward markets may be shallow or absent. Examples include currencies of smaller African, Central Asian, or Pacific island nations.

4. Frontier

   Currencies from frontier markets with limited or restricted convertibility. These may be pegged to a major currency, subject to official exchange rates, or carry transfer restrictions that make them impractical for cross-border settlement. Examples include currencies from markets classified as frontier by MSCI or FTSE Russell.

5. Historical

   Currencies no longer in active use. This includes legacy eurozone currencies superseded by the Euro (DEM, FRF, ITL, ESP, and others), as well as currencies from defunct states or monetary unions. Historical currencies are retained in the system for back-office reconciliation and historical analytics on older trade populations.

Monetary Natures

Monetary nature classifies currencies by the underlying economic mechanism that gives them value. The classification matters for collateral eligibility, regulatory capital treatment, and the applicability of pricing models.

Open the Monetary Natures window from *Reference Data* → *Classifications*.

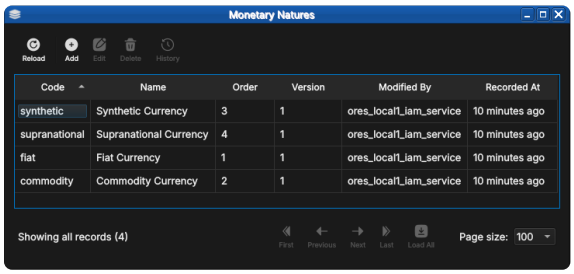


Figure 8.13: The Monetary Natures list showing all four standard values — synthetic, supranational, fiat, and commodity — with their display order and provenance.

There are four standard values:

1. Fiat

   Government-issued currency that derives its value from legal tender status rather than from a physical commodity. The issuing central bank or treasury manages supply through monetary policy. The overwhelming majority of currencies in the ISO 4217 standard are fiat — USD, EUR, GBP, JPY, and so on.

2. Commodity

   Currency backed by or representing a claim on a physical commodity. In practice this covers the ISO 4217 commodity codes:

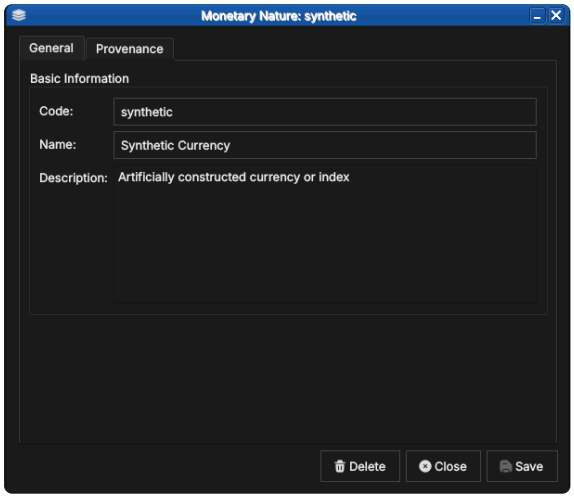


Figure 8.14: The detail dialog for the `synthetic` monetary nature, showing the code, name, and description.

`XAU` (gold), `XAG` (silver), `XPT` (platinum), and `XPD` (palladium). Pricing is driven by spot commodity markets rather than interest rate differentials.

3. Synthetic

   Artificially constructed currency or index that does not correspond to any single government or commodity. Examples include basket currencies and internal transfer pricing units used within large financial institutions.

4. Supranational

   Currency issued by a multi-national authority rather than a single sovereign. The primary example is `XDR` (Special Drawing Rights), issued by the International Monetary Fund as an international reserve asset and valued as a basket of USD, EUR, CNY, JPY, and GBP — reviewed and revised by the IMF every five years.

Shell commands

The ORE Studio interactive shell provides quick currency access without opening the Qt UI:

```
# List all currencies (paginated)
currencies get

# Add a new currency:
#   <iso_code> <name> <numeric_code> <symbol> <
    fractions_per_unit>
#   <change_reason_code> <change_commentary>
currencies add XTS "Test Currency" 963 T$ 100 system.test "
    manual example"

# Delete a currency by ISO code
currencies delete XTS

# Show history for a currency
```

```
currencies history USD
```

CLI commands

The `ores.cli` command-line tool provides a richer interface for automation. All currency commands live under `ores.cli refdata currencies`:

```
# List all currencies as a table
ores.cli refdata currencies list --tenant "$ORES_TENANT" --
    format table

# List a specific currency as JSON
ores.cli refdata currencies list --tenant "$ORES_TENANT" --
    format json --key EUR

# Add a new currency
ores.cli refdata currencies add \
  --tenant "$ORES_TENANT" \
  --iso-code XTS --name "Test Currency" \
  --numeric-code 963 --modified-by operator \
  --currency-type fiat

# Delete a currency by ISO code
ores.cli refdata currencies delete --tenant "$ORES_TENANT" --
    iso-code XTS

# Export all currencies to a file
ores.cli refdata currencies export --tenant "$ORES_TENANT" --
    output currencies.json
```

Conclusion

The chapter set out to show that managing a currency well follows from understanding its identity, the extensions that make it usable, its interface, and its programmatic equivalents in turn, and it has traced exactly that path. Its own birth-to-retirement lifecycle set the background the rest of the chapter operates against. ISO 4217 supplies the standard identity — code, numeric code, name, and minor units — while ORE Studio's extensions cover what the standard leaves out: market tiers, monetary natures for digital assets, display formatting, and rounding conventions. Building on that, the Qt interface took the record from the list view down to a single detail dialog and through the editing and history workflows, where reverting to an earlier version demonstrated the bitemporal audit trail keeping the full record intact. The three auxiliary classification tables then fixed the vocabulary a currency may reference, and the shell and CLI reproduced the same operations for scripted use. Currencies are the template: subsequent entity chapters follow this same progression.

See also

- Currency Lifecycle — the domain-knowledge note this chapter's Currency lifecycle section is drawn from.
- ISO 4217 Currency Codes — the international standard for currency identification.
- SIX Financial Information — the ISO 4217 maintenance agency.
- MSCI Market Classification — framework for developed, emerging, and frontier market designations.

9

Currency Pairs

A currency on its own, as the previous chapter set out, is a self-contained identity: a code, a name, a set of display and rounding rules. The moment two currencies are set against one another, a second entity comes into being, one whose identity is relational rather than intrinsic. This chapter examines that entity, the *currency pair*, and the separate but tightly coupled record of market conventions that governs how quotes on that pair are read, rounded, and dated.

Overview

The chapter follows the same progression as the previous one, adapted to a case where two closely related entities, rather than one, share the stage. It opens with the domain foundation in Currency pairs and market conventions, distinguishing what a pair *is* from the market conventions layered on top of it, and setting out the classification scheme, the base/quote precedence rule, and the pip and tick mechanics that later sections draw on. It then turns to the first entity, walking from The Currency Pairs window through Currency Pair Details, editing, and history, in the same pattern the Currencies chapter established. It then does the same for the second entity in Currency Pair Conventions, and closes by explaining how the two windows are cross-navigable from one another. The Conclusion draws the pair of entities back together.

Currency pairs and market conventions

A currency pair names an exchange rate relationship between two currencies rather than a currency itself: it answers the question of how many units of one currency, the *quote* currency, are needed to buy one unit of the other, the *base* currency. Market convention writes this as `CCY1/CCY2`, base over quote, and, crucially, treats the two di-

rections as the same market rather than as independent instruments: `EUR/USD` and `USD/EUR` are not two pairs but one pair quoted two ways, and a well-behaved system permits only one canonical direction to exist as a record. Which currency of the two takes the base position is itself conventional rather than arbitrary, governed by a fixed precedence order with the euro at its head, followed by sterling, the Australian and New Zealand dollars, the US dollar, and then the remaining currencies — so market participants write `EUR/USD`, never `USD/EUR`, even though both describe the same underlying rate.

Currency pairs are not a homogeneous population; the market classifies them by liquidity and tradability, a taxonomy independent of the mere fact that both legs are valid ISO 4217 currencies. A *major* pairs the US dollar with one of the six other currencies with the deepest global markets — the euro, sterling, the yen, the Swiss franc, and the Australian, Canadian, and New Zealand dollars — and carries the tightest spreads and the most continuous liquidity of any pair in the market. A *minor*, sometimes called a cross, pairs two of those major non-dollar currencies against one another with no dollar leg at all, a pair such as `EUR/GBP` or `GBP/JPY`; because the dollar is absent, such a rate is frequently derived by triangulating through it rather than quoted directly. An *exotic* pairs a major currency against the currency of a smaller or developing economy, carrying markedly wider spreads, thinner liquidity, and greater sensitivity to domestic political and economic events. A further distinct classification exists alongside this one for commodity-backed instruments, where one leg is not a sovereign currency at all but a precious metal such as gold or silver traded under currency-like conventions; ORE Studio labels this fourth case *commodity*. It is worth noting explicitly that classification and raw trading volume diverge in at least one well-known respect: the Scandinavian currencies, the Danish krone, the Norwegian krone, and the Swedish krona, trade in somewhat lower volume than the seven majors and are sometimes loosely called exotic on that basis, yet institutionally they belong to the same deep-liquidity tier as the majors and should be classified as minors, not exotics, when paired with the dollar.

Once a pair's identity and classification are settled, a further set of market conventions governs how its rate is actually quoted and handled operationally, and these live on a separate record from the pair itself. A *pip* is the standard unit of rate movement for a pair, and its size is not universal: most pairs quote to four decimal places, so that a single pip equals `0.0001` of the rate, while pairs with a yen leg quote to only two decimal places, making a pip equal to `0.01`. The *pip factor* is this decimal scaling made explicit and storable, the multiplier that converts a count of pips into an absolute rate move, and getting it wrong does not produce a merely inaccurate result but one whose sign and order of magnitude are both wrong, since it governs how forward points are added to or subtracted from a spot rate. Layered on top of the pip convention is the *tick size*, the smallest increment by which a quoted rate is actually permitted to move,

stated in units of pips and capable of varying by pair, by trading venue, and by instrument; a pip defines the denomination, a tick defines the smallest coin in that denomination actually in circulation. Finally, settlement timing follows its own convention: an FX rate, notwithstanding the informal name *spot rate*, is very rarely a rate for immediate delivery but rather a forward rate for the earliest date on which funds can actually settle, arrived at by advancing from today by the number of business days each currency's home market needs to clear a payment, one day for the US dollar and typically two for most other currencies, with the pair as a whole taking the longer of its two legs' requirements — which is why most pairs are conventionally said to settle two business days after the trade date.

A word of scope is due before moving on. Real FX markets also distinguish pairs by *deliverability* — whether a trade in that pair settles through physical delivery of both currencies or, for currencies subject to capital controls, is instead settled in cash in a third currency against a published fixing, the non-deliverable forward arrangement. ORE Studio explored modelling this distinction directly on the currency pair record and withdrew it again during this development cycle: a single flag proved unable to express the reality that some pairs support more than one settlement arrangement, sometimes depending on the trading centre involved, and a proper treatment needs its own structure rather than a field bolted onto the pair. Deliverability and non-deliverable settlement are, at the time of writing, not represented anywhere in ORE Studio's reference data and are not described further in this chapter.

With the domain foundation in place, the base/quote relationship and its precedence rule, the classification taxonomy, and the pip, tick, and settlement-timing conventions that will recur throughout the entity descriptions below, the chapter turns from what these concepts mean to where a user goes to work with the records that carry them.

The Currency Pairs window

Open the Currency Pairs window from the *Reference Data* menu. It lists all currency pairs defined in the tenant, with one row per pair.

Each row shows the pair code (with the base and quote currencies' flag icons composited into the same cell), the base and quote currency codes individually, the classification badge, the version number, the identity of the last modifier, and when the record was last recorded. The toolbar buttons reload the list and adjust the display; a further *Conventions* button, described in Currency Pair Conventions below, cross-navigates to the companion window. The list is paginated; use the page controls at the bottom right to navigate. Double-clicking a row opens the **Currency Pair Details** dialog.

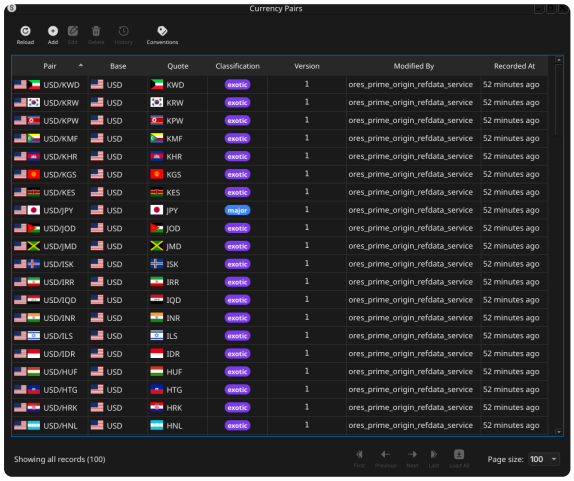


Figure 9.1: The Currency Pairs window, showing the full list of currency pairs in the tenant. Each row shows the composited base+quote flag, the individual Base and Quote codes, the Classification badge, version, and provenance columns.

Currency Pair Details

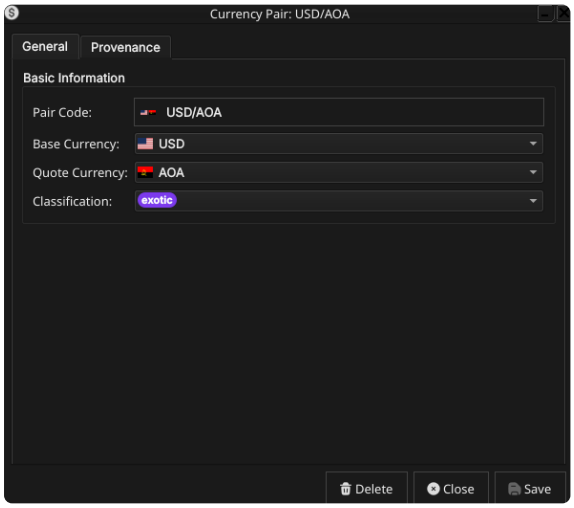


Figure 9.2: The Currency Pair Details dialog for USD/AOA, showing the Pair Code field with both flags, the locked Base and Quote Currency combos, and the Classification combo.

The dialog carries a compact set of fields. **Pair Code** is not typed directly; it is computed live from the **Base Currency** and **Quote Currency** combos while the record is still being created, and the field itself remains read-only throughout. **Base Currency** and **Quote Currency** are flagged combo boxes, each showing the relevant currency's flag alongside its ISO code, populated from the currencies already on file. **Classification** is a combo offering the four values described above — major, minor, exotic, and commodity.

Once a pair has been saved for the first time, its Base Currency and Quote Currency combos lock: the values remain visible, flags and all, but can no longer be changed, and Pair Code is therefore permanently fixed from the moment of creation onward. This reflects the domain reality that a currency pair's identity *is* its two legs; changing either leg would not modify the pair but create a different one under a different natural key.

Two checks run at creation time, both driven by the base-currency-precedence rule described in Currency pairs and market conven-

tions. Attempting to create a pair whose code already exists is rejected outright, since that would silently create a new version of the existing record rather than a genuinely new one. Attempting to create the *inverse* of a pair that already exists — for example, adding `USD/EUR` when `EUR/USD` is already on file — is rejected for the same underlying reason: the two directions describe one market, not two, and only the canonically ordered direction may exist as a record.

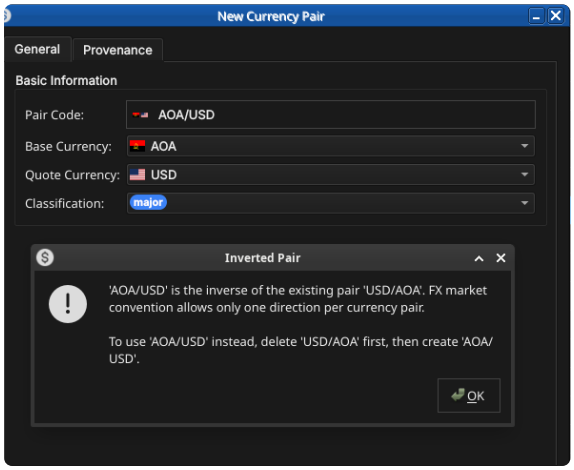


Figure 9.3: With USD/AOA already on file, attempting to create AOA/USD is rejected: the warning names both the rejected code and the existing pair it inverts, and explains the delete-then-recreate workaround for the rare case the direction genuinely needs to change.

The three action buttons at the bottom, **Delete**, **Close**, and **Save**, apply to the pair as a whole. The dialog also has a **Provenance** tab, read-only, showing the audit metadata for the current version; see the Reference Data — Provenance section for a full description of the fields common to all reference data entities.

Editing a currency pair

To edit a currency pair, open it in the Currency Pair Details dialog, change its **Classification** — the only field still open to editing once Base Currency and Quote Currency have locked — and click **Save**. As with every reference-data entity, ORE Studio prompts for a change reason before the record is written; select a **Reason** from the drop-down, add optional **Commentary**, and click **Save** to confirm. Every save creates a new version rather than overwriting the previous one.

Currency pair history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**. Select a version to populate the detail panel; the **Changes** tab shows which fields differ between the selected version and its predecessor, and the **Revert** button reinstates any historical version as a new version, preserving the full audit chain. The mechanics are identical to currency history, described in full in the previous chapter's Currency history section.

Currency Pair Conventions

Where the currency pair record establishes *identity*, base, quote, and classification, a second, separate record establishes the *market conventions* that govern how a pair's rate is quoted, rounded, and dated: pip factor, tick size, decimal places, the advancing calendar, the business day convention, and two boolean flags governing forward-date generation. The two records stand in a strict one-to-one relationship, keyed by the same pair code, and are kept as distinct entities rather than merged into one for the same reason the Currencies chapter kept Rounding Types and Market Tiers separate from the Currency record itself: identity and convention change at different rates and for different reasons, and separating them keeps each record's audit history meaningful on its own terms.

Open the Currency Pair Conventions window from the *Reference Data* menu, or from the *Conventions* toolbar button on the Currency Pairs window described above.

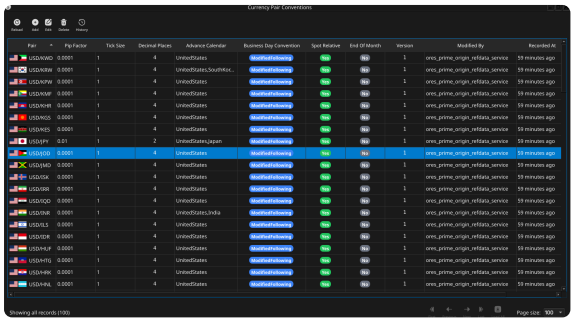


Figure 9.4: The Currency Pair Conventions window. USD/JPY (row eight) shows the JPY-cross convention — pip factor `0.01`, two decimal places — contrasting with the =0.0001=/four-decimal-place convention every other row here shares.

Each row shows the pair code with its composited flag icons, pip factor, tick size, decimal places, advance calendar, business day convention badge, and the spot-relative and end-of-month flags rendered as Yes/No labels, followed by the usual version and provenance columns. Double-clicking a row opens the **Currency Pair Convention Details** dialog.

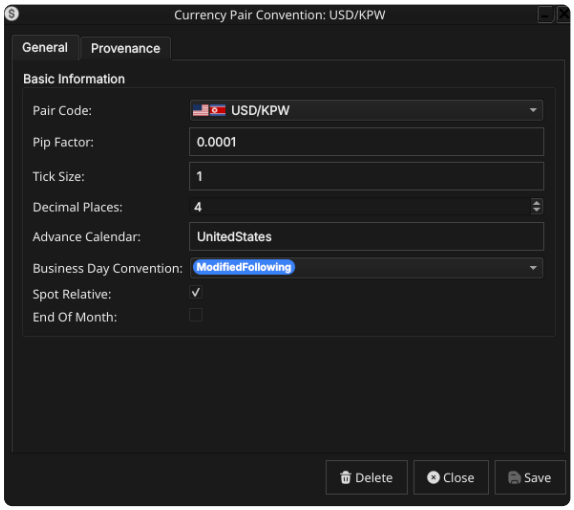


Figure 9.5: The Currency Pair Convention Details dialog for USD/KPW, showing the locked Pair Code combo with its flags, Pip Factor, Tick Size, Decimal Places, Advance Calendar, Business Day Convention, and the Spot Relative/End Of Month checkboxes.

Pair Code is a flagged combo, but unlike the Currency Pair Details

dialog it does not derive its value from two separate legs; instead it lets the user pick directly from the pairs already on file, and, as with Base and Quote Currency above, is selectable only while creating the record and locks, still showing its flags, once saved. Attempting to create a second convention record for a pair code that already has one is rejected, on the same reasoning as the duplicate- pair check on the identity entity: a pair has at most one convention record, and the correct action is to edit the existing one rather than create a second.

The remaining fields follow directly from the domain concepts set out earlier in the chapter. **Pip Factor** and **Tick Size** express the pip and tick mechanics described in Currency pairs and market conventions; a typical non-yen pair carries a pip factor of `0.0001`, a yen cross `0.01`. **Decimal Places** fixes how many digits the rate displays, consistent with the same pip convention — four for most pairs, two for yen crosses. **Advance Calendar** names the calendar used when advancing from the spot date to a forward date. **Business Day Convention** selects among the standard date-rolling rules — Following, Modified Following, Preceding, Modified Preceding, Unadjusted, Half-Month Modified Following, and Nearest — that determine how a date falling on a holiday is adjusted onto a business day. **Spot Relative** indicates whether forward dates for this pair are generated relative to the spot date rather than the trade date, and **End Of Month** indicates whether the end-of-month date-rolling convention applies.

Editing a currency pair convention

Editing follows the same pattern as every other reference-data entity in ORE Studio: open the record, change any field still open to editing once Pair Code has locked, and click **Save**. A change reason is required before the save is written, exactly as for currency pairs above.

Currency pair convention history

History and revert work identically to currency pair history: open the history icon or right-click and choose **History**, select a version, review the **Changes** tab, and use **Revert** to reinstate an earlier version as a new one.

Conclusion

The chapter set out to treat the currency pair as a relational entity distinct from, but built upon, the currencies documented in the previous chapter, and to separate that identity from the market conventions layered on top of it. The base-currency-precedence rule fixed how a pair's two legs are ordered and why only one of the two possible directions may ever be recorded; the major/minor/exotic/ commodity

taxonomy classified the population of pairs by liquidity and tradability, independent of the raw fact that both legs are valid currencies; and the pip, tick, and settlement-timing conventions supplied the vocabulary the second entity's fields draw upon. The Qt interface then walked both entities from list to detail to history, in the pattern the previous chapter established, and showed how the two windows cross-navigate via the Conventions toolbar button. Unlike currencies, currency pairs and their conventions do not yet have shell or command-line equivalents; that gap is tracked as future work rather than glossed over here.

See also

- Currency pairs — the domain-knowledge hub this chapter draws on.
- Currencies — the previous chapter, covering the individual currency legs a pair is built from.
- Currency Pair (Investopedia) — an accessible introduction to currency pair quoting conventions.

10

Countries

This chapter examines the *country* as a reference-data entity in OreStudio. Countries are identified by the international ISO 3166-1 standard; the chapter sets out that standard and the gaps it leaves, the extensions OreStudio layers on top, and the complete lifecycle of managing country records — viewing, editing, deleting, and auditing them — through the Qt interface, the interactive shell, and the command-line tool.

Overview

The chapter advances the argument that managing a country well means first understanding what identifies it, then the interface that governs its lifecycle, and finally the programmatic equivalents — and it proceeds in that order. It begins by establishing the identity of a country in ISO 3166-1 and its limitations, which sets out the codes that name a country and the gaps in the standard that motivate OreStudio's extensions; this is the premise the rest of the chapter builds on. From there it surveys the body of records in The Countries window, explaining the list view, paging, and reloading, before narrowing to a single record in Country Details and its General and Provenance tabs. It then turns to changing a record under audit — Editing, Deleting, and Country history — culminating in reverting to an earlier version, the point at which the bitemporal audit trail does its work. Finally it shows the same operations automated from the shell and the CLI, and the Conclusion draws these steps together.

ISO 3166-1 and its limitations

The international standard for country codes is ISO 3166-1. It defines three parallel codes for each country plus a short name:

- a two-letter alphabetic code, `alpha-2` (`US`, `GB`, `FR`) — the most widely used form, and OreStudio's primary identifier;

- a three-letter alphabetic code, `alpha-3` (`USA`, `GBR`, `FRA`), which is more mnemonic and collision-resistant;
- a three-digit `numeric` code (`840`, `826`, `250`), drawn from the UN M49 series and language-independent;
- the country's short name (e.g. *United States*).

ISO 3166-1 is near-universal, but it has gaps that matter in practice:

- It carries only a **short name**. The full official name (*United States of America*, *French Republic*) — needed for legal documents and reports — is not part of the standard.
- The country set is **political and changes over time**. Codes are added, withdrawn, and occasionally **reassigned**: a code freed by one country can later be given to another, so a bare code is not a stable historical key. ISO publishes transitional reservations, but consumers must still track the changes.
- It conflates **countries with territories and dependencies** and takes no position consistent with every user's expectations on contested or partially-recognised entities.
- It says nothing about **visual identity** — there is no flag or emblem in the standard.
- It does not model **subdivisions** (states, provinces); those are a separate standard, ISO 3166-2.
- A block of codes (`XA` to `XZ`) is reserved for **private use**, useful for fictional or internal entities that must not collide with real codes.

OreStudio uses the ISO 3166-1 fields as primary and extends the model with an **official name** and an optional **flag image**. The standard fields remain authoritative; the extensions are supplementary.

The Countries window

Open the Countries window from the *Reference Data* menu. It lists all countries defined in the tenant, one row per country, each row carrying the flag, alpha-2 and alpha-3 codes, numeric code, name, official name, version, last modifier, and when the record was last recorded.

The list is paginated; use the page controls at the bottom right to navigate the full set of countries. The toolbar buttons reload the list and open the add, edit, delete, and history actions.

Double-clicking a row opens the **Country Details** dialog.

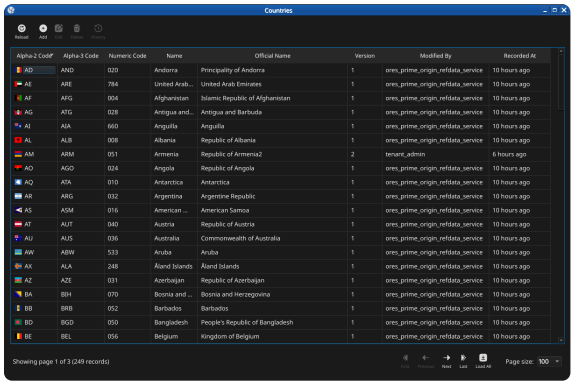


Figure 10.1: The Countries window, showing the full list of countries in the tenant. Each row shows the flag, alpha-2 and alpha-3 codes, numeric code, name, official name, version number, the identity of the last modifier, and when the record was last recorded. The status bar shows the current page and total record count.

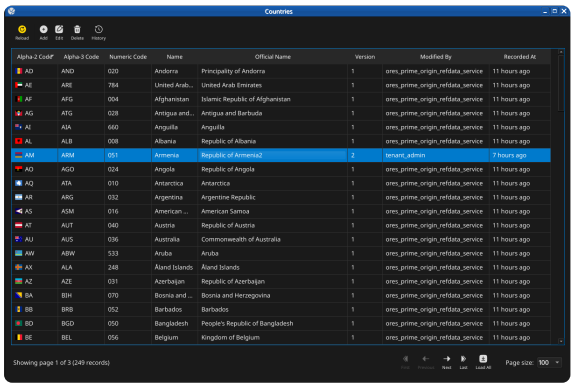


Figure 10.2: The toolbar Reload button refetches the current page from the server; recently-changed rows are briefly highlighted.

Country Details

The Country Details dialog has two tabs — **General** and **Provenance** — and an **Icon** group for the flag. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the country as a whole.

General

Figure 10.3: The General tab for a country. It shows the alpha-2 and alpha-3 ISO 3166-1 codes, the numeric code, the short name, the official name, and the country flag icon.

The General tab carries the core identity of the country:

- **Alpha-2 Code** — the two-letter ISO 3166-1 code. This is the primary key and cannot be changed after creation.

- **Alpha-3 Code** — the three-letter ISO 3166-1 code.
- **Numeric Code** — the ISO 3166-1 numeric code (e.g. `840` for the United States).
- **Name** — the country's short name (e.g. *United States*).
- **Official Name** — the full official name (e.g. *United States of America*), an OreStudio extension to the standard.
- **Icon** — the country flag, an OreStudio extension. Click the flag to choose a different image from the system's image dataset.

Provenance

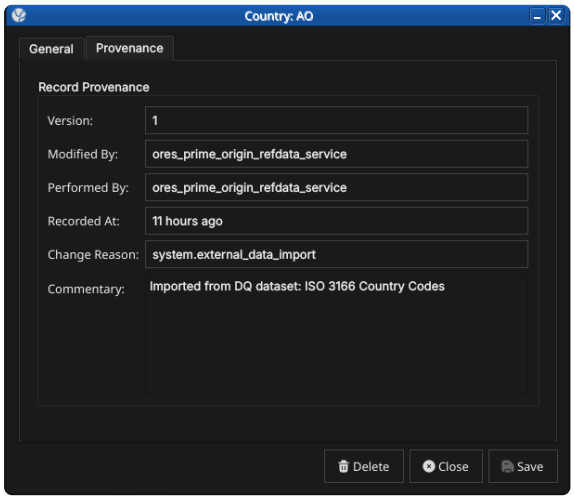


Figure 10.4: The Provenance tab showing record metadata: version number, the service that last modified the record, who performed the operation, when it was recorded, the change reason code, and a free-text commentary.

The Provenance tab is read-only and shows the audit metadata for the current version. See the Reference Data — Provenance section for a full description of the provenance fields common to all reference data entities.

Editing a country

To edit a country, open it in the Country Details dialog, make your changes, and click **Save**. Before the record is written, OreStudio prompts for a change reason — the same *Change Reason Required* dialog shown in the Currencies chapter, and not repeated here. Select a **Reason** from the drop-down and add optional **Commentary**, then confirm. Every save creates a new version; the previous version is never overwritten.

Deleting a country

To delete a country, open it and click **Delete**. OreStudio asks for confirmation before the record is closed off.

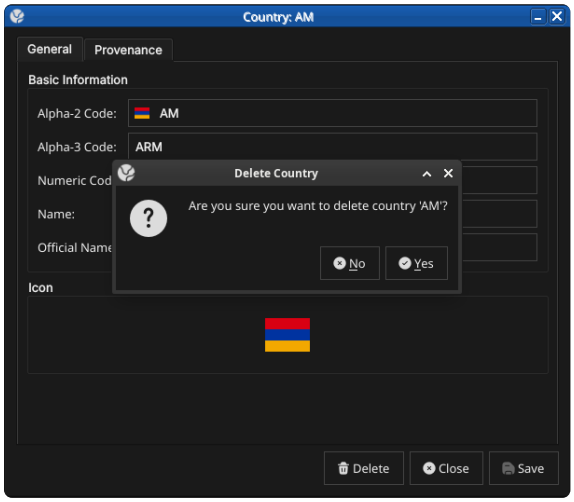


Figure 10.5: The confirmation prompt shown before a country is deleted. Deletion is a soft close — the record's history is preserved and remains visible in the History dialog.

Country history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

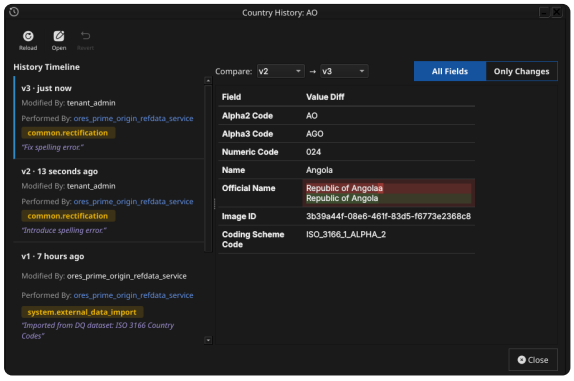


Figure 10.6: The History dialog for country `AO` (Angola), comparing version 2 against version 3. The **Only Changes** toggle narrows the field list to what actually differs between the two selected versions — here, a spelling correction to the Official Name.

Pick the two versions to compare from the **Compare** drop-downs. The **All Fields** / **Only Changes** toggle switches between showing every field and showing only the fields that differ between the two selected versions. The **Revert** button reinstates any historical version as a **new** version, preserving the full audit chain — the old version is never rewritten.

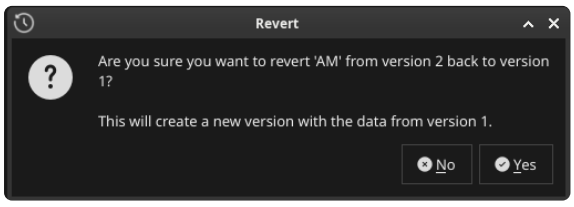


Figure 10.7: Choosing Revert on a past version asks for confirmation before staging the revert.

The revert opens the chosen version's values in an editable detail dialog; saving (with a change reason) writes them as the new current version.

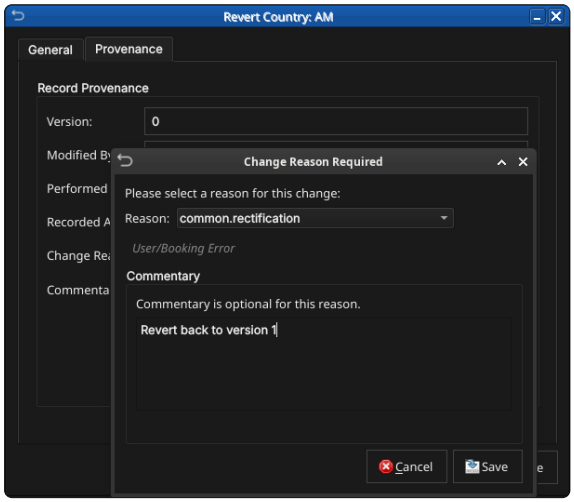


Figure 10.8: The reverted values open in the detail dialog ready to save as a new version, exactly like any other edit.

Shell commands

The ORE Studio interactive shell provides quick country access without opening the Qt UI:

```
# List all countries (paginated)
countries get

# Add a new country:
#   <alpha2> <alpha3> <numeric> <name> <official_name>
#   <change_reason_code> <change_commentary>
countries add XX XXX 999 "Test Country" "Test Country Official"
     system.new_record "manual example"

# Delete a country by alpha-2 code
countries delete XX

# Show history for a country
countries history GB
```

CLI commands

The `ores.cli` command-line tool provides a richer interface for automation. All country commands live under `ores.cli refdata countries`:

```
# List all countries as a table
ores.cli refdata countries list --tenant "$ORES_TENANT" --
    format table

# List a specific country as JSON
ores.cli refdata countries list --tenant "$ORES_TENANT" --
    format json --key US

# Add a new country
ores.cli refdata countries add \
  --tenant "$ORES_TENANT" \
  --alpha2-code XX --alpha3-code XXX --numeric-code 999 \
```

```
  --name "Test Country" --official-name "The Republic of Test
    Country" \
  --modified-by super_admin --change-reason-code system.
    new_record

# Delete a country by alpha-2 code
ores.cli refdata countries delete --tenant "$ORES_TENANT" --
    alpha2-code XX
```

Conclusion

The chapter set out to show that managing a country well follows from understanding its identity, its interface, and its programmatic equivalents in turn, and it has traced exactly that path. ISO 3166-1 supplies the identity — alpha-2, alpha-3, and numeric codes with a short name — while OreStudio's extensions add the official name and the flag, filling the gaps the standard leaves. Building on that, the Qt interface took the record from the list view down to a single detail dialog and through the editing, deletion, and history workflows, where reverting to an earlier version demonstrated the bitemporal audit trail keeping the full record intact rather than overwriting it. The shell and the CLI then reproduced the same operations for scripted use, closing the loop between interactive and automated management. Country is deliberately the same shape as the currency chapter, and the entity chapters that follow adopt the same progression.

See also

- ISO 3166 Country Codes — the international standard for country identification.
- UN M49 Standard Country Codes — the numeric-code series ISO 3166-1 draws on.
- Currencies — the template chapter this one mirrors.

11

Business Centres

This chapter examines the *business centre* as a reference-data entity in ORE Studio. Business centres are identified by the FpML `businessCenterScheme` code list; the chapter sets out that standard, the role a business centre plays in identifying a holiday calendar, and the complete lifecycle of managing business centre records — viewing, editing, deleting, and auditing them — through the Qt interface.

Overview

The chapter advances the argument that managing a business centre well means first understanding what it identifies, then the interface that governs its lifecycle. It begins by establishing the identity of a business centre in The FpML business centre scheme, which sets out the 4-character codes that name a business centre and the coding-scheme scoping that governs their uniqueness; this is the premise the rest of the chapter builds on. From there it surveys the body of records in The Business Centres window, explaining the list view, paging, and reloading, before narrowing to a single record in Business Centre Details and its General and Provenance tabs. It then turns to changing a record under audit — Editing, Deleting, and Business centre history — culminating in reverting to an earlier version, the point at which the bitemporal audit trail does its work. Finally the Conclusion draws these steps together and names what is deliberately still missing.

The FpML business centre scheme

A business centre is a 4-character code drawn from the FpML `businessCenterScheme` coding scheme[1]: either the real geographical location whose calendar governs a bad business day (`USNY` for New York, `GBLO` for London), or an FpML-format rate-publication calendar code used for fixing-day offsets. The scheme's own description is precise about the distinc-

[1] `http://www.fpml.org/coding-scheme/business-center`. A business centre's defining role in ORE Studio is exactly the one the FpML description names: it identifies **which** holiday calendar a business day-adjustment is performed against. The business centre itself is only the identifier — resolving a code to its actual set of holiday dates is a downstream concern the pricing/analytics layer carries out, not something the business centre record stores. Every

tion: the geographical codes are "implicitly locatable and used for identifying a bad business day for the purpose of payment and rate calculation day adjustments", while the rate-publication codes "are used in the context of the fixing day offsets" — two different jobs sharing the same 4-character code shape.

The Business Centres window

Open the Business Centres window from the *Reference Data* menu. It lists all business centres defined in the tenant, one row per centre, each row carrying the code, source, description, city, coding scheme, country (with its flag), version, last modifier, and when the record was last recorded.

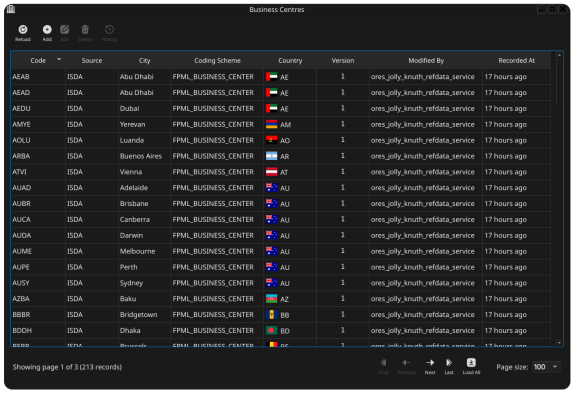


Figure 11.1: The Business Centres window, showing a page of business centres imported from the FpML reference data. Each row shows the code, source, city, coding scheme, country flag, version, last modifier, and when the record was last recorded; the status bar shows the current page and total record count.

The list is paginated; use the page controls at the bottom right to navigate the full set of business centres. The toolbar buttons reload the list and open the add, edit, delete, and history actions.

Double-clicking a row opens the **Business Centre Details** dialog.

Business Centre Details

The Business Centre Details dialog has two tabs — **General** and **Provenance**. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the business centre as a whole.

General

The General tab carries the core identity of the business centre:

- **Code** — the FpML business centre code (e.g. `USNY`). This is the primary key and cannot be changed after creation.
- **Source** — the data source identifier (e.g. *FpML*, *ISDA*, *Internal* for platform-seeded centres).
- **Description** — a short human-readable description of the centre.
- **City** — the city name, derived from the description for centres imported from the FpML reference data.

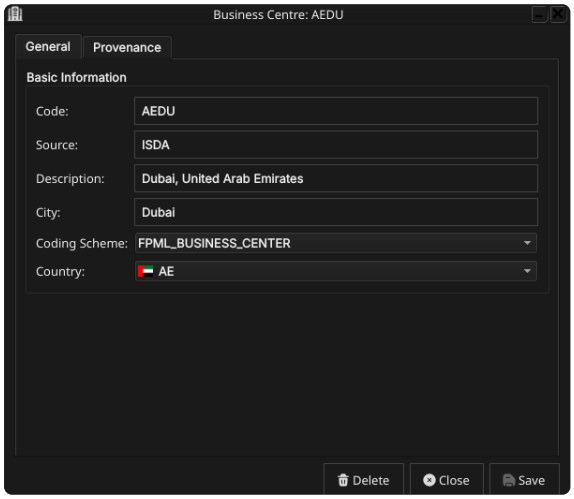


Figure 11.2: The General tab for business centre `AEDU` (Dubai), showing its code, source, description, city, the populated Coding Scheme combo, and the flagged Country combo.

- **Coding Scheme** — the coding scheme this centre's code belongs to, chosen from a combo populated with every coding scheme known to the system (e.g. `FPML_BUSINESS_CENTER`, `NONE`). A value is required — the server rejects a save with no coding scheme selected.
- **Country** — an optional link to a country, chosen from a flagged combo. Selecting a country renders its flag both in this combo and in the **Country** column of the list window.

Provenance

The Provenance tab is read-only and shows the audit metadata for the current version. See the Reference Data — Provenance section for a full description of the provenance fields common to all reference data entities.

Editing a business centre

To edit a business centre, open it in the Business Centre Details dialog, make your changes, and click **Save**. Before the record is written, ORE Studio prompts for a change reason — the same *Change Reason Required* dialog shown in the Currencies chapter, and not repeated here. Select a **Reason** from the drop-down and add optional **Commentary**, then confirm. Every save creates a new version; the previous version is never overwritten.

Deleting a business centre

To delete a business centre, open it and click **Delete**. ORE Studio asks for confirmation before the record is closed off.

Deletion is a soft close — the record's history is preserved and remains visible in the History dialog.

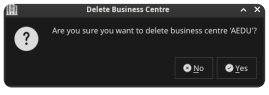


Figure 11.3: The confirmation prompt shown before business centre `AEDU` is deleted.

Business centre history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

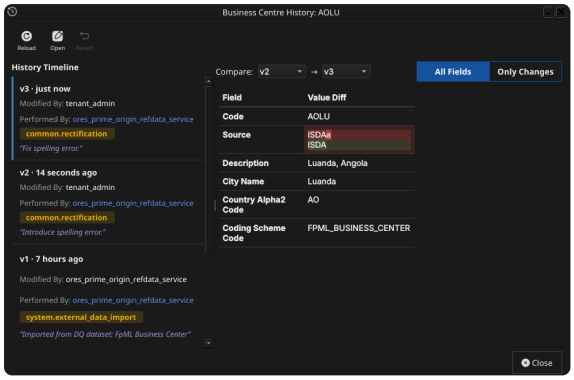


Figure 11.4: The History dialog for business centre `AOLU` (Luanda), comparing version 2 against version 3. The Only Changes toggle narrows the field list to what actually differs — here, a spelling correction to the Source field.

Pick the two versions to compare from the **Compare** drop-downs. The **All Fields** / **Only Changes** toggle switches between showing every field and showing only the fields that differ between the two selected versions. The **Revert** button reinstates any historical version as a **new** version, preserving the full audit chain — the old version is never rewritten. Reverting a business centre follows exactly the same confirm-then-save-with-a-reason flow described for Countries; it is not repeated here.

Conclusion

The chapter set out to show that managing a business centre well follows from understanding what it identifies before turning to its interface, and it has traced exactly that path. The FpML `businessCenterScheme` supplies the identity — a 4-character code naming either a real geographical calendar location or a rate-publication calendar, scoped to a coding scheme but unique per tenant regardless of which scheme it came from — while ORE Studio's own country link adds a purely cosmetic flag icon on top. Building on that, the Qt interface took the record from the list view down to a single detail dialog and through the editing, deletion, and history workflows, where reverting to an earlier version demonstrated the same bitemporal audit trail already familiar from the Currencies and Countries chapters. Business centre does not yet have shell or command-line equivalents, nor Wt web UI or HTTP REST API support; those gaps are tracked as backlog work rather than glossed over here.

See also

- Business Centre — the domain-knowledge hub this chapter draws on.
- Currencies — the template chapter this one mirrors.

- Countries — the country a business centre may optionally link to for its flag icon.
- Books — a book's rates centre is a business centre, referenced by code.

12

Portfolios

This chapter examines the *portfolio* as a reference-data entity in ORE Studio. A portfolio is the logical grouping node that organises books for risk management; the chapter sets out what a portfolio represents alongside the book it sits next to, the interface that governs its lifecycle, and the meaning of its fields — a self-referencing hierarchy, a purpose classification, and an operational status.

Overview

The chapter advances the argument that managing a portfolio well means first understanding what it represents relative to the book it sits alongside, then the interface that governs it. It begins by establishing Portfolios and books — the core distinction between the two, and the hierarchy portfolios alone carry — before turning to What is a Portfolio? and the fields a portfolio record actually carries. From there it surveys the body of records in The Portfolios window before narrowing to a single record in Portfolio Details and its General and Provenance tabs. It then turns to changing a record under audit — Editing, Deleting, and Portfolio history — culminating in reverting to an earlier version, the point at which the bitemporal audit trail does its work. Before concluding, it turns to the lookup entity backing the portfolio's own purpose classification field in Purpose Types. The Conclusion draws these steps together.

Portfolios and books

A portfolio is a trading concept, not an accounting one: it is an arbitrary hierarchy that traders and desks shape however makes sense for risk management, and it is never reflected into the general ledger. This sets it apart from a book, which must map onto the firm's ledger and is created by Finance rather than by the trading system itself. A useful shorthand for the pair is that books are the *physical* representa-

tion of trading activity and portfolios the *logical* one — or, borrowing a filesystem analogy, a portfolio is a folder and a book is a file. A portfolio holds books and other portfolios; it never holds deals directly, just as a folder holds files and other folders but is not itself a file.

This split matters because of where hierarchy lives. In the real-world accounting sense a book can have child books through ledger setup, but ORE Studio deliberately does not model that book-to-book hierarchy — the book record has no parent-book field, and there is no plan to add one. Instead, every layer of grouping that the ledger's own structure would otherwise express — a global root grouping, retired branches, sub-groupings by desk or region — is expressed once, through the portfolio tree. A portfolio carries a self-referencing parent, so portfolios can nest inside other portfolios to any depth, and every book links to exactly one portfolio. Duplicating that structure on the book side would add a second tree to keep in sync for no benefit, since the trading system's only job with respect to books is to hold a view of the ledger, not to manage the ledger's own hierarchy.

Desks group books into portfolios for reasons such as instrument type, valuation complexity, trader seniority, or region — a Head of Desk's portfolio, for instance, can be the superset of all their traders' individual portfolios, nested arbitrarily deep. With the relationship between the two entities established, the rest of this chapter turns from that relationship to the portfolio record itself — what fields it carries and the interface that governs it.

What is a Portfolio?

A *portfolio* is a named grouping node within a party's organisational structure, identified by a system-generated id and a human-readable **Name** — e.g. "Global Rates" or "APAC Credit". Every portfolio belongs to exactly one party, and, per Portfolios and books above, may optionally have a parent portfolio, letting portfolios nest to represent a trading organisation's actual reporting lines. The parent relationship is not yet exposed as an editable field in the Portfolio Details dialog described below — a portfolio's place in the hierarchy is set at creation and read through the record's data rather than edited from a combo box, the same gap the Book chapter's account of `owner_unit_id` describes for the equivalent business-unit field on a book.

A portfolio carries a **purpose type** — its classification by intent, chosen from a fixed set of values: *Risk* (the default, for day-to-day risk-management groupings), *Regulatory* (portfolios kept for regulatory-reporting purposes), *Client Reporting*, and *Internal*. It also carries an optional **aggregation currency** — the currency P&L and risk figures roll up into at this node — and an optional free-text **description**. A **Virtual** flag distinguishes a node that exists purely for on-demand reporting, not persisted into trade attribution, from an ordinary port-

folio node; and, like a book, a portfolio carries a **status** reflecting its position in an operational lifecycle — *Active*, *Inactive*, *Closed*, *Frozen*, or *Pending*. Unlike a book's status, whose real-world multi-party opening and closing workflow the Book chapter documents in detail, portfolio status is a simpler field with no equivalent formal workflow documented yet — it is set and changed directly on the record.

With the record's fields established, the rest of this chapter turns from what a portfolio contains to where you go to work with one.

The Portfolios window

Open the Portfolios window from the *Reference Data* menu. It lists every portfolio defined in the tenant, with one row per portfolio.

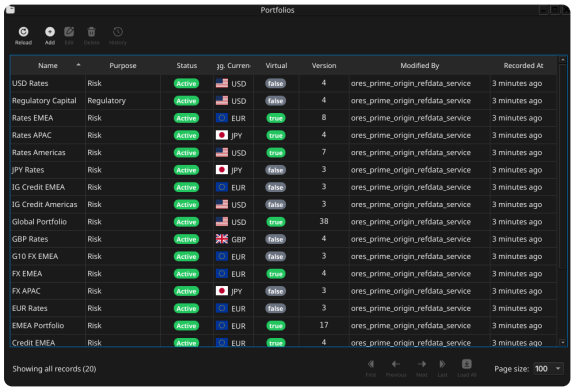


Figure 12.1: The Portfolios window showing the full list of portfolios in the tenant.

Each row shows the name, purpose type, status (as a colour-coded badge), aggregation currency (with flag icon), the virtual flag (as a colour-coded badge), version number, the identity of the last modifier, and when the record was last recorded. The toolbar buttons reload the list and open the add, edit, delete, and history actions. The list is paginated; use the page controls at the bottom right to navigate. Double-clicking a row opens the **Portfolio Details** dialog.

Portfolio Details

The Portfolio Details dialog has two tabs — **General** and **Provenance**. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the portfolio as a whole.

General

The General tab carries the fields of the portfolio:

- **Id** — the portfolio's identifier. This is the primary key and cannot be changed after creation.
- **Name** — the portfolio's display name.

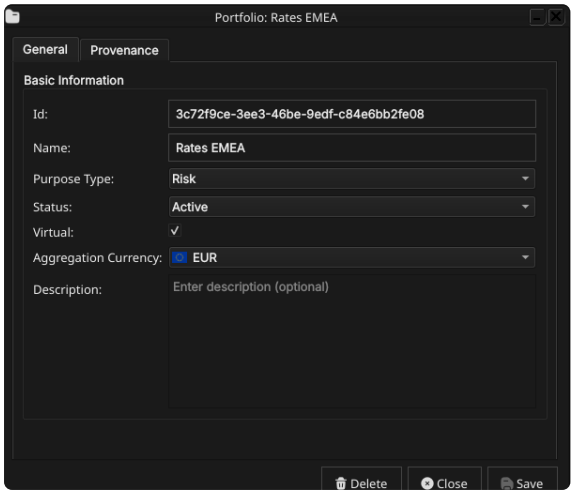


Figure 12.2: Portfolio Details — General tab, showing the id, name, purpose type, status, aggregation currency, virtual flag, and description fields.

- **Purpose Type** — the portfolio's classification by intent, chosen from a combo box populated with the tenant's configured purpose types and shown as plain text once selected. See What is a Portfolio? above for what each value means.

- **Status** — the portfolio's position in its operational lifecycle, chosen from a fixed combo box (Active, Inactive, Closed, Frozen, Pending) and shown as a colour-coded badge once selected.

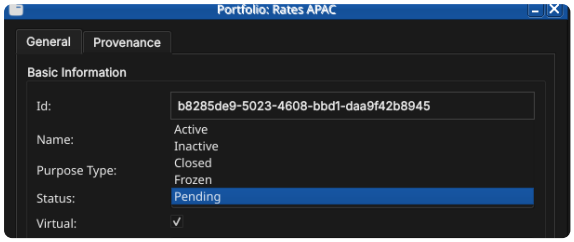


Figure 12.3: The Status combo box open, showing all five status values.

- **Aggregation Currency** — the currency P&L and risk figures roll up into at this node, chosen from a combo box populated with ISO currency codes, each shown with its flag icon; c.f. the same combo widget in the Currencies chapter.

- **Virtual** — a checkbox marking a node that exists purely for on-demand reporting, shown as a colour-coded badge in the list view.

- **Description** — free-text, optional.

Provenance

The Provenance tab is read-only and shows the audit metadata for the current version. See the Reference Data — Provenance section for a full description of the provenance fields common to all reference data entities.

Editing a portfolio

To edit a portfolio, open it in the Portfolio Details dialog, make your changes, and click **Save**. Before the record is written, OreStudio prompts for a change reason — the same *Change Reason Required* dialog shown in the Currencies chapter, and not repeated here. Select a **Reason** from the drop-down and add optional **Commentary**, then confirm. Every save creates a new version; the previous version is never overwritten.

Deleting a portfolio

To delete a portfolio, open it and click **Delete**. OreStudio asks for confirmation before the record is closed off. Deletion is a soft close — the record's history is preserved and remains visible in the History dialog.

Portfolio history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

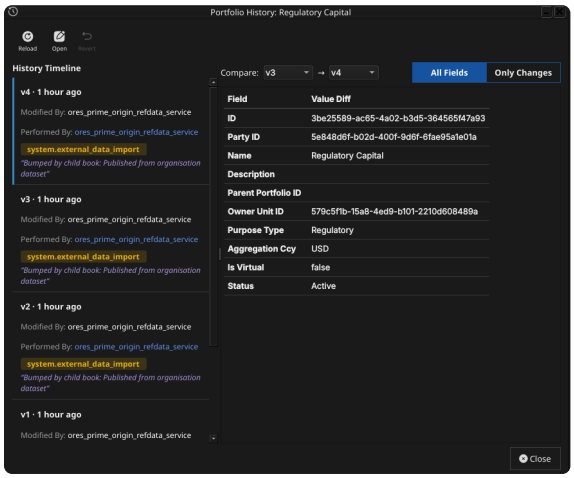


Figure 12.4: The Portfolio History dialog, comparing two versions of a portfolio.

Pick the two versions to compare from the **Compare** drop-downs. The **All Fields** / **Only Changes** toggle switches between showing every field and showing only the fields that differ between the two selected versions. The **Revert** button reinstates any historical version as a **new** version, preserving the full audit chain — the old version is never rewritten.

Purpose Types

Where the portfolio record's **Purpose Type** field (see What is a Portfolio? above) captures a single classification chosen from a combo box, a second, separate window manages the set of values that combo

box offers. Purpose types are reference data in their own right — a system-tenant-managed lookup entity, seeded on provisioning with four values (*Risk*, *Regulatory*, *Client Reporting*, *Internal*) but editable and extensible like any other reference-data record — kept as a distinct entity for the same reason the Currency Pairs chapter kept Currency Pair Conventions separate from the Currency Pair record itself: the set of valid classifications changes at a different rate, and for different reasons, than any individual portfolio that references one.

Open the Purpose Types window from the *Reference Data* menu.

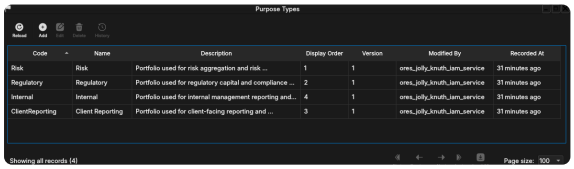


Figure 12.5: The Purpose Types window, listing the four seeded values (Risk, Regulatory, Client Reporting, Internal) with their display order.

Each row shows the code, name, description, display order, version, and the usual provenance columns. Double-clicking a row opens the **Purpose Type Details** dialog.

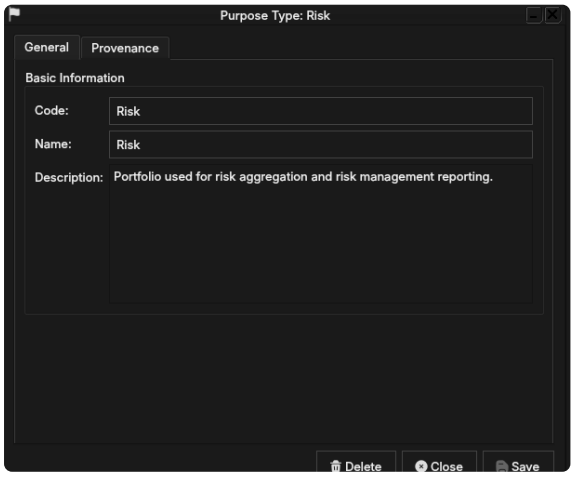


Figure 12.6: The Purpose Type Details dialog for the Risk purpose type, showing its code, name, and description.

Code is the purpose type's unique identifier and, like a currency's ISO code, locks once the record is created. **Name** is the human-readable label shown in the Portfolio Details' Purpose Type combo. **Description** is free-text, explaining what the classification means in practice — the four seeded values' descriptions are the source for the meanings given in What is a Portfolio? above. **Display Order** is an integer controlling the combo box's ordering, lowest first — shown in the list window's Display Order column; the Details dialog does not yet expose it (tracked in Audit and standardize the display_order field across lookup/code-table entities).

Editing, deleting, and history follow the same pattern as every other reference-data entity in ORE Studio and are not repeated here in full: open the record, change a field, and **Save**, confirming a change reason as described for currencies in the Currencies chapter; deletion is a soft close preserving history; and the history icon opens the same Compare/Revert workflow described in Portfolio history

above. Purpose types have no shell or command-line access — a gap shared by every reference-data entity in ORE Studio, not specific to purpose types, and tracked separately as part of ongoing commissioning work.

Conclusion

The chapter set out to show that managing a portfolio well follows from understanding what it represents relative to a book and the interface that governs its lifecycle, and it has traced exactly that path. A portfolio is the logical grouping node in a party's organisational structure — a folder to a book's file — identified by an id and name, optionally nested under a parent portfolio, and carrying a purpose type, an aggregation currency, a virtual flag, and a status. Building on that, the Qt interface took the record from the list view down to a single detail dialog — its currency picker showing flags, its status and virtual flag shown as badges — and through the editing, deleting, and history workflows, where reverting to an earlier version demonstrated the same bitemporal audit trail described in the Reference Data chapter. The chapter then turned to the purpose type lookup entity backing the portfolio's own classification field — a small reference-data record with the same list/detail/ history interface, kept separate for the same reasons a currency pair convention is kept separate from a currency pair. Shell and command-line access to portfolios and purpose types, and an editable parent-portfolio field in the Details dialog, are not yet implemented; all are tracked separately as part of ongoing commissioning work.

See also

- Reference Data — the shared data quality and audit framework this chapter builds on.
- Currencies — the aggregation currency a portfolio references, and the Change Reason Required dialog.
- Book — the entity a portfolio groups, and the fuller account of a lifecycle workflow's real-world weight.

13

Books

This chapter examines the *book* as a reference-data entity in OreStudio. A book is the internal ledger unit that owns positions; the chapter sets out what a book represents, the interface that governs its lifecycle, and the meaning of its fields — a flag-icon currency picker, an operational status with its own lifecycle, and its trading-book/banking-book classification.

Overview

The chapter advances the argument that managing a book well means first understanding what it represents, then the interface that governs its lifecycle — and it proceeds in that order. It begins by establishing what a book is in What is a Book?, distinguishing a book from the counterparties and portfolios it sits alongside, and by working through the real-world Book lifecycle a book moves through — from its multi-party opening to closure — that the Status field on-screen only partially reflects; this is the premise the rest of the chapter builds on. From there it surveys the body of records in The Books window before narrowing to a single record in Book Details and its General and Provenance tabs. It then turns to changing a record under audit — Editing, Deleting, and Book history — culminating in reverting to an earlier version, the point at which the bitemporal audit trail does its work. The Conclusion draws these steps together.

What is a Book?

A *book* is the internal ledger unit that owns positions within a party's trading organisation. Every trade books into exactly one book; a book aggregates the trades it owns for risk, P&L, and reporting purposes. A book is, first and foremost, an accounting concept rather than a purely trading-system one: it must reconcile with the firm's general ledger, and OreStudio's own view of a book is downstream

of that ledger rather than the other way round.

Books sit within a party's organisational structure alongside business units and portfolios, and it helps to keep the two apart: a **portfolio** groups related books for a trading strategy or desk, and can itself be composed of other portfolios; a **book** is the leaf-level unit that actually carries positions. A useful shorthand is that books are the physical representation and portfolios the logical one — or, borrowing a filesystem analogy, a portfolio is a folder and a book is a file. In the real-world accounting sense a book can have child books through ledger setup, but ORE Studio deliberately does not model that book-to-book hierarchy: a book has no parent-book field. Instead, all hierarchy in ORE Studio is expressed through the portfolio tree — portfolios can nest inside other portfolios, and every book links to exactly one portfolio — so the folder/file split above is also the whole of the tool's hierarchy, not just an analogy for it.

Every book carries a **functional currency** — the accounting currency its positions are reported in. Trades themselves can happen in whatever currency the deal was struck in, but the ledger needs one settled currency per book to carry balances and P&L in; that currency is designated by the ledger, not chosen freely by the desk. A book also carries a **GL account reference**, which is simply the general ledger account in the firm's accounting system that this book's activity maps to — the trading system's record of a position only means something once it can be traced back to a specific line in the firm's books of account. Alongside that sits a **cost centre**, the internal code that activity in this book is charged against for cost allocation and management reporting; a full treatment of how functional currency, GL accounts, and cost centres compose into the wider accounting hierarchy belongs to a future ledger-focused chapter, not this one.

A book also carries a **status** — its current operational state, such as active, closed, or frozen — chosen from the set of statuses configured for the tenant. See Book lifecycle below for what drives a book between these states.

Perhaps the most consequential classification a book carries is whether it is a **trading book** or a **banking book** — a distinction that traces back to the Basel III/IV capital adequacy framework, and specifically to a component of it called FRTB (the Fundamental Review of the Trading Book).

The distinction is about *intent*, not instrument type: a position held with the intent of benefiting from short-term price movements, market making, or hedging other trading-book positions belongs in the trading book; a position held to maturity or for longer-term investment purposes belongs in the banking book. The same instrument — a bond, a swap — can sit in either book depending on why it is held.

The distinction matters because the two books are subject to entirely different capital regimes. Trading-book positions are capitalised for **market risk** — the risk that market prices move against an open position — recognising that these positions may be actively

traded and revalued daily. Banking-book positions are instead capitalised for **credit risk** — the risk that a counterparty fails to pay — reflecting that these positions are typically held to term. Getting this classification wrong, or moving positions between the two books opportunistically to reduce capital charges, is precisely what regulators police most closely; switching a position's classification after the fact is heavily restricted for that reason.

Book lifecycle

A book moves through a much heavier real-world process than a simple open/closed flag suggests: a deliberate opening workflow, a live phase in which deals move and access is periodically reviewed, and a closing process gated on the book being flat. This section walks through that process in its own terms; c.f. the **Status** field on the General tab below, which is OreStudio's simplified, editable reflection of it.

Opening a book

A new book is not something a desk sets up for itself. Bringing a book live is a deliberately heavyweight, multi-party process, with four separate functions involved:

- The **desk** initiates the request — it knows it needs a new book (a new product line, a new trading strategy) but cannot create or enrich one itself.
- A **controller** reviews and approves the initiation request — an independent check that exists specifically to prevent a desk from authorising its own new book.
- Three functions then **enrich** the book in parallel, each contributing the data their function owns: **Finance** wires the book into the ledger — its functional currency, cost centre, and monthly balance carry-forward; **Market Risk** sets its regulatory book type (Trading or Banking) and assigns it to a rates centre for consistent revaluation data; **Operations** sets up the book's allowed-currency and allowed-product constraints. Because all three enrich the book at the same time rather than in sequence, each must independently approve its own contribution before the book can move on.
- Only once every approval is in does the book's access review begin; once that completes, the book is set up on the booking systems and becomes **Active**.

While a book is active

A live book is not static:

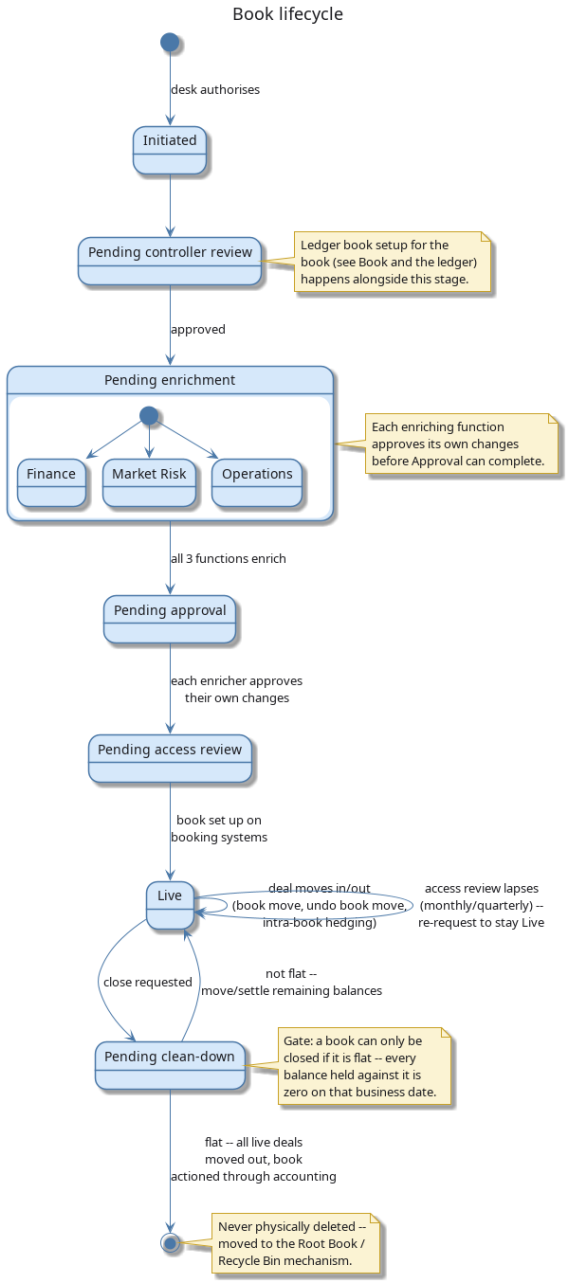


Figure 13.1: The book lifecycle: the multi-party opening workflow, the live-book loop (deal moves, periodic access review), and the flat-check gate on closing.

- Deals move between books routinely — freely if the move stays within the same legal entity and branch, or via a close-and-rebook if not. An erroneous move can be undone.

- Access to a book is not permanent: it must be periodically reviewed (monthly or quarterly, depending on the tenant's policy). If nobody renews a user's access, it lapses automatically and must be re-requested.

- A book can be temporarily **Frozen** — taken out of normal trading use without being closed down entirely.

Closing a book

A book can only be closed once it is **flat** — every balance held against it is zero on that business date. Closure is executed through accounting: all remaining live deals are moved to a different book, after which the book's status moves to **Closed**. A closed book is never physically deleted; it is retained in the system for its audit history, exactly like any other historical version.

With the concept and its lifecycle established, the rest of this chapter turns to the interface OreStudio provides for it — starting with the Books window, which lists every book governed by the process just described.

The Books window

Open the Books window from the *Trading* menu. It lists all books defined in the tenant, with one row per book.

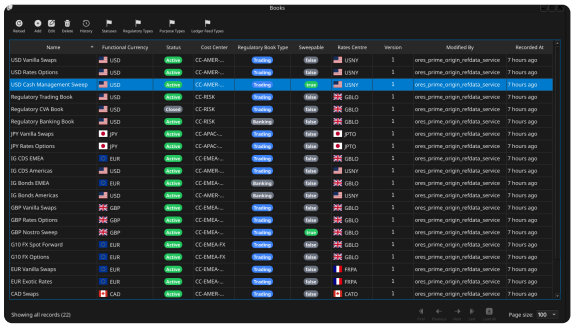


Figure 13.2: The Books window, showing the full list of books in the tenant. Each row shows the name, functional currency (with flag icon), status (as a colour-coded badge), cost centre, regulatory book type (as a colour-coded Trading/Banking badge), version number, the identity of the last modifier, and when the record was last recorded. The status bar shows the current page and total record count.

The toolbar buttons reload the list and open the add, edit, delete, and history actions. The list is paginated; use the page controls at the bottom right to navigate. Double-clicking a row opens the **Book Details** dialog.

Book Details

The Book Details dialog has two tabs — **General** and **Provenance**. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the book as a whole.

General

The General tab carries the fields of the book:

- **Id** — the book's identifier. This is the primary key and cannot be changed after creation.
- **Name** — the book's display name. Unlike the id, the name can be changed after creation — real trading systems allow a book to be renamed while its id and GL mapping remain the durable identity.

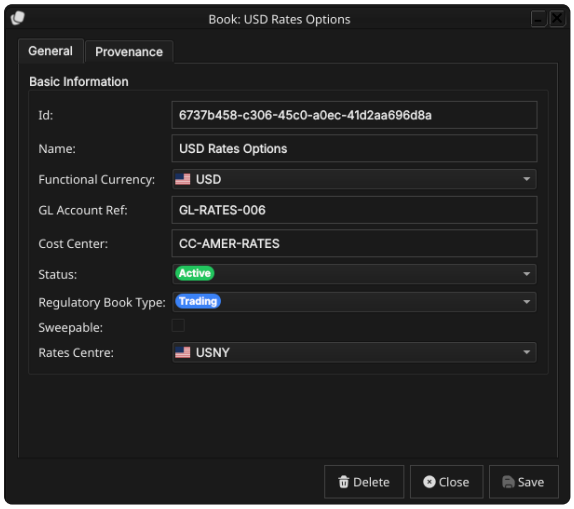


Figure 13.3: The General tab for a book. It shows the id, name, functional currency, GL account reference, cost centre, status, and regulatory book type.

- **Functional Currency** — the currency this book reports in, designated by the ledger, chosen from a combo box populated with ISO currency codes, each shown with its flag icon.

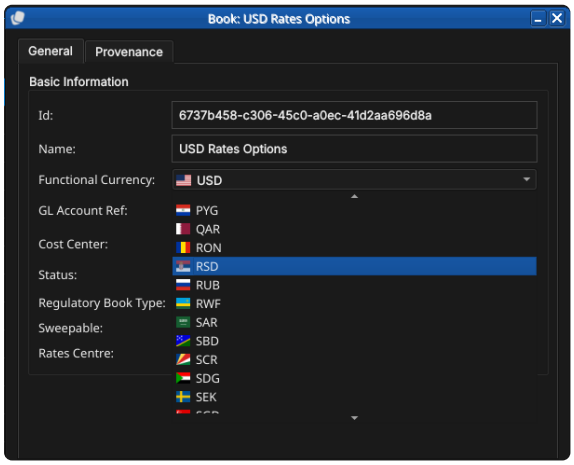


Figure 13.4: The Functional Currency combo box open, showing the list of ISO currency codes each with its flag icon; c.f. the same combo widget in the Currencies chapter. Selecting a currency updates the closed box to show the chosen code and flag.

- **GL Account Ref** — free-text reference to the general ledger account this book maps to.

- **Cost Center** — free-text internal cost allocation code.

- **Status** — the book's position in its operational lifecycle, chosen from a combo box populated from the tenant's configured book statuses and shown as a colour-coded badge once selected. See Book lifecycle above for what each value means.

- **Regulatory Book Type** — a combo box choosing the book's Basel classification, *Trading* or *Banking*, also shown as a colour-coded badge. See Book lifecycle and What is a Book? above for what the distinction means.

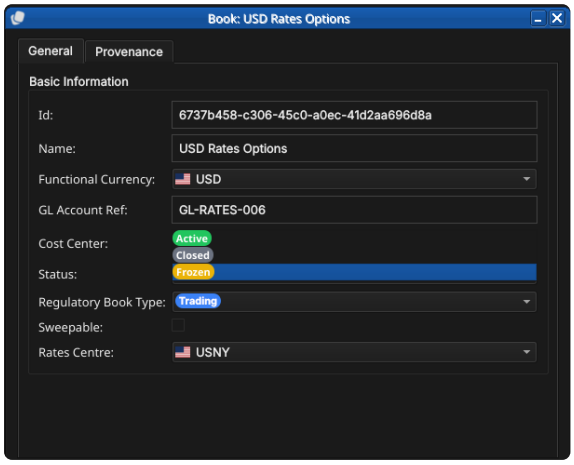


Figure 13.5: The Status combo box open, showing the badge-coloured status values.

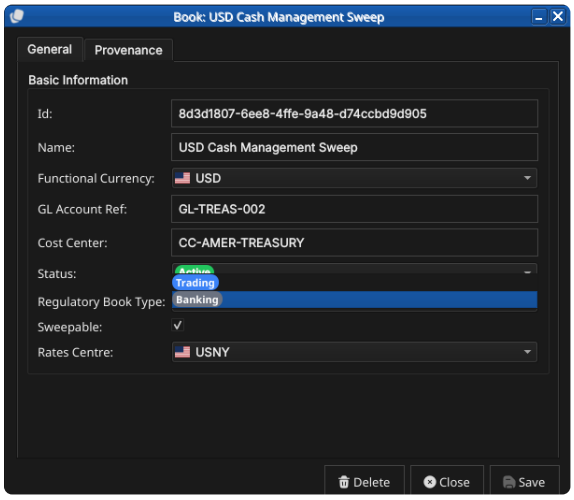


Figure 13.6: The Regulatory Book Type combo box open, showing both values (Trading, Banking) as coloured badges.

Provenance

The Provenance tab is read-only and shows the audit metadata for the current version. See the Reference Data — Provenance section for a full description of the provenance fields common to all reference data entities.

Editing a book

To edit a book, open it in the Book Details dialog, make your changes, and click **Save**. Before the record is written, OreStudio prompts for a change reason — the same *Change Reason Required* dialog shown in the Currencies chapter, and not repeated here. Select a **Reason** from the drop-down and add optional **Commentary**, then confirm. Every save creates a new version; the previous version is never overwritten.

Deleting a book

To delete a book, open it and click **Delete**. OreStudio asks for confirmation before the record is closed off. Deletion is a soft close —

the record's history is preserved and remains visible in the History dialog.

Book history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

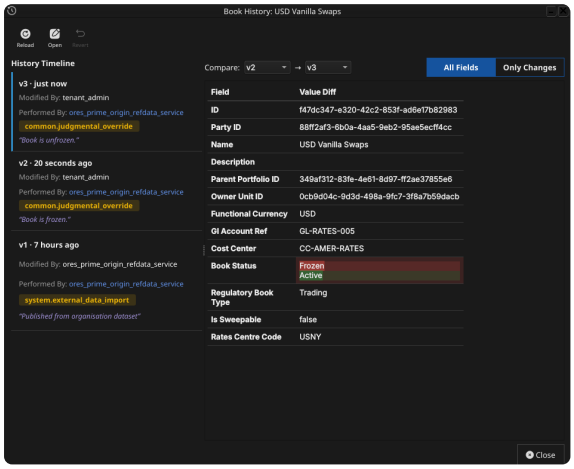


Figure 13.7: The History dialog for book USD Vanilla Swaps, comparing version 2 against version 3. The Only Changes toggle narrows the field list to the single field that actually differs — Book Status, Frozen to Active.

Pick the two versions to compare from the **Compare** drop-downs. The **All Fields** / **Only Changes** toggle switches between showing every field and showing only the fields that differ between the two selected versions. The **Revert** button reinstates any historical version as a **new** version, preserving the full audit chain — the old version is never rewritten.

Conclusion

The chapter set out to show that managing a book well follows from understanding what it represents and the interface that governs its lifecycle, and it has traced exactly that path. A book is the leaf-level ledger unit that owns positions, identified within its party by an id and name and carrying a functional currency, a GL mapping, a cost centre, a status, and a trading/non-trading flag. Building on that, the Qt interface took the record from the list view down to a single detail dialog — its currency picker showing flags, its status governed by the book lifecycle — and through the editing, deleting, and history workflows, where reverting to an earlier version demonstrated the same bitemporal audit trail described in the Reference Data chapter. Shell and command-line access to books is not yet implemented; it is tracked separately as part of the shell and CLI commissioning work.

See also

- Reference Data — the shared data quality and audit framework this chapter builds on.
- Currencies — the functional currency a book references, and the Change Reason Required dialog.

14

Parties

This chapter examines the *party* as a reference-data entity in ORE Studio. A party is an internal legal entity — the organisation itself and each of its subsidiaries — and, barring the tenant's own house root, every party sits somewhere beneath another in a single corporate hierarchy tree; the chapter sets out that hierarchy rule, the identifiers and contact information a party carries, and the complete lifecycle of managing party records through the Qt interface.

Overview

The chapter advances the argument that managing a party well means first understanding what it represents and how it fits into the organisation's structure, then the interface that governs its lifecycle. It begins by establishing what a party is in What is a Party?, then shows that a party's type and status are themselves configurable lookups with their own windows in Party types and statuses. From there it surveys the body of records in The Parties window before narrowing to a single record in Party Details and its five tabs — General, Identifiers (where the ten schemes a party can carry turn out to be a configurable lookup in their own right), Contact Information, Hierarchy (where the house-root rule introduced early in the chapter finally gets its screen), and Provenance. It then turns to changing a record under audit — Editing, Deleting, and Party history — culminating in reverting to an earlier version, the point at which the bitemporal audit trail does its work. The Conclusion draws these steps together and names what is deliberately still missing.

What is a Party?

A *party* is an internal legal entity participating in financial transactions — the organisation's own top-level entity and each of its

subsidiaries, branches, and other group-structure members. See the Tenants chapter's Parties section for the conceptual model this chapter builds on: the house/counterparty distinction, the administrative system party, and how a user's position in the party hierarchy determines their data visibility. This chapter does not repeat that model; it picks up from it to document the Qt interface for managing individual party records.

A party is, first and foremost, a stand-in for a real legal entity — a registered company, a bank, a branch — and real legal entities are notoriously hard to identify consistently: the same organisation can appear under different names, registration numbers, and codes across different jurisdictions, counterparties, and vendor systems, with no single authority reconciling them all.

Because no single identifier scheme covers every counterpart a party might need to be recognised by, a party can carry more than one — see Identifiers below for the full set of schemes OreStudio recognises, including the LEI and BIC schemes GLEIF publishes.

Every party carries a **full name** (its official registered legal name) and a **short code** (a brief mnemonic used elsewhere in the system for quick reference), both of which participate in the party's natural key alongside its surrogate id. A party also carries a **codename** — a globally unique, immutable, human-readable identifier (adjective_noun, e.g. `swift_falcon`) auto-generated on creation and used internally as the party's per-party message-queue prefix and scheduled-job namespace; it plays no role in the Qt UI and is not user-editable.

A party's **type** classifies what kind of entity it is (e.g. *Corporate*), chosen from the tenant's configured party types, and its **status** tracks where it sits in its own operational lifecycle (e.g. *Active*), chosen from the tenant's configured party statuses — both shown as colour-coded badges once selected. A party also carries a **business center** — an FpML business center code identifying the entity's primary location, the same code list documented in the Business Centres chapter.

Party types and statuses

A party's **type** and **status** are not fixed enumerations built into OreStudio; they are themselves tenant-configured reference-data lookups, each with its own list and detail window under the *Reference Data* menu, alongside the Party Types and Party Statuses entries. Both follow the same small shape — a **Code** (the value stored against the party), a **Name** (the label shown in the combo and, once selected, on the colour-coded badge), an optional **Description**, and a **Display Order** controlling where the value falls in the combo's drop-down list — plus the standard Provenance tab described in Provenance below.

Editing, deleting, and history for party type and party status records follow the same pattern described later in this chapter for the party record itself — a change reason on save, a soft close on delete, and

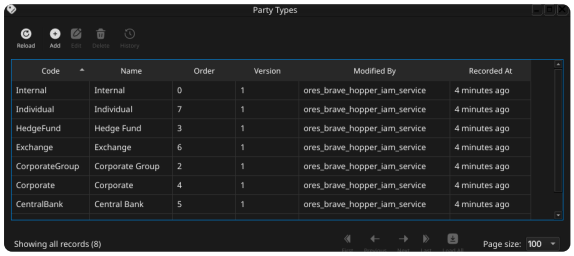


Figure 14.1: The Party Types window, listing the tenant's configured party types by code, name, order, version, last modifier, and when the record was last recorded.

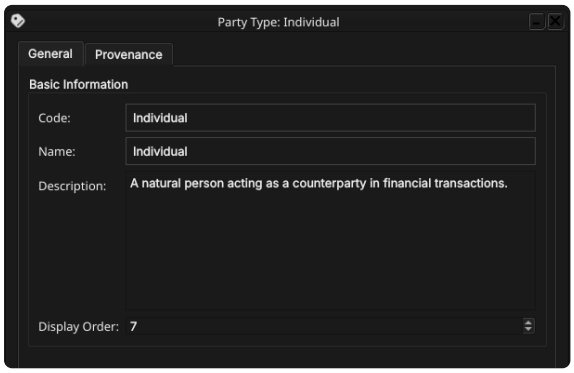


Figure 14.2: The Party Type Details dialog, showing its General tab (Code, Name, Description, Display Order) and Provenance tab.

a full version history with revert. Because both lookups are shared across every party (and, for status, every counterparty too — see the Counterparties chapter), changing a code or renaming a value here is immediately visible wherever that type or status is displayed.

With type and status established as configurable lookups in their own right, the rest of this chapter turns to the interface OreStudio provides for the party record itself — starting with the Parties window, which lists every party in the tenant.

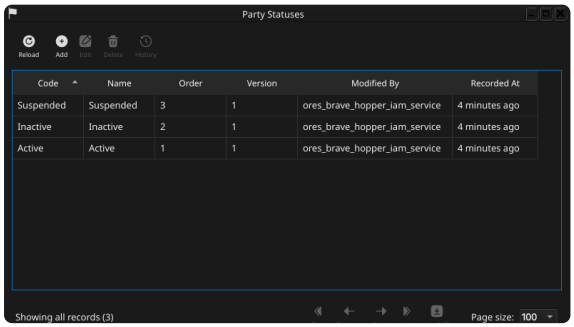


Figure 14.3: The Party Statuses window, listing the tenant's configured party statuses by code, name, order, version, last modifier, and when the record was last recorded.

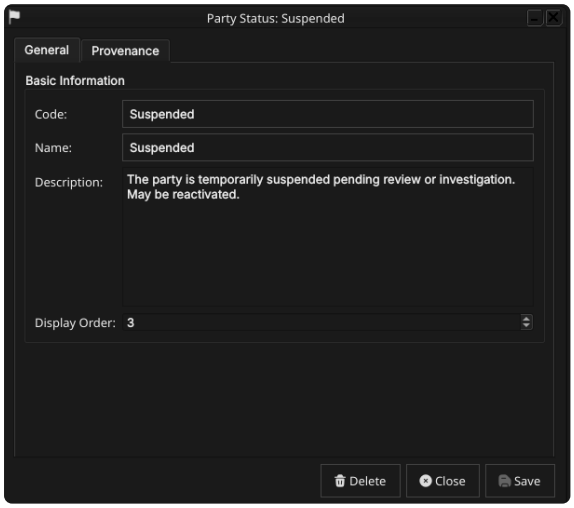


Figure 14.4: The Party Status Details dialog, showing its General tab (Code, Name, Description, Display Order) and Provenance tab.

The Parties window

Open the Parties window from the *Reference Data* menu. It lists all parties defined in the tenant, one row per party, each row carrying the short code, full name, party type, status, business center, version, last modifier, and when the record was last recorded.

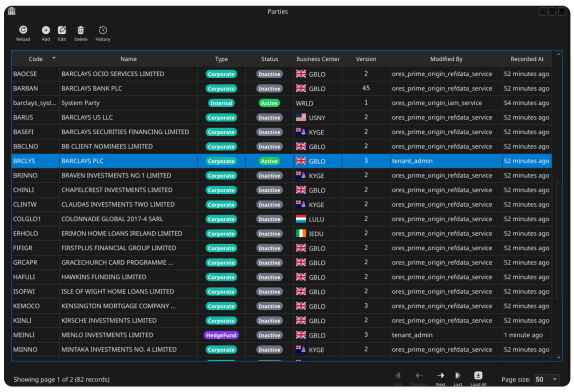


Figure 14.5: The Parties window, showing a page of parties. Each row shows the short code, name, party type and status badges, business center, version, last modifier, and when the record was last recorded; the status bar shows the current page and total record count.

The list is paginated; use the page controls at the bottom right to navigate. The toolbar buttons reload the list and open the add, edit, delete, and history actions. Double-clicking a row opens the **Party Details** dialog.

Party Details

The Party Details dialog has five tabs — **General**, **Identifiers**, **Contact Information**, **Hierarchy**, and **Provenance**. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the party as a whole.

General

The General tab carries the core fields of the party:

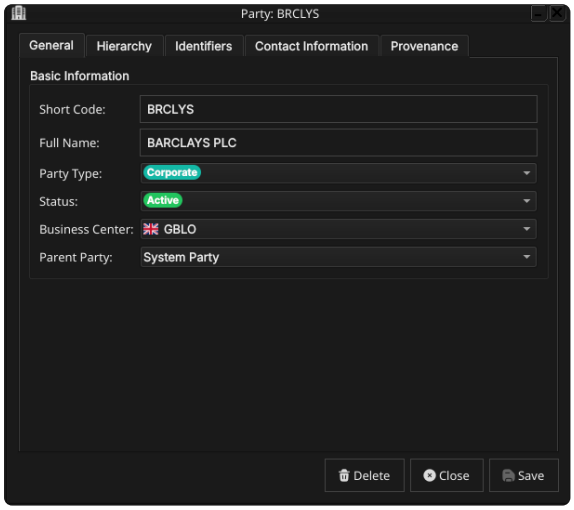


Figure 14.6: The General tab for a party, showing its short code, full name, the populated Party Type and Status combos, the flagged Business Center combo, and the Parent Party combo.

- **Short Code** — a brief mnemonic code. This is part of the natural key and cannot be changed after creation.

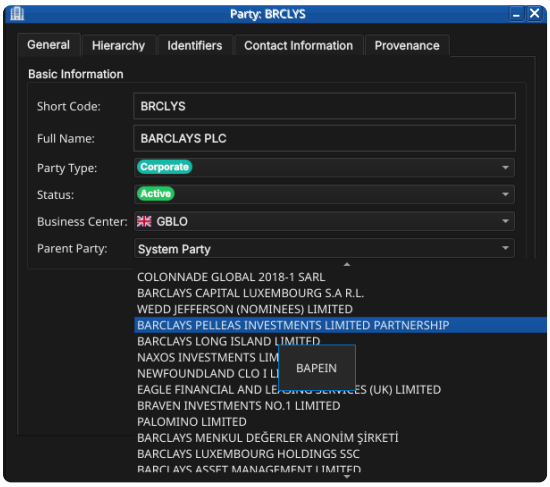
- **Full Name** — the party's official registered legal name.

- **Party Type** — the party's classification, chosen from a combo populated from the tenant's configured party types and shown as a colour-coded badge once selected.

- **Status** — the party's operational status, chosen from a combo populated from the tenant's configured party statuses and shown as a colour-coded badge once selected.

- **Business Center** — the party's primary location, chosen from a flagged combo of business center codes (see Business Centres).

- **Parent Party** — the party's immediate parent in the hierarchy, chosen from a combo listing every other party by full name, plus a leading **No Parent** entry. When editing an existing party, the party being edited is excluded from its own list of candidate parents, since a party can never be its own ancestor. **No Parent** is a genuinely valid choice — it is how the tenant's house root itself is represented — but saving a *second* party with no parent is rejected by the database's single-root constraint; see Hierarchy below for the full rule and how it plays out on that tab.



Identifiers

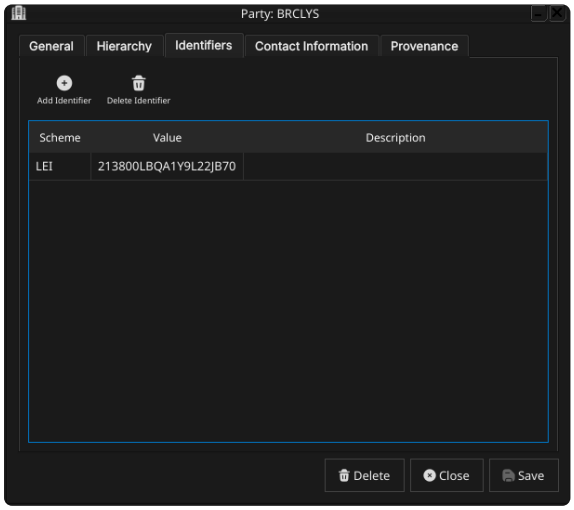


Figure 14.7: The Identifiers tab, listing the external identifiers held against a party.

The Identifiers tab is an editable table of external identifiers held against the party. Each row carries an **Identification Scheme** (chosen from the tenant's configured schemes), an **Identifier Value** (the actual code or string within that scheme), and an optional free-text **Description**. A party may hold at most one identifier per scheme, but may hold identifiers under as many different schemes as apply to it. Use the row toolbar to add, edit, or remove identifier rows; changes to this tab are saved together with the rest of the party record when **Save** is clicked.

The ten schemes OreStudio recognises span a spectrum from globally standardised to purely internal, reflecting how fragmented legal entity identification remains in practice: even the LEI, the closest thing to a universal answer, only reaches back to 2012 and still leaves gaps that jurisdiction-specific and venue-specific schemes fill.

Scheme	Standard / issuer	What it identifies
`LEI`	ISO 17442 / GLEIF	Any legal entity, globally, mandated for MiFID II/EMIR/Dodd-Frank/Basel III reporting.
`BIC`	ISO 9362 / SWIFT	Banks and financial institutions on the SWIFT network.
`MIC`	ISO 10383 / SWIFT	Trading venues (e.g. `XNYS`, `XLON`), where a party also acts as a venue.
`NATIONAL_ID`	Jurisdiction-specific	Passport, tax ID, or national ID card — covers MiFID II client identification.
`CEDB`	CFTC (US)	CFTC Entity Directory code for non-LEI entities in swap data reporting.
`NATURAL_PERSON`	Not standardised	Individuals (employee ID, trader ID); value interpreted contextually.
`ACER`	EU Agency for Energy Regulation	Non-LEI energy market participants, for REMIT reporting.
`DTCC_PARTICIPANT_ID`	DTCC (US)	Member firms in US clearing and settlement systems.
`MPID`	FINRA (US)	Broker-dealers and ATSs in US equities markets (also known as AII).
`INTERNAL`	Proprietary	A party's own OMS, CRM, or clearing-system client ID.

GLEIF and the Legal Entity Identifier (LEI)
The Global Legal Entity Identifier Foundation (GLEIF) is a not-for-profit organisation that oversees the global LEI system on behalf of financial regulators. A **Legal Entity Identifier (LEI)** is a 20-character alphanumeric code that uniquely identifies a legal entity participating in financial markets — a bank, a corporation, a fund, a branch. LEIs are mandated by major regulatory frameworks including MiFID II, EMIR, Dodd-Frank, and Basel III for trade reporting and counterparty identi-

fication.
GLEIF publishes the full registry of LEI entities and their corporate hierarchies (parent-child relationships) as open data. OreStudio uses the GLEIF dataset to seed counterparties and, when a party's root LEI is selected during tenant provisioning, to populate the initial party hierarchy from real organisational data — see Hierarchy below.
A **BIC** (Bank Identifier Code, ISO 9362) is an 8 or 11 character code identifying a specific financial institution, used in SWIFT messaging for settlement routing. GLEIF publishes a LEI-to-BIC mapping dataset that OreStudio also imports, allowing settlement systems to resolve a counterparty's BIC from its LEI.

These ten schemes are seeded on tenant provisioning, but the table above is not a fixed list either — the **Identification Scheme** combo itself is populated from the Party Id Schemes lookup, its own reference-data entity with a list and detail window under the *Reference Data* menu. Each scheme record carries a **Code** (`LEI`, `BIC`, and so on — the value stored against each identifier row), a **Name**, an optional **Description**, a **Coding Scheme** field (an external coding-scheme identifier, used where the value needs to resolve against an outside standard such as an FpML scheme URI), a **Display Order**, and a **Max Cardinality** field (an optional cap on how many identifiers under this scheme a single party may hold, left `(unset)` for schemes with no such limit), plus the standard Provenance tab.

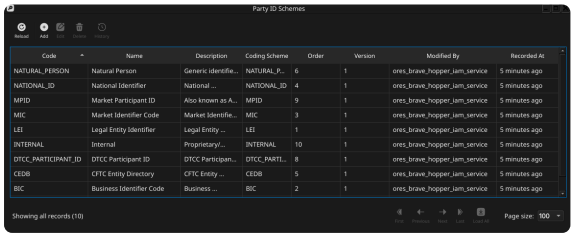


Figure 14.8: The Party Id Schemes window, listing the tenant's configured identification schemes by code, name, description, coding scheme, order, version, last modifier, and when the record was last recorded.

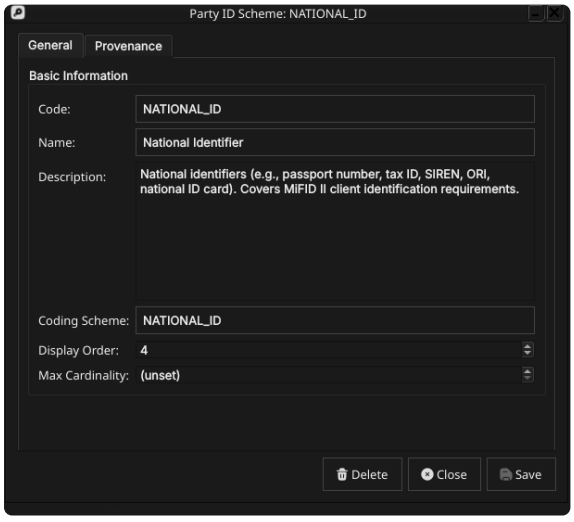


Figure 14.9: The Party Id Scheme Details dialog, showing its General tab (Code, Name, Description, Coding Scheme, Display Order, Max Cardinality) and Provenance tab.

A tenant can add, edit, or retire schemes here rather than being limited to the ten in the table above; editing, deleting, and history

for a scheme record follow the same change-reason, soft-close, and revert pattern as the party record itself, described later in this chapter.

Contact Information

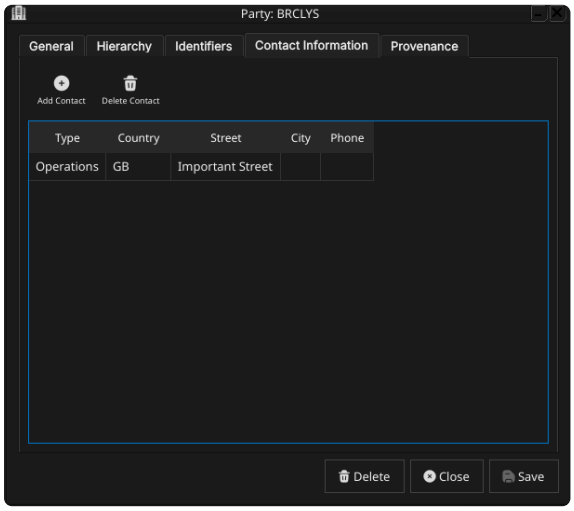


Figure 14.10: The Contact Information tab, listing the addresses and contact points held against a party.

The Contact Information tab is an editable table of addresses and contact points held against the party. Each row carries a **Contact Type** (e.g. *Legal*, *Operations*, *Settlement*, *Billing* — distinguishing, for instance, a registered legal address from an operational one) and an optional street address, city, state, postal code, phone, and email, plus an optional **Country** chosen from a flagged combo of ISO 3166-1 alpha-2 country codes. A party may hold more than one contact record, one per contact type. Use the row toolbar to add, edit, or remove rows; changes to this tab are saved together with the rest of the party record when **Save** is clicked.

Hierarchy

The Hierarchy tab is read-only and shows this party's position in the tenant's single corporate tree: its chain of ancestor parties up to the root, and its immediate children, if any. This is the screen that makes the party hierarchy rule introduced earlier in the chapter concrete, and it is worth spelling that rule out in full here.

Every tenant has, at most, two parties with no parent: the system party created automatically during provisioning (administrative only, never part of the business hierarchy), and — among the *operational* parties that actually own trades, books, and analytics results — exactly one **house root**, the organisation's own top-level legal entity. This is enforced structurally, not just by convention: the database carries a unique index that permits at most one operational party per tenant with no parent. Once that one slot is taken by the house root, every other operational party must supply a parent — a second attempt to create a rootless operational party is rejected outright.

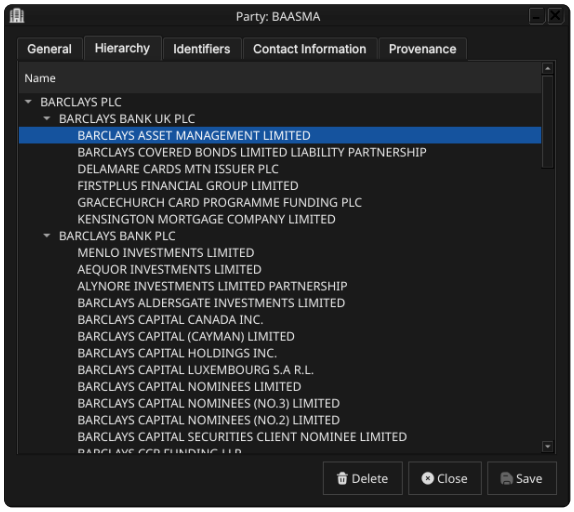


Figure 14.11: The Hierarchy tab for a party, showing its chain of ancestors up to the tenant's root party and its immediate children.

The practical consequence for the General tab's Parent Party combo (see General above) is that it does offer a **No Parent** choice, but the database enforces the rest: saving a second party with no parent is rejected outright, so in practice **No Parent** only succeeds for the tenant's one house root. A business entity newly onboarded into ORE Studio almost always needs to attach somewhere in the one tree that already exists for its tenant — a party can also be created from the shell's `provision party` command, which is not bound by any particular UI's field requirements.

In practice, most parties never go through the manual Create dialog at all: the shell-driven GLEIF (Global Legal Entity Identifier Foundation) reference data pipeline resolves an entity's real-world corporate-hierarchy relationships and assigns each imported party's parent automatically, level by level, from that external data. The manual Create dialog exists for the smaller number of parties added or corrected by hand outside that pipeline — and this Hierarchy tab is where the result of either path, automatic or manual, becomes visible for a given party.

Provenance

The Provenance tab is read-only and shows the audit metadata for the current version. See the Reference Data — Provenance section for a full description of the provenance fields common to all reference data entities.

Editing a party

To edit a party, open it in the Party Details dialog, make your changes, and click **Save**. Before the record is written, ORE Studio prompts for a change reason — the same *Change Reason Required* dialog shown in the Currencies chapter, and not repeated here. Select a **Reason** from the drop-down and add optional **Commentary**, then confirm. Every

save creates a new version; the previous version is never overwritten.

Deleting a party

To delete a party, open it and click **Delete**. ORE Studio asks for confirmation before the record is closed off. Deletion is a soft close — the record's history is preserved and remains visible in the History dialog.

Party history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

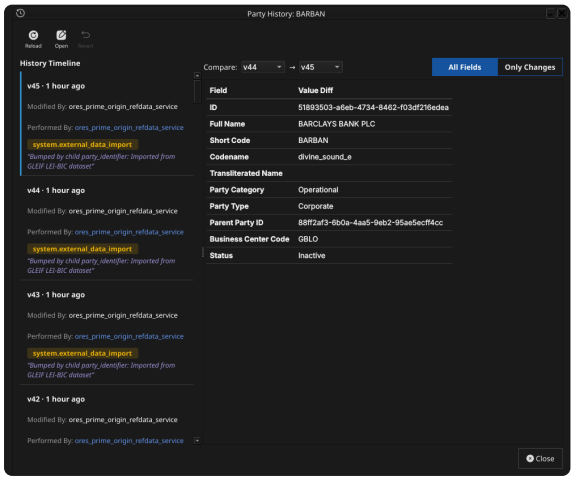


Figure 14.12: The History dialog for party BARCLAYS BANK PLC, comparing version 44 against version 45. The **Only Changes** toggle narrows the field list to what actually differs between the two selected versions.

Pick the two versions to compare from the **Compare** drop-downs on the right. The **All Fields** / **Only Changes** toggle switches between showing every field and showing only the fields that differ between the two selected versions. The **Revert** button reinstates any historical version as a **new** version, preserving the full audit chain — the old version is never rewritten.

Conclusion

The chapter set out to show that managing a party well follows from understanding what it represents and how it fits into the organisation's structure, and it has traced exactly that path. A party is an internal legal entity identified by a short code and full name, carrying a type and status, and sitting beneath a parent party in a single corporate hierarchy tree — with the sole exception of the tenant's own house root; the Parent Party combo's **No Parent** choice exists precisely for that one narrow exception, with the database rejecting any second attempt to use it. Building on that, the Qt interface took the record from the list view through its General, Identifiers, Contact

Information, Hierarchy, and Provenance tabs, and through the editing, deleting, and history workflows, where reverting to an earlier version demonstrated the same bitemporal audit trail described in the Reference Data chapter. The shell's `provision party` command covers bulk/GLEIF onboarding (see Hierarchy above), but general-purpose shell/CLI commands for adding, editing, deleting, or viewing the history of an individual party are not yet implemented — tracked separately rather than glossed over here. See the Counterparties chapter for the external-facing mirror of this one.

See also

- Tenants — the multi-tenant, multi-party conceptual model (house vs counterparty, the system party, hierarchy-based data visibility) this chapter assumes and builds on.
- Reference Data — the shared data quality and audit framework this chapter builds on.
- Currencies — the template chapter this one mirrors, including the Change Reason Required dialog.
- Counterparties — the external-facing mirror of this chapter.
- Business Centres — the business center a party links to, and the country flag a contact record may carry.
- Temporal Composite Entity Versioning — the architecture behind the Identifiers and Contact Information child tables.

15

Counterparties

This chapter examines the *counterparty* as a reference-data entity in ORE Studio. A counterparty is the external mirror of a party — sharing almost the same shape but modelling the opposite side of a relationship — and the chapter sets out where the two genuinely diverge, the identifiers and contact information a counterparty carries, and the complete lifecycle of managing counterparty records through the Qt interface.

Overview

The chapter advances the argument that managing a counterparty well means first understanding how it differs from the party chapter this one mirrors, then the interface that governs its lifecycle. It begins by establishing what a counterparty is in What is a Counterparty?, leaning on the Parties chapter for everything the two entities share rather than repeating it. From there it sets out The counterparty hierarchy — a real structural difference from a party's single house-root rule — before surveying the body of records in The Counterparties window and narrowing to a single record in Counterparty Details and its General, Identifiers, Contact Information, Hierarchy, and Provenance tabs. It then turns to changing a record under audit — Editing, Deleting, and Counterparty history — culminating in reverting to an earlier version, the point at which the bitemporal audit trail does its work. The Conclusion draws these steps together.

What is a Counterparty?

A *counterparty* is an external legal entity the organisation trades with — a bank, a broker-dealer, a corporate, a fund — as distinct from a party, which represents the organisation's own internal structure. The two entities are deliberately near-identical in shape — full name, short code, party type, status, business center, identifiers, contact in-

formation, a hierarchy field — because both are answering the same underlying question (what legal entity is this, and how do we recognise it consistently), just for opposite sides of a trade. Everything about legal-entity identification — GLEIF, the LEI standard, why identification is hard, the identifier schemes OreStudio recognises — is covered once in the Parties chapter's What is a Party? section and not repeated here.

Two field-level differences are worth calling out because they reflect a genuine modelling choice, not an oversight. First, a counterparty has no `codename` — the auto-generated, immutable, per-party message-queue and scheduled-job identifier a party carries — because that machinery exists to namespace a party's **own** processing (its queues, its scheduled reports), and a counterparty has none of its own to namespace. Second, a counterparty's **full name** is explicitly **not** unique within a tenant, unlike a party's: the same external legal name can legitimately recur across distinct branches or booking entities of the same institution, each with its own short code, so OreStudio indexes full name for search without constraining it to be distinct.

The counterparty hierarchy

A counterparty's `parent_counterparty_id` lets counterparties form group structures too — a subsidiary counterparty pointing at its parent institution — but the resemblance to a party's hierarchy ends there. A party's hierarchy is a single tree per tenant with exactly one designated house root, enforced by a database constraint; a counterparty's hierarchy has no such constraint at all. Any number of counterparties may have no parent, because each represents an independent external organisation — Barclays' counterparties include many unrelated banks and corporates, and there is no reason to force them into one shared tree the way a tenant's own internal structure is.

The Parent Counterparty combo reflects that directly: alongside every other counterparty, it offers a leading **No Parent** entry, and choosing it is a completely ordinary, unconstrained choice here — unlike a party's equivalent **No Parent** choice, nothing in the database restricts how many counterparties may have no parent. A standalone institution with no group structure to nest under is created or edited through this dialog like any other, simply by leaving Parent Counterparty set to **No Parent**.

The Counterparties window

Open the Counterparties window from the *Reference Data* menu. It lists all counterparties defined in the tenant, one row per counterparty, each row carrying the short code, full name, party type, status, business center, version, last modifier, and when the record was last

recorded — the same column layout as the Parties window.

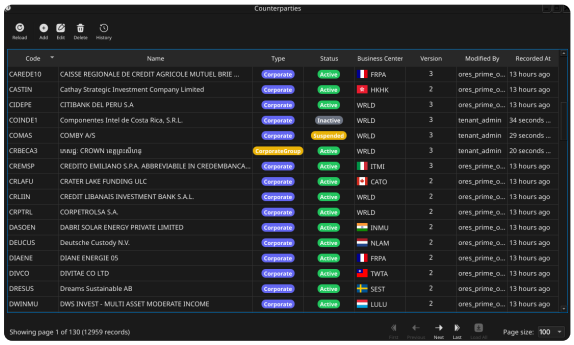


Figure 15.1: The Counterparties window, showing a page of counterparties. Each row shows the short code, name, party type and status badges, business center, version, last modifier, and when the record was last recorded; the status bar shows the current page and total record count.

The list is paginated; use the page controls at the bottom right to navigate. The toolbar buttons reload the list and open the add, edit, delete, and history actions. Double-clicking a row opens the **Counterparty Details** dialog.

Counterparty Details

The Counterparty Details dialog has five tabs — **General**, **Identifiers**, **Contact Information**, **Hierarchy**, and **Provenance** — the same set as Party Details. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the counterparty as a whole.

General

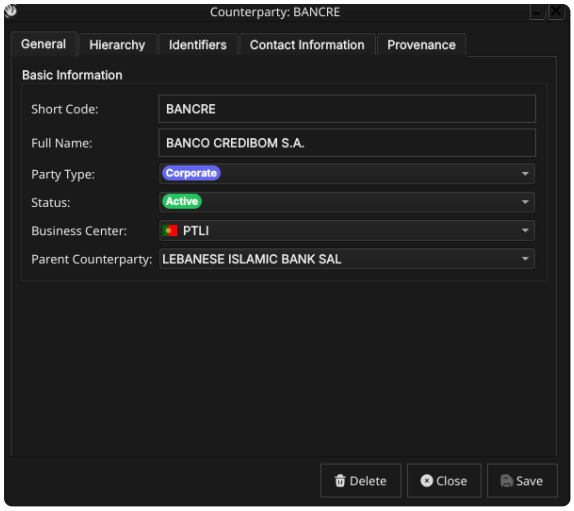
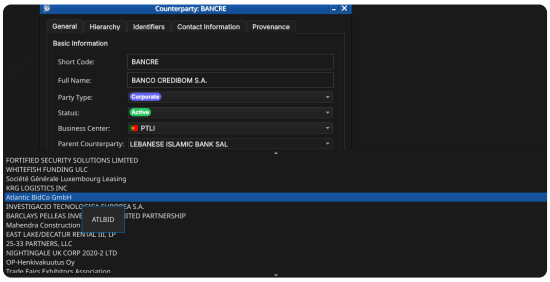


Figure 15.2: The General tab for a counterparty, showing its short code, full name, the populated Party Type and Status combos, the flagged Business Center combo, and the Parent Counterparty combo.

The General tab carries the core fields of the counterparty:

- **Short Code** — a brief mnemonic code. This is part of the natural key and cannot be changed after creation.
- **Full Name** — the counterparty's registered legal name. Unlike a party's, this is not required to be unique — see What is a Counterparty? above.

- **Party Type** — the counterparty's classification, chosen from the same tenant-configured party types a party uses, shown as a colour-coded badge once selected.
- **Status** — the counterparty's operational status, chosen from the same tenant-configured party statuses a party uses, shown as a colour-coded badge once selected.
- **Business Center** — the counterparty's primary location, chosen from a flagged combo of business center codes (see Business Centres).
- **Parent Counterparty** — the counterparty's immediate parent, chosen from a combo listing a leading **No Parent** entry followed by every other counterparty by full name. When editing an existing counterparty, the counterparty being edited is excluded from its own list of candidate parents. See The counterparty hierarchy above for why **No Parent** is a completely ordinary choice here, unlike the equivalent choice for a party.



Identifiers

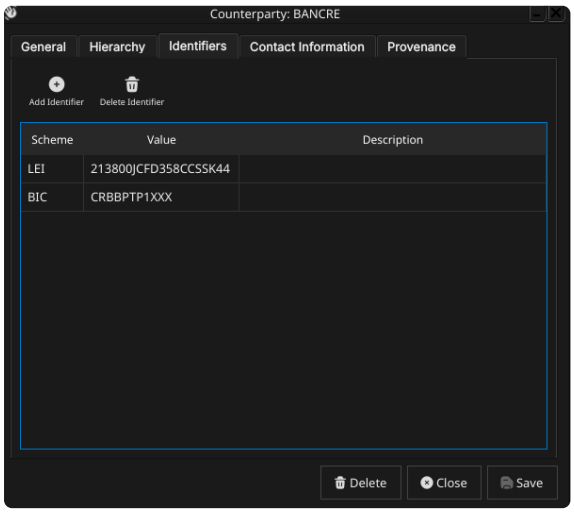


Figure 15.3: The Identifiers tab, listing the external identifiers held against a counterparty.

The Identifiers tab works exactly as a party's does — the same editable table, the same **Identification Scheme**, **Identifier Value**, and **Description** fields, the same ten schemes (LEI, BIC, and so on) — see Parties' Identifiers section for the full scheme table; it is not repeated

here. A counterparty may hold at most one identifier per scheme, but as many different schemes as apply to it.

Contact Information

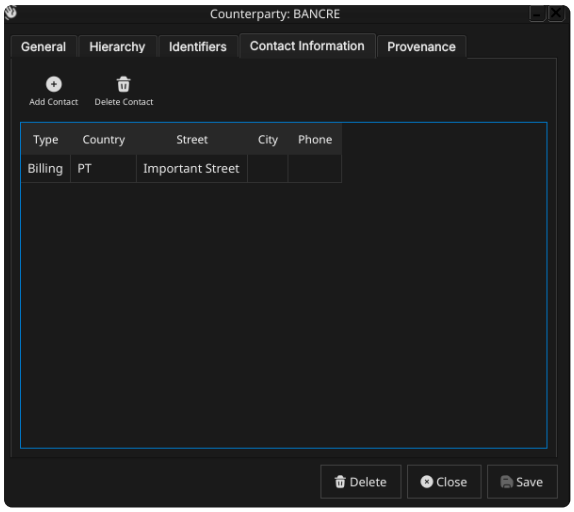


Figure 15.4: The Contact Information tab, listing the addresses and contact points held against a counterparty.

The Contact Information tab is, again, identical in shape to a party's: **Contact Type**, an optional street address, city, state, postal code, phone, and email, plus an optional flagged **Country** combo. A counterparty may hold more than one contact record, one per contact type.

Hierarchy

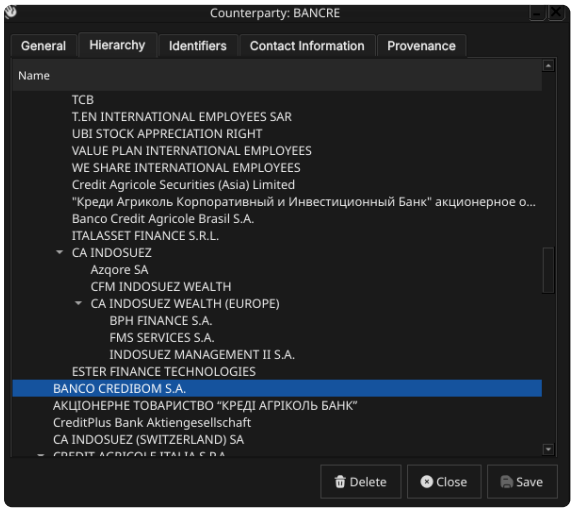


Figure 15.5: The Hierarchy tab for a counterparty, showing its chain of ancestors (if any) and its immediate children (if any).

The Hierarchy tab is read-only and shows this counterparty's position in whatever group structure it belongs to, if any — its chain of ancestor counterparties and its immediate children. Unlike a party's Hierarchy tab, there is no guarantee of a single connected tree reaching back to one root: an unrelated counterparty with no parent sim-

ply shows no ancestors at all. See The counterparty hierarchy above for why.

Provenance

The Provenance tab is read-only and shows the audit metadata for the current version. See the Reference Data — Provenance section for a full description of the provenance fields common to all reference data entities.

Editing a counterparty

To edit a counterparty, open it in the Counterparty Details dialog, make your changes, and click **Save**. Before the record is written, ORE Studio prompts for a change reason — the same *Change Reason Required* dialog shown in the Currencies chapter, and not repeated here. Select a **Reason** from the drop-down and add optional **Commentary**, then confirm. Every save creates a new version; the previous version is never overwritten.

Deleting a counterparty

To delete a counterparty, open it and click **Delete**. ORE Studio asks for confirmation before the record is closed off. Deletion is a soft close — the record's history is preserved and remains visible in the History dialog.

Counterparty history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

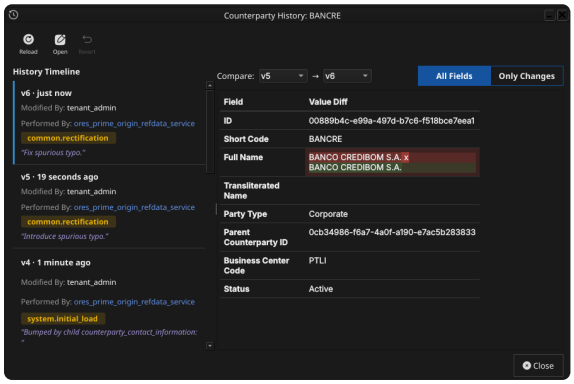


Figure 15.6: The History dialog for a counterparty, comparing two versions. The All Fields/Only Changes toggle narrows the field list to what actually differs between the two selected versions.

Pick the two versions to compare from the **Compare** drop-downs. The **All Fields** / **Only Changes** toggle switches between showing every field and showing only the fields that differ between the two selected versions. The **Revert** button reinstates any historical version

as a **new** version, preserving the full audit chain — the old version is never rewritten.

Conclusion

The chapter set out to show that managing a counterparty well follows from understanding how it differs from the party it mirrors, and it has traced exactly that path. A counterparty shares a party's full name, short code, type, status, business center, identifiers, and contact information almost verbatim, but diverges in two real ways: it carries no codename, since it has no processing of its own to namespace, and its hierarchy has no single-root constraint, since each counterparty represents an independent external organisation rather than a branch of the tenant's own structure. Building on that, the Qt interface took the record from the list view through its General, Identifiers, Contact Information, Hierarchy, and Provenance tabs, and through the editing, deleting, and history workflows, where reverting to an earlier version demonstrated the same bitemporal audit trail described in the Reference Data chapter. Shell and command-line access to counterparties is not yet implemented beyond the `provision tenant` GLEIF import path.

See also

- Parties — the internal mirror of this chapter; legal-entity identification, GLEIF/LEI, and the full identifier scheme table all live there rather than being duplicated here.
- Reference Data — the shared data quality and audit framework this chapter builds on.
- Currencies — the template chapter this one mirrors, including the Change Reason Required dialog.
- Business Centres — the business center a counterparty links to.
- Temporal Composite Entity Versioning — the architecture behind the Identifiers and Contact Information child tables.